

Statistics for Social Scientists Who Are A Little Afraid of Them

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Preface

Important Notes

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Attributions

Several chapters in this book are adapted from either Danielle Navarro's excellent textbook. The citation for that textbook is: Navarro, D. (2018). Learning statistics with R: A tutorial for psychology students and other beginners (version 0.6). The website is <https://compcogscisydney.org/learning-statistics-with-r/>.

Other chapters draw from Crump and colleagues online book. The citation for that book is: Crump, M.J.C., Navarro, D.J., & Suzuki, J. (2019, June 5). Answering Questions with Data: Introductory Statistics for Psychology Students. The website is: <https://www.crumplab.com/statistics/>

Both books are also published under the same creative commons license.

- In Chapter 1, I have used large swaths of text from the Navarro and Crump texts. I skipped the bits about reliability and validity (important, but usually this is covered in other courses) and I've added some different examples that make more sense to me!

- In Chapter 2, I have also used quite a bit from Navarro and Crump, though I've created a slightly different order and examples.

1 Introduction: Why Write This Book?

This book is aimed at social scientists who are relatively new to statistics – likely for an advanced undergraduate student or a new graduate student. After teaching statistics for psychology for nearly two decades, I’ve learned that a lot of my students come into my class with a fair amount of math phobia, and many of them believe that learning statistics is a “necessary evil” to do what they want to do with their careers. While, certainly, some students won’t necessarily engage much with statistics in their future jobs, I believe that effective and responsible social scientists must be able to understand, appreciate, and critique statistics. I have an odd sense of fondness for my students who are reluctant learners of statistics!

I also believe that I am somewhat unique among textbook writers in that I was not very good at statistics when I was in graduate school. I realize that’s an odd admission for someone who is writing an entire book on the topic. I had to work very hard at statistics because it didn’t “just make sense” to me. I had to do a lot of thinking, experimentation, and practice to really understand the math behind statistics. If you are someone who has a natural affinity for statistics, the topic was going to be a breeze for you no matter what. But if you are someone who finds they don’t like statistics, or if you feel that you have to work a lot harder than your classmates, I hope this book will help you make some connections and feel more confident in your skills! My approach in this book is that plenty of students are pretty rusty in terms of mathematical notation and concepts, so some plain definitions and practical examples can sometimes help clarify how numbers in a data set translate to useful information and answers from statistical tests. Hopefully together, we can move the needle to take you from feeling nervous or reluctant about statistics to someone who feels confident. Or maybe you will even realize that you love statistics once you understand them... stranger things have happened!

1.0.1 Data Sets

Throughout this book, I will provide some example data sets we can work through. All of these are fabricated, though some may be inspired by actual studies. I often do this to make the sample size manageable, or to ensure we get particular results so we can more easily dig into the math behind them. Although I wanted to generalize this book to fit most social sciences, my own PhD is in Psychology, so I tend to gravitate towards examples that fit that best. However, I try to find some examples that might speak to other social sciences. I think it is helpful for students to learn statistics in a realm that may not directly translate to their own field. The regression class I took in graduate school was in an Agricultural department, and I feel that trying to translate our findings about haystacks and recumbent cows into variables that related

to psychology was helpful to deepen my understanding. So, if you are not a psychologist, know that you are benefiting in some ways by having to take the extra step of considering how they relate to your own field.

All data sets are available as .csv files, which can easily be imported into most statistical analysis programs (SPSS, R, Stata). You can find all of these datasets [here](#); I also will link specific data sets when they crop up in the textbook.

2 Fundamentals of Statistics

2.1 Learning Objectives

- Understand what we mean when we talk about “statistics” and what their goal is for explaining the world around us.
- Understand why it is important for social scientists to understand statistics, even when it is not an active part of their career goals.
- Understand how statistics provide a link between theory and data in explaining social scientific phenomena.
- Differentiate between different scales of measurement (nominal, ordinal, ratio, interval).
- Differentiate between different types of research designs, and what types of questions each design can answer.

2.2 What are statistics?

To the surprise of many students, statistics is a fairly significant part of a social science education. To the surprise of nobody, statistics is very rarely the *favorite* part of one’s social science education. After all, if you really love the idea of statistics, you’d be enrolled in a statistics class, not a social science class. In my own experience of teaching statistics, there is usually a sizable portion of students who are quite unhappy to find themselves in my class. So I think a good place for us to start is addressing some of the common questions and myths people have about statistics.

To me, the most basic question is: What are statistics? Many of my students had their last experience with math in a calculus class, which often feels very abstract to them. As social scientists, it’s hard to think of why we might need a derivative, or why it might be important how quickly a cone fills with fluid. Statistics is much more prevalent in society, so I think students would greatly benefit from getting exposure to statistics in high school or college. But nobody has appointed me Queen of the World yet, so it is what it is.

I particularly like Oxford Language’s definition of statistics: “The practice or science of collecting and analyzing numerical data in large quantities, especially for the purpose of inferring proportions in a whole from those in a representative sample.” There are several aspects of this definition that are worth exploring further.

First, note that the definition references “numerical data” – while certainly qualitative data has its value, statistics join the scene once we want to quantify anything. Some statistics can be quite simple, such as percentages. Other statistics are more complex, like the logistic regression I cover towards the end of this book. But ultimately, we do need to have numbers and some variance in the numbers in order to make statistics valuable for answering our questions.

Second, the definition notes that we need “data in large quantities.” What is large here? That depends! But statistics are not terribly useful to us if we only have 5 data points. A common rule of thumb is to have a minimum of 30 data points for each condition we want to study, but ideally we’d like more than that.

Third, I like that this definition explains that the ultimate goal of statistics is “inferring proportions in a whole from those in a representative sample.” We’ll dig into this point in Chapter 2, but for now, it suffices to say that the entire goal of conducting statistics is to take a sample, and use that to learn about what might be happening in a broader population. But for this to be trustworthy, we need to be thoughtful about who is and who is not included in our sample.

In a nutshell, statistics allow us to take numerical information from a sample, and use that to understand what might be happening in a larger population of subjects.

2.3 Why do we need statistics?

Given this definition, why are social scientists so obsessed with statistics? Just about any empirical article will have mysterious statistics plastered all over it, but usually the authors don’t bother to explain why these statistics are so important to the points they are making. Students in social science often believe that research is a bit silly, because the field is just telling us something that is *common sense*.

It does seem like that. But the best answer for why we need statistics is that humans are not trustworthy when it comes to understanding social science phenomena. Social scientists are susceptible to many biases, temptations and frailties humans suffer from. Statistics serves as a safeguard from ourselves. What seems like “common sense” is really a dependence on gut instincts, persuasive verbal arguments, and human reasoning. But humans were not really designed to solve scientific problems; we were designed to handle day-to-day inferences that helped us avoid being devoured by tigers or whatever. The modern world has fewer tiger-based problems (except for zookeepers I guess) and so relying on people to gauge reality based on gut feelings prevents us from making wise guesses about how people and societies work. In fact, in some ways, our brains are set up in a way that makes us quite bad at predicting human characteristics and behaviors.

2.3.0.1 The curse of belief bias

People are mostly pretty smart. We're certainly much smarter than the other species we share the planet with (though many people might disagree). Our minds are amazing things, and we seem to be capable of the most incredible feats of thought and reason. That doesn't make us perfect though. And among the many things that psychologists have shown over the years is that we really to find it hard to be neutral, to evaluate evidence impartially and without being swayed by pre-existing biases. A good example of this is the **belief bias effect** in logical reasoning: if you ask people to decide whether a particular argument is logically valid (i.e., the conclusion would be true if the premises were true), we tend to be influenced by the believability of the conclusion, even when we shouldn't. For instance, here's a valid argument where the conclusion is believable:

No cigarettes are inexpensive (Premise 1)
Some addictive things are inexpensive (Premise 2)
Therefore, some addictive things are not cigarettes (Conclusion)

And here's a valid argument where the conclusion is not believable:

No addictive things are inexpensive (Premise 1)
Some cigarettes are inexpensive (Premise 2)
Therefore, some cigarettes are not addictive (Conclusion)

The *logical structure* of argument #2 is identical to the structure of argument #1, and they're both valid. However, in the second argument, there are good reasons to think Premise 1 is incorrect, and as a result, it's probably the case that the Conclusion is incorrect. But that's irrelevant to the topic at hand: an argument is deductively valid if the conclusion is a logical consequence of the premises. That is, a valid argument doesn't have to involve true statements.

Meanwhile, here's an invalid argument where the conclusion is believable:

No addictive things are inexpensive (Premise 1)
Some cigarettes are inexpensive (Premise 2)
Therefore, some addictive things are not cigarettes (Conclusion)

And here's a valid argument where the conclusion is not believable:

No cigarettes are inexpensive (Premise 1)
Some addictive things are inexpensive (Premise 2)
Therefore, some cigarettes are not addictive (Conclusion)

Suppose people are perfectly able to set aside their pre-existing biases about what is true and what isn't, and purely evaluate an argument on its logical merits. We'd expect 100% of people to say that the valid arguments are valid, and 0% of people to say that the invalid arguments are valid. So if you ran an experiment looking at this, you'd expect to see data like this:

Table 2.1: Expected and logical conclusions from evaluating the validity of an argument

	Conclusion feels true	Conclusion feels false
Argument is valid	100% say "valid"	100% say "valid"
Argument is invalid	0% say "valid"	0% say "valid"

However, in the classic study by Evans, Barston and Pollard (1983) they found the following results:

Table 2.2: The actual results of Evans et al (1983)

	Conclusion feels true	Conclusion feels false
Argument is valid	92% say "valid"	46% say "valid"
Argument is invalid	92% say "valid"	8% say "valid"

This table highlight the problem with trusting our gut instincts. Humans rely a lot on how the argument *feels*, so when a valid argument *feels* false, 54% of people are fooled into saying it's not valid. Likewise, when an invalid conclusion *feels* true, only 8% of people will realize that it is, in fact, not valid. When reality matches the feeling, people do pretty well. But when there is a mismatch, people are much worse at evaluating the validity of a statement.

The good news here is that we have a great tool for solving this problem, and the name of that tool is "statistics." Relying on statistics helps us get away from depending too much on common sense or what "feels true" to us when talking about human behavior. Statistics *typically* keep us honest about what data does and does not tell us. I won't go so far as to say that statistics *always* keep us honest, though! There are a few things that might prevent us from finding the truth.

First of all, data often has some interesting nooks and crannies that hide interesting secrets from us. One thing I absolutely love about statistics is that often we can use it to solve puzzles, but we might need to be creative in how we apply it. As one example from my own research, I was once tasked with determining whether a particular assessment predicted managerial performance. I had manager scores on the assessment, and several years of performance data (so, I had performance data from 2015, 2016, and so on). Managers started at different times, so they each had some performance data that was missing. Initially, there was no correlation between assessment performance and job performance for each year. However, when I rearranged the data so that it reflected their performance based on *how long they were in*

the role (so a manager who started in 2015 would have their performance in 2016 as “Year 1 performance”, whereas a manager who started in 2018 would have their performance in 2019 as “Year 1 performance”), suddenly it became crystal clear that managers who did poorly on their assessment absolutely tanked in job performance in their third year. I needed to do some critical thinking about what “time” represented in the data to unlock its secrets.

Second, there are some secrets we can’t unlock with our data. For my previous example, I was able to show that the assessment was predictive of performance. Why did managers with poor scores crash out in their third year, instead of in their first or second? Who knows! I certainly could speculate, but the data I had available wasn’t going to answer that question (and frankly, it would be essentially impossible to collect data that would provide a definitive answer to that question).

Third, statistics is a “garbage in, garbage out” situation. If I have crummy data, I can certainly use statistics to get an answer... but that answer is only as good as the data I’m using. We’ll talk more about this in future chapters, but statistics is only part of the puzzle; issues like good methodology and appropriate sampling have major effects on our findings too.

Finally, there are many ways that social scientists can misrepresent data or mislead their audience on what the data means. Sometimes this might be unintentional, but “lying with statistics” is a topic we will explore later in this text. It’s a major ethical concern, because we make a lot of important decisions based on data and statistics, and misrepresentation of the data can have major repercussions.

2.3.1 Why do I have to know about statistics?

Often, my students like to gripe that they don’t plan to be directly involved with statistics. I have plenty of students who plan to go into fields where they are not going to do much research – maybe they will go on to treat patients in a clinic, or work in companies on developing training, or work in schools designing support programs for students. So if you don’t plan on *doing* statistics, why bother to learn them?

First, while you might be able to outsource a lot of the “doing of statistics” to someone else, you need to at least be a competent consumer of statistics. If you plan on designing studies, you will need to know how different designs related to statistics. If you plan on reading research, you’ll be able to understand and critique that research if you know statistics. Hopefully, knowing statistics will also help alert you to any “fishiness” in the data that might suggest errors or malfeasance.

Second, knowing statistics also ensures that if someone is trying to pull a fast one on you, you’ll know. I once sat through a vendor presentation where the presenter clearly did not know he was dealing with a room full of statisticians, and he got torn to shreds for trying to sell us a measure that was absolute baloney.

Finally, it's *expensive* to hire someone to do your stats! Most organizations aren't interested in spending that money if they can have an expert on staff. So, you'll be more employable if you can at least speak competently about statistics, even if that's not the focus of your job. Most workplaces are becoming more data-savvy, so this is a good skill to have in modern society. And yes, I know that artificial intelligence can serve as one way around knowing statistics. Unfortunately, at this time most available AI platforms are quite mediocre at handling numbers and data. So, even if you plan to use AI to help you, it is not dependable and you need to be ready to double check its answers.

2.4 There's more to research methods than statistics

So far, most of what I've talked about is statistics, and so you'd be forgiven for thinking that statistics the only thing that's important here. Admittedly, about 99% of student *fears* relate to the statistics part of the course, so I've focused on the stats in this discussion, and hopefully I've convinced you that statistics matters, and more importantly, that it's not to be feared. That being said, it's pretty typical for introductory research methods classes to be very stats-heavy because you almost always find yourself needing statistics before you need the other research methods training. Why? Frankly, the statistics tend to be the easier part of understanding and conducting research. Methodology is harder to master. They both matter. I really do want to stress that research design is just as important as data analysis, and this book does spend a fair amount of time on it. However, while statistics has a kind of universality, and provides a set of core tools that are useful for most types of psychological research, the research methods side isn't quite so universal. There are some general principles that everyone should think about, but a lot of research design is very idiosyncratic, and is specific to the area of research that you want to engage in.

2.4.1 A brief introduction to research design

In this chapter, we're going to start thinking about the basic ideas that go into designing a study, collecting data, checking whether your data collection works, and so on. It won't give you enough information to allow you to design studies of your own, but it will give you a lot of the basic tools that you need to assess the studies done by other people. However, since the focus of this book is much more on data analysis than on data collection, I'm only giving a very brief overview. Note that this chapter is "special" in two ways. Firstly, it's much more psychology-specific than the later chapters. Secondly, it focuses much more heavily on the scientific problem of research methodology, and much less on the statistical problem of data analysis. Nevertheless, the two problems are related to one another, so it's traditional for stats textbooks to discuss the problem in a little detail. This chapter relies heavily on @Campbell1963 for the discussion of study design, and @Stevens1946 for the discussion of scales of measurement.

2.4.2 Theoryland and Dataland: Creating an Operational Definition

One aspect of social science that's unique from other sciences is that most of what we measure isn't tangible. In "hard" science, there are many opportunities to measure things like weight, chemical makeup, hardness, speed. We often can observe these qualities in materials. Social science is rarely afforded this privilege. While some qualities might be directly observable (such as eye color), other variables we assume exist. Take "intelligence" as an example – this is a quality we think exists, but we can't directly observe it (the size and weight of someone's brain is not the same thing as "intelligence"). Even trying to define what "intelligence" is and how it looks when someone has a lot of it is a subject of great debate.

Ultimately, I like to think that social scientists live in two different worlds called Theoryland and Dataland. "Intelligence" exists in Theoryland because it theoretically exists, but we can't directly measure it. But we need some terminology to understand how things work in Theoryland.

First, we can think of "intelligence" as a **theoretical construct**. In social science, we use constructs all the time – intelligence, love, socioeconomic status – this are all things we think exist but can't be directly observed or handled. So instead, we need to come up with a **measure** – a method or tool we use to make our observations. This might be a question on a survey, or a behavioral observation, or a brain scan. **An operational definition** is the logical connection between the measure and the theoretical construct. If I want to create an operational definition of the construct "intelligence," I might do that by giving someone an IQ test, or asking someone to solve a puzzle in a certain amount of time. Ultimately, how a subject performs on this measure provides us with a **variable** – that is, the "data" we end up with in our data set. The data set exists in our other home, Dataland, where we are able to observe information that reflects (but is not identical to) our theoretical construct.

In other words, this operational definition is essentially a bridge between Theoryland (where our construct exists) and Dataland (where we apply a measure and get data that tells us about what's going on in the real world around us). Sometimes this bridge is well-designed and solid, and other times it might not be.

Take, for example, the construct of "age." We aren't going to be able to directly observe someone's age by being there at their birth and precisely measuring from then until the instant we collect data. Instead, we have to operationalize what we mean by age. It seems straightforward, but it really isn't! For instance, as an Industrial-Organizational psychologist who studies working adults, usually I assume that it is okay to measure age in years. But if you're a Developmental psychologist, that's way too crude, and so you often measure age in *years and months* (if a child is 2 years and 11 months, this is usually written as "2;11"). If you're interested in newborns, you might want to measure age in *days since birth*, maybe even *hours since birth*. In other words, the way in which you specify the allowable measurement values is important and dependent upon the type of questions you ask and the type of research you intend to do.

Looking at this a bit more closely, you might also realize that the concept of “age” isn’t actually all that precise. In general, when we say “age” we implicitly mean “the length of time since birth”. But that’s not always the right way to do it. Suppose you’re interested in how newborn babies control their eye movements. If you’re interested in kids that young, you might also start to worry that “birth” is not the only meaningful point in time to care about. If Baby Alice is born 3 weeks premature and Baby Bianca is born 1 week late, would it really make sense to say that they are the “same age” if we encountered them “2 hours after birth”? In one sense, yes: by social convention, we use birth as our reference point for talking about age in everyday life, since it defines the amount of time the person has been operating as an independent entity in the world, but from a scientific perspective that’s not the only thing we care about. When we think about the biology of human beings, it’s often useful to think of ourselves as organisms that have been growing and maturing since conception, and from that perspective Alice and Bianca aren’t the same age at all. So you might want to define the concept of “age” in two different ways: the length of time since conception, and the length of time since birth. When dealing with adults, it won’t make much difference, but when dealing with newborns it might.

Moving beyond these issues, there’s the question of methodology. What specific “measurement method” are you going to use to find out someone’s age? As before, there are lots of different possibilities:

- You could just ask people “how old are you?” The method of self-report is fast, cheap and easy, but it only works with people old enough to understand the question, and some people lie about their age.
- You could ask an authority (e.g., a parent) “how old is your child?” This method is fast, and when dealing with kids it’s not all that hard since the parent is almost always around. It doesn’t work as well if you want to know “age since conception”, since a lot of parents can’t say for sure when conception took place. For that, you might need a different authority (e.g., an obstetrician). For adults, parents might be harder to use as an authority, so maybe a friend or partner is a better choice... but we are trusting that they actually know the right answer to our question.
- You could look up official records, like birth certificates. This is time consuming and annoying, but it has its uses (e.g., if the person is now dead).

As you can see, the ways we think about and measure age have different pros and cons that are going to be influenced by our research question. Some operational definitions are clearly poor (asking a two-year-old their age is sketchy at best) but there is no perfect way to measure “age” that will work for all research contexts. At the very least, when we try to create a decent operational definition of a variable, it involves:

- Being precise about what you are trying to measure: For instance, does “age” mean “time since birth” or “time since conception” in the context of your research?

- Determining what method you will use to measure it: Will you use self-report to measure age, ask a parent, or look up an official record? If you're using self-report, how will you phrase the question?
- Defining the set of the allowable values that the measurement can take: Note that these values don't always have to be numerical, though they often are. When measuring age, the values are numerical, but we still need to think carefully about what numbers are allowed. Do we want age in years, years and months, days, hours? Etc. For other types of measurements (e.g., gender), the values aren't numerical. But, just as before, we need to think about what values are allowed. If we're asking people to self-report their gender, what options do we allow them to choose between? Is it enough to allow only "male" or "female"? Do you need an "other" option? Or should we not give people any specific options, and let them answer in their own words? And if you open up the set of possible values to include all verbal response, how will you interpret their answers?

Operationalization is a tricky business, and there's no "one, true way" to do it. The way in which you choose to operationalize the informal concept of "age" or "gender" into a formal measurement depends on what you need to use the measurement for. Often you'll find that the community of scientists who work in your area have some fairly well-established ideas for how to go about it. In other words, operationalization needs to be thought through on a case by case basis. Nevertheless, while there are a lot of issues that are specific to each individual research project, there are some aspects to it that are pretty general.

2.4.3 Scales of Measurement

As we think about operational definitions and their link to data, we also need to consider what type of data we have available. A very useful concept for distinguishing between different types of variables is what's known as **scales of measurement**. Sorry to spoil my punchline later on, but understanding scales of measurement is *vital* for knowing which statistics you'll need to answer your research questions!

2.4.4 Nominal scale

A **nominal scale** variable (also referred to as a **categorical** variable) is one in which there is no particular actual "amount" attached to the variable. Eye color is a classic example of a nominal scale – we usually think of this in categories like "green" or "brown" but there's no amount of anything attached to this variable. Sometimes you might have a number attached to a nominal scale, but often this number doesn't actually mean anything. For example, zip codes or postal codes are numbers that don't reflect an amount of anything; the zip code 55433 isn't "more zippy" than the zip code 11201 – they are just categories. Or if we're measuring someone's religion, if we code "Buddhist" as a 1 and "Muslim" as a 2 in our data set, the higher value

doesn't indicate that Islam is better or more religious than Buddhism; the numbers assigned are arbitrary and just reflect that people have checked a different box on their survey.

Nominal scales are fairly limited in the types of mathematical manipulations we can do with them. It doesn't make sense for us to talk about the "average religion" of a sample to be 1.75. At most, we can report on the frequencies of different groups, like reporting that 25 people in our sample report being Buddhists.

2.4.5 Ordinal scale

Ordinal scale variables have a bit more structure than nominal scale variables, but not by a lot. An ordinal scale variable is one in which there is a natural, meaningful way to order the different possibilities, but you can't do anything else. The usual example given of an ordinal variable is "finishing position in a race". You *can* say that the person who finished first was faster than the person who finished second, but you *don't* know how much faster. As a consequence we know that 1st was faster than 2nd, and we know that 2nd was faster than 3rd, but the difference between 1st and 2nd might be much larger than the difference between 2nd and 3rd.

Here's an more psychologically interesting example. Suppose I'm interested in people's attitudes to climate change, and I ask them to pick one of these four statements that most closely matches their beliefs:

- (1) Temperatures are rising, because of human activity
- (2) Temperatures are rising, but we don't know why
- (3) Temperatures are rising, but not because of humans
- (4) Temperatures are not rising

Notice that these four statements actually do have a natural ordering, in terms of "the extent to which they agree with the current science". Statement 1 is a close match, statement 2 is a reasonable match, statement 3 isn't a very good match, and statement 4 is in strong opposition to the science. So, in terms of the thing I'm interested in (the extent to which people endorse the science), I can order the items as $1 > 2 > 3 > 4$. Since this ordering exists, it would be very weird to list the options like this...

1. Temperatures are rising, but not because of humans
2. Temperatures are rising, because of human activity
3. Temperatures are not rising

4. Temperatures are rising, but we don't know why

...because it seems to violate the natural “structure” to the question.

So, let's suppose I asked 100 people these questions, and got the following answers:

Table 2.3: Possible results of asking about climate change

	Number
(1) Temperatures are rising, because of human activity	51
(2) Temperatures are rising, but we don't know why	20
(3) Temperatures are rising, but not because of humans	10
(4) Temperatures are not rising	19

When analyzing these data, it seems quite reasonable to try to group (1), (2) and (3) together, and say that 81 of 100 people were willing to *at least partially* endorse the science. And it's *also* quite reasonable to group (2), (3) and (4) together and say that 49 of 100 people registered *at least some disagreement* with the dominant scientific view. However, it would be entirely bizarre to try to group (1), (2) and (4) together and say that 90 of 100 people said...what? There's nothing sensible that allows you to group those responses together at all.

That said, notice that while we *can* use the natural ordering of these items to construct sensible groupings, what we *can't* do is average them. For instance, in my simple example here, the “average” response to the question is 1.97. If you can tell me what that means, I'd love to know. Because that sounds like gibberish to me!

2.4.6 Interval scale

In contrast to nominal and ordinal scale variables, **interval scale** and ratio scale variables are variables for which the numerical value is genuinely meaningful. In the case of interval scale variables, the *differences* between the numbers are interpretable, but the variable doesn't have a “natural” zero value. One good example of this is SAT scores. SAT scores range from 400-1600, so it's impossible to even get a zero on this test, and if you somehow managed to do so, it's not an indication you have “no intelligence.” But the difference between a score of 1000 and 1100 is the same as the distance between 1100 and 1200. Another example is years – the distance between 1990 and 2000 is ten years, and the distance between 2000 and 2010 is also ten years. The year of 0 doesn't mean “no time at all” – it's just where we started counting years the way we continue to number them in modern times.

2.4.7 Ratio scale

The fourth and final type of variable to consider is a **ratio scale** variable. Just like with an interval variable, the numbers on a ratio scale indicate a fairly precise measure of the amount of a variable. It differs from interval scales in two important ways:

- A zero on a ratio scale is a true zero, in that it indicates an absence of that quality.
- You can reasonably multiply and divide the values on this scale.

As an example, consider reaction time. If, somehow, someone was able to get zero milliseconds as their reaction time, it really does mean “no time at all.” We can also do multiplication and division with reaction time – someone who takes 250 milliseconds to respond is twice as fast as someone who takes 500 milliseconds. Or if I’m studying snacking behavior, I might count how many candies someone takes out of a dish. In this case, someone could take zero, and that would mean “no snacking.” And someone who eats 10 candies has eaten twice as many candies as someone who eats 5 candies.

I find that students have a hard time distinguishing interval and ratio scales, so the good news is we largely treat them the same in how we conduct statistics in social sciences, but it’s good to know the difference between both scales in case it comes up.

It’s also worth noting that we can measure the same variable with several different scales depending on how we create our measure. Age is a great example of this. If I ask someone to write in how many years old they are, that would be a ratio scale (a zero means I somehow have given my survey to a newborn who has been alive for zero years; someone who is 90 is three times older than someone who is 30). However, I could also **polytomize** my variable, where I take age and split it into several options, such as asking people to check a box that describes their age range (18 to 25; 26 to 30; 31 to 45...). Or I could **dichotomize** my variable by asking them to choose between two age options (“40 and under” versus “over 40”). In this case, I’ve taken what is a ratio variable, and I’ve turned it into an ordinal variable.

We generally avoid polytomizing or dichotomizing a variable because doing that loses information. I can always take a ratio variable and turn it into an ordinal variable after the fact, but I can’t do the reverse. If someone checks the “26 to 30” box for their age, I will have no way of knowing what their exact age is, so I won’t be able to talk about differences between 27 and 28-year-old respondents. Occasionally, there might be a good reason to dichotomize or polytomize a variable – one example of this is whether someone passes or fails a test. While the score on the test is a ratio variable, there is an important distinction between someone who gets a 59% on the test (failing) and someone who gets a 60% on the test (passing). But doing this basically means that people who get a 30% versus a 59% on the test are more similar to each other than two people who get a 59% and a 60% on the test – dichotomizing their score into a pass/fail ordinal variable collapses people in the groups into broad buckets, which is only helpful in certain circumstances.

2.4.8 Continuous versus discrete variables

There's a second kind of distinction that you need to be aware of, regarding what types of variables you can run into. This is the distinction between continuous variables and discrete variables. The difference between these is as follows:

- A **continuous variable** is one in which, for any two values that you can think of, it's always logically possible to have another value in between.
- A **discrete variable** is, in effect, a variable that isn't continuous. For a discrete variable, it's sometimes the case that there's nothing in the middle.

These definitions probably seem a bit abstract, but they're pretty simple once you see some examples. For instance, response time is continuous. If Alan takes 3.1 seconds and Ben takes 2.3 seconds to respond to a question, then it's possible for Cameron's response time to lie in between, by taking 3.0 seconds. And of course it would also be possible for David to take 3.031 seconds to respond, meaning that his response time would lie in between Cameron's and Alan's. And while in practice it might be impossible to measure response time that precisely, it's certainly possible in principle. Because we can always find a new value in between any two other ones, we say that response time is continuous.

Discrete variables occur when this rule is violated. For example, nominal scale variables are always discrete: there isn't a type of transportation that falls "in between" trains and bicycles, not in the strict mathematical way that 2.3 falls in between 2 and 3. So transportation type is discrete. Similarly, ordinal scale variables are always discrete: although "2nd place" does fall between "1st place" and "3rd place", there's nothing that can logically fall in between "1st place" and "2nd place". Interval scale and ratio scale variables can go either way. As we saw above, response time (a ratio scale variable) is continuous. Temperature in degrees Celsius (an interval scale variable) is also continuous. However, the year you went to school (an interval scale variable) is discrete. There's no year in between 2002 and 2003. The number of questions you get correct on a true-or-false test (a ratio scale variable) is also discrete: since a true-or-false question doesn't allow you to be "partially correct", there's nothing in between 5/10 and 6/10. The table summarizes the relationship between the scales of measurement and the discrete/continuity distinction. Cells with a tick mark correspond to things that are possible. I'm trying to hammer this point home, because (a) some textbooks get this wrong, and (b) people very often say things like "discrete variable" when they mean "nominal scale variable". It's very unfortunate.

Table 2.4: The relationship between the scales of measurement and the discrete/continuity distinction. Cells with an x correspond to things that are possible.

	continuous	discrete
nominal		x
ordinal		x

	continuous	discrete
interval	x	x
ratio	x	x

2.4.9 Some complexities

Okay, I know you're going to be shocked to hear this, but ...the real world is much messier than this little classification scheme suggests. Very few variables in real life actually fall into these nice neat categories, so you need to be kind of careful not to treat the scales of measurement as if they were hard and fast rules. It doesn't work like that: they're guidelines, intended to help you think about the situations in which you should treat different variables differently. Nothing more.

So let's take a classic example, maybe *the* classic example, of a psychological measurement tool: the **Likert scale**. The humble Likert scale is the bread and butter tool of all survey design. You yourself have filled out hundreds, maybe thousands of them, and odds are you've even used one yourself. Suppose we have a survey question that looks like this:

Which of the following best describes your opinion of the statement that “all pirates are freaking awesome” ...

and then the options presented to the participant are these:

- (1) Strongly disagree
- (2) Disagree
- (3) Neither agree nor disagree
- (4) Agree
- (5) Strongly agree

This set of items is an example of a 5-point Likert scale: people are asked to choose among one of several (in this case 5) clearly ordered possibilities, generally with a verbal descriptor given in each case. However, it's not necessary that all items be explicitly described. This is a perfectly good example of a 5-point Likert scale too:

- (1) Strongly disagree
- (2)
- (3)
- (4)
- (5) Strongly agree

Likert scales are very handy, if somewhat limited, tools. The question is, what kind of variable are they? They're obviously discrete, since you can't give a response of 2.5. They're obviously

not nominal scale, since the items are ordered; and they're not ratio scale either, since there's no natural zero.

But are they ordinal scale or interval scale? One argument says that we can't really prove that the difference between "strongly agree" and "agree" is of the same size as the difference between "agree" and "neither agree nor disagree". In fact, in everyday life it's pretty obvious that they're not the same at all. So this suggests that we ought to treat Likert scales as ordinal variables. On the other hand, in practice most participants do seem to take the whole "on a scale from 1 to 5" part fairly seriously, and they tend to act as if the differences between the five response options were fairly similar to one another. As a consequence, a lot of researchers treat Likert scale data as if it were interval scale. It's not interval scale, but in practice it's close enough that we usually think of it as being **quasi-interval scale**.

2.5 The role of variables: predictors and outcomes

Okay, I've got one last piece of terminology that I need to explain to you before moving away from variables. Normally, when we do some research we end up with lots of different variables. Then, when we analyse our data we usually try to explain some of the variables in terms of some of the other variables. It's important to keep the two roles "thing doing the explaining" and "thing being explained" distinct. So let's be clear about this now. Firstly, we might as well get used to the idea of using mathematical symbols to describe variables, since it's going to happen over and over again. Let's denote the "to be explained" variable Y , and denote the variables "doing the explaining" as X_1 , X_2 , etc.

Now, when we doing an analysis, we have different names for X and Y , since they play different roles in the analysis. The classical names for these roles are **independent variable** (IV) and **dependent variable** (DV). The IV is the variable that you use to do the explaining (i.e., X) and the DV is the variable being explained (i.e., Y). The logic behind these names goes like this: if there really is a relationship between X and Y then we can say that Y depends on X , and if we have designed our study "properly" then X isn't dependent on anything else. However, I personally find those names horrible: they're hard to remember and they're highly misleading, because (a) the IV is never actually "independent of everything else" and (b) if there's no relationship, then the DV doesn't actually depend on the IV. And in fact, because I'm not the only person who thinks that IV and DV are just awful names, there are a number of alternatives that I find more appealing.

For example, in an experiment the IV refers to the **manipulation**, and the DV refers to the **measurement**. So, we could use **manipulated variable** (independent variable) and **measured variable** (dependent variable).

Table 2.5: The terminology used to distinguish between different roles that a variable can play when analyzing a data set.

role of the variable	classical name	modern name
“to be explained”	dependent variable (DV)	Measurement
“to do the explaining”	independent variable (IV)	Manipulation

We could also use **predictors** and **outcomes**. The idea here is that what you’re trying to do is use X (the predictors) to make guesses about Y (the outcomes). This is summarized in the table:

Table 2.6: Alternative terminology used to distinguish between different roles that a variable can play when analyzing a data set.

role of the variable	classical name	modern name
“to be explained”	dependent variable (DV)	outcome
“to do the explaining”	independent variable (IV)	predictor

Because in this book we’ll be talking about many different research designs other than experiments, I will use “predictors” and “outcomes” as the terminology going forward.

Speaking of experiments... it’s time to tackle research designs!

2.6 Research Designs

It took me a while in graduate school to actually understand what my professors meant when they talked about “research designs.” It sounds fancier than it is – all it means is that certain methods allow us to answer certain questions. The type of design we use also affects what statistics we should use, so it’s an important distinction for the topic of statistics. The types of questions we ask also tend to be affected by how much attention our research topic has garnered so far... it’s almost like thinking about the life span of a variable you plan to study.

2.6.1 Descriptive Research: “What is Y ?”

One of the very first steps in learning about a variable is defining it! Researchers design new constructs all the time, and part of learning about that construct is understanding what is and what is not part of that construct. One variable I studied in graduate school was something called “workplace incivility” – this is behavior that is not so intense that you can call it abuse, but it’s low-grade disrespectful or rude behavior. It could include verbal behavior (“That’s

quite an... interesting outfit you're wearing today!") or non-verbal behavior (rolling your eyes, refusing to respond to a question).

Early in the study of workplace incivility, researchers needed to learn a bit about what types of things people find uncivil at work, and what types of behaviors were frequent or infrequent. To learn more about this construct, researchers conducted **descriptive research** – research that helped describe and define this variable. Some of this research was fully qualitative, such as asking people what they considered to be rude and disrespectful (but not abusive) behavior at work. Or researchers might conduct a case study to explore a specific incident of incivility. Some research was quantitative, where researchers quantified what types of behaviors happened most frequently. Statistics in this kind of research tend to be pretty minimal – often we use statistics like frequencies or mean/median/mode to help us get a handle on what that variable is. In order to study incivility, we need to understand how it differs from related but distinct variables, such as “workplace abuse” or “workplace aggression.”

2.6.2 Correlational Research: “What is related to Y?”

Once researchers had a handle on a good definition for workplace incivility, they started to ask what relates to that variable. Who is likely to experience it? Who is likely to instigate it? What are some emotions and behaviors that relate to working in a place with a lot of incivility? This research is known as **correlational research**. The name is a little unfortunate – despite the word “correlation” being in the name, you don’t necessarily use a statistical correlation to test it. When you hear people say “Correlation is not causation” usually they are talking about something that was studied using a correlational *design*.

In correlational research, we are able to talk about other variables that are associated with it, but we *cannot make assertions about causality*. For example, imagine I give employees a survey asking them about how much they experience incivility in their workplace, and I discover that people who are very low in agreeableness report more incivility. I can’t say that being disagreeable causes incivility in the workplace! Maybe it’s true that grumpy people invite uncivil behavior from others. Or perhaps disagreeable people interpret neutral behavior as uncivil. Perhaps being treated rudely causes people to become more disagreeable. Or maybe there’s a third variable that causes someone to both be disagreeable and to be treated rudely by others. Who knows! The point is that all I can say is that these variables are associated, but I can’t claim to know anything about causality.

2.6.3 Experimental Research: “What causes Y? What does Y cause?”

In some cases, the variable life cycle can progress to delving into causality. We might want to know what causes our variable, or what outcomes are a result of our variable. *Causality can only be established through experimental methods*.

The key features of **experimental research** is that the researcher controls all aspects of the study, especially what participants experience during the study. In particular, the researcher manipulates or varies something (IVs), and then allows the outcome variable (DV) to vary naturally. The idea here is to deliberately vary the something in the world (IVs) to see if it has any causal effects on the outcomes. Moreover, in order to ensure that there's no chance that something other than the manipulated variable is causing the outcomes, everything else is kept constant or is in some other way "balanced" to ensure that they have no effect on the results. In practice, it's almost impossible to *think* of everything else that might have an influence on the outcome of an experiment, much less keep it constant. The standard solution to this is **randomization**: that is, we randomly assign people to different groups, and then give each group a different treatment (i.e., assign them different values of the predictor variables). We'll talk more about randomization later in this book, but for now, it's enough to say that what randomization does is minimize (but not eliminate) the chances that there are any systematic difference between groups.

As an example, let's say I want to examine whether incivility causes higher levels of stress. I create a lab study where participants are invited in to work on a task, and I train my research assistant to either be civil or uncivil to the participants. Each session, my assistant flips a coin before the study starts to determine whether to be civil (heads) or uncivil (tails). Everything else about the study remains the same. At the beginning and at the end of the session, participants have their cortisol measured using a cheek swab. If I see that cortisol levels rise for the participants treated uncivilly but not for participants who are treated civilly, I can be reasonably confident that incivility does indeed cause stress.

However, hopefully you can see there might be some ethical problems with this – it's not ideal that I'm stressing out my participants by being mean to them in a study, and this can have all kinds of negative effects on them and their opinion about research and deception. So, there might be some variables where doing a true experiment might be impossible or unethical (such as the effects of smoking – it would not be ethical to randomly assign our participants to either become smokers or non-smokers).

In these cases, we might choose to use a **quasi-experiment**. In this case, we still might introduce a manipulation of some sort, but our groups are not randomly assigned. In my prior example, maybe we work with a company where they have two different plants, and we know they are planning to hire a new manager at each plant –one who is uncivil and one who is civil. We watch to see whether the plant with the uncivil manager demonstrates higher levels of stress. However, the lack of random assignment makes it harder for us to confidently assert causality – the people who work in each plant might be different, or something else might happen at the plant that could cause higher levels of stress, like an accident or layoffs. Quasi-experiments are not as effective as true experiments in understanding causality, but in some cases, its the best we can do.

2.6.4 Meta-Analysis: What do we know so far about X and Y?

The last type of research worth discussing here is the meta-analysis. The statistics for this are beyond what we will cover in this book, but suffice it to say that when we've established a lot of studies looking at the relationship between a few variables, a meta-analysis is a way to get an overview of what all the studies have to say about the relationship, and it allows us to come up with an estimate of the correlation between variables across all studies, as well as some variables that might affect the relationship. Meta-analysis relies on other studies, so this can only happen when a decent amount of literature on the topic has built up, so we're likely to see this later in the life cycle of a variable.

2.6.5 Putting it all together

I've been yapping for a long time at this point about a lot of things other than statistics. So why am I making you read all of this when the goal of this book is to teach you statistics?

This all ties together in that understanding your variables and your research design will help you know what test you should use to answer your research questions. Once we start diving into specific tests in [Section 7.1](#), I'll get into more specifics. But for now, it's worthwhile to let you know:

- Some designs are incompatible with some statistics (you will basically never do a Pearson correlation on data from an experiment).
- Which statistics you use depend on how you scale your variables. If you measure age as a ratio variable versus an ordinal variable, that will determine what statistic is most appropriate for your data.

This will all start to come together soon, but if I gave you a test on all this stuff right now and you feel like you could maybe squeak out a C on that test, you are where you need to be! I know it seems odd, but for the first few chapters of this book, you're just going to be simmering in new information, and you'll find that all of a sudden, it starts to connect and make sense. So don't feel too bad if you're overwhelmed! We have a ways to go before we can start pulling all the strings together. And don't be afraid to revisit this chapter to refresh yourself!

3 Samples and Populations

3.1 Learning Objectives

- Understand what a population is.
- Understand what a sample is, and some of the ways we might draw a sample from a population.
- Evaluate how a particular sampling method might lead a sample to be a poor representation of a population.
- Understand the concept of a distribution of sample means, and understand how the central limit theorem means that the sample mean distribution will be a normal curve regardless of the population distribution.
- Understand the concept of a “model” and how models can be used to adjust our understanding of social science phenomena.

If you recall back in Section 2.1, I noted that the definition of statistics is “The practice or science of collecting and analyzing numerical data in large quantities, especially for the purpose of inferring proportions in a whole from those in a representative sample.” In this chapter, we’re going to dive into the last part of that definition – our ability to infer something from a sample.

Also, recall in Section 2.4.2 I introduced you to the concept of Theoryland and Dataland. There, we talked about how in Theoryland, we might have a construct we’d like to measure, but that we have to bridge the gap between Theoryland and Dataland by creating an operational definition of our construct, which allows us to measure that variable in the real world and obtain data about that variable.

That analogy is useful in this chapter too, as we think about how we can bridge the gap between a sample we obtain in the real world (Dataland) and what that sample can tell us about the population at hand (Theoryland). In a nutshell, when we talk about “inferring” something from our sample, this is our bridge between Theoryland and Dataland in this context.

3.1.1 Defining a population

A sample is a concrete thing (i.e., it exists in Dataland). You can open up a data file, and there's the data from your sample. A **population**, on the other hand, is a more abstract idea (i.e., it exists in Theoryland). It refers to the set of all possible people, or all possible observations, that you want to draw conclusions about, and is generally **much** bigger than the sample... so big that it is not possible for us to find everyone in this population if we want to conduct research. In an ideal world, the researcher would begin the study with a clear idea of what the population of interest is, since the process of designing a study and testing hypotheses about the data that it produces does depend on the population about which you want to make statements. However, that doesn't always happen in practice: usually the researcher has a fairly vague idea of what the population is and designs the study as best they can on that basis.

Sometimes it's easy to state the population of interest. For instance, when researchers conduct political polls, the population is typically "all eligible voters at the a time of the study" – millions of people. So, if I want to learn more about voters, I can probably find a way to sample a few thousand of them to learn more about their opinions. However, a lot of research questions don't necessarily focus on such a specific population. Suppose I run an experiment using 100 undergraduate students as my participants. My goal, as a professor, is to try to discover what teaching techniques are most effective for students. So, which of the following would count as "the population":

- All of the undergraduate psychology students at my institution?
- Undergraduate students in general, anywhere in the world?
- Psychology undergraduate students anywhere in the world?
- People who could potentially become psychology students?
- Anyone currently alive?
- Any human being, past, present or future?

Each of these defines a real group of mind-possessing entities and it's not at all clear which one ought to be the true population of interest.

3.1.2 Simple random samples

Regardless of how we define the population, the critical point is that the sample is a subset of the population, and our goal is to use our knowledge of the sample to draw inferences about the properties of the population. The relationship between the two depends on the **procedure** by which the sample was selected. This procedure is referred to as a **sampling method**, and it is important to understand why it matters.

To keep things simple, imagine we have a bag containing 10 chips. Each chip has a unique letter printed on it, so we can distinguish between the 10 chips. The chips come in two colors, black and white.

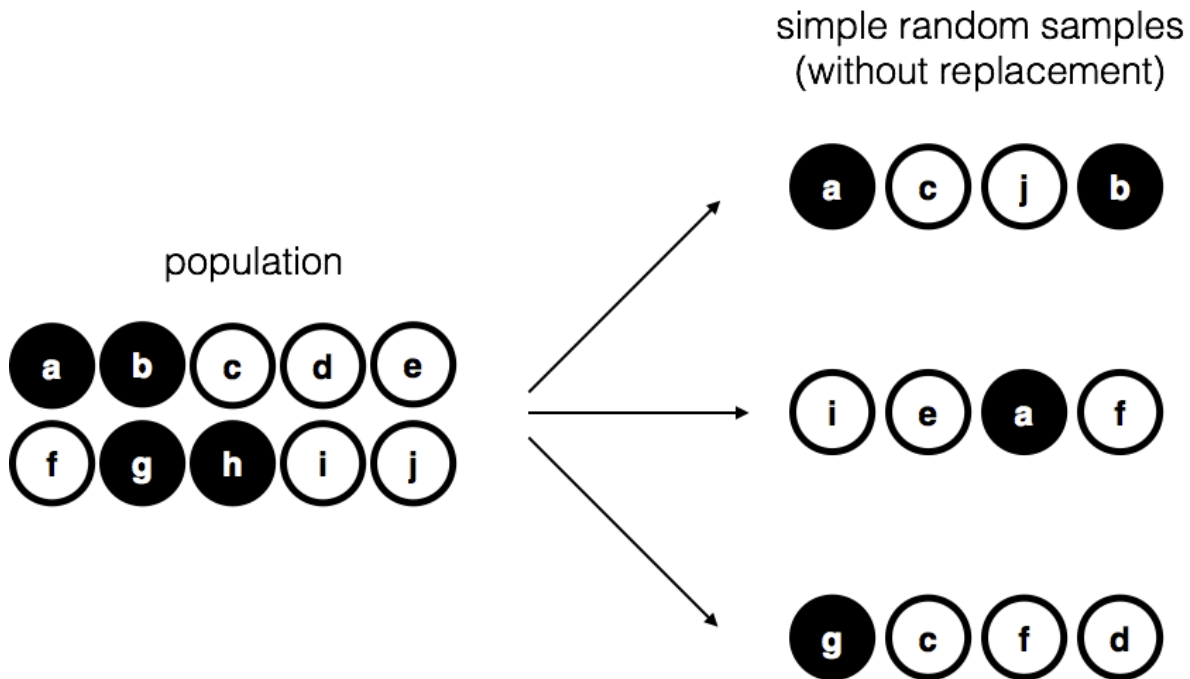


Figure 3.1: Simple random sampling without replacement from a finite population

This set of chips is the population of interest, and it is depicted graphically on the left of Figure 3.1.

As you can see from looking at the picture, there are 4 black chips and 6 white chips, but of course in real life we wouldn't know that unless we looked in the bag. Now imagine you run the following “experiment”: you shake up the bag, close your eyes, and pull out 4 chips without putting any of them back into the bag. First out comes the *a* chip (black), then the *c* chip (white), then *j* (white) and then finally *b* (black). If you wanted, you could then put all the chips back in the bag and repeat the experiment, as depicted on the right hand side of Figure 3.1. Each time you get different results, but the procedure is identical in each case. The fact that the same procedure can lead to different results each time, we refer to it as a **random** process. However, because we shook the bag before pulling any chips out, it seems reasonable to think that every chip has the same chance of being selected. A procedure in which every member of the population has the same chance of being selected is called a **simple random sample**. The fact that we did **not** put the chips back in the bag after pulling them out means that you can't observe the same thing twice, and in such cases the observations are said to have been sampled **without replacement**.

To help make sure you understand the importance of the sampling procedure, consider an alternative way in which the experiment could have been run. Suppose instead that I opened the bag, and decided to pull out four black chips without putting any of them back in the bag. This **biased** sampling scheme is depicted in Figure 3.2.

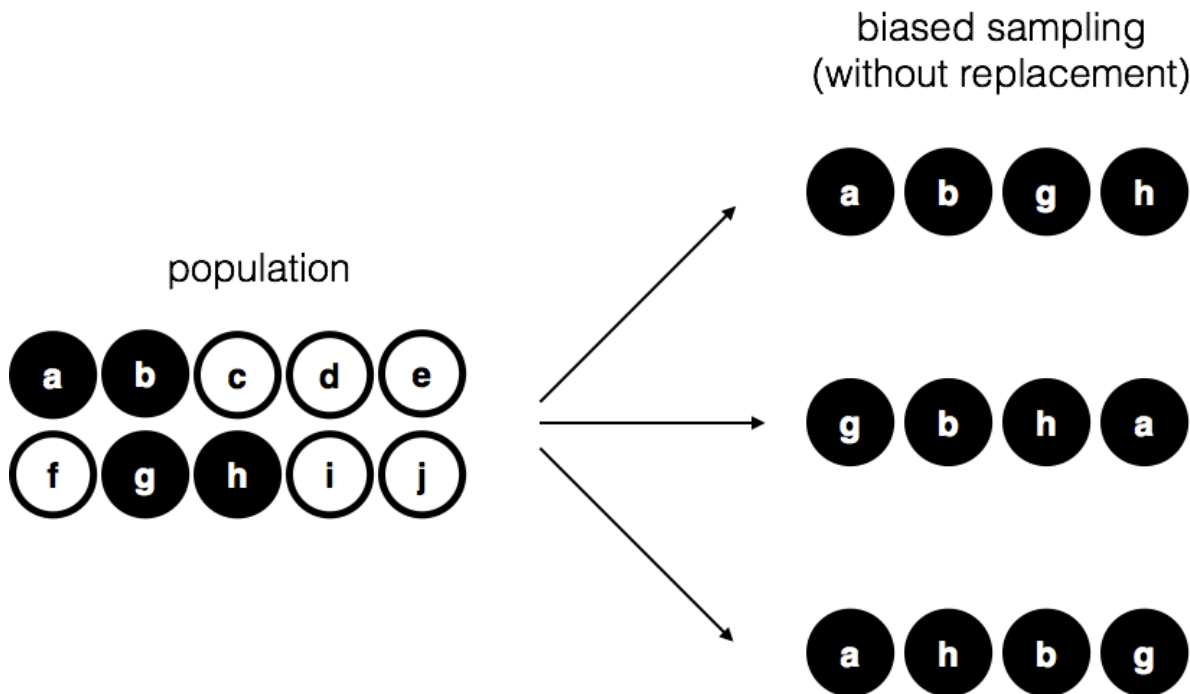


Figure 3.2: Biased sampling without replacement from a finite population

Now consider the evidentiary value of seeing 4 black chips and 0 white chips. Clearly, my biased sampling scheme is problematic. If I admit that I peeked in the bag ahead of time, you should know to be skeptical of my findings. However, if I don't share that my sample is biased, or if I don't realize that this is a bad way to sample, you may come away with the incorrect assumption that there are only black chips in the bag. For this reason, statisticians prefer it when we are transparent about how we sample our population, and ideally, we want a simple random sample because it makes the data analysis **much** easier.

A third procedure is worth mentioning. This time around we close our eyes, shake the bag, and pull out a chip. This time, however, we record the observation and then put the chip back in the bag. Again we close our eyes, shake the bag, and pull out a chip. We then repeat this procedure until we have 4 chips. Data sets generated in this way are still simple random samples, but because we put the chips back in the bag immediately after drawing them it is referred to as a sample **with replacement**. The difference between this situation and the first one is that it is possible to observe the same population member multiple times, as illustrated in Figure 3.3.

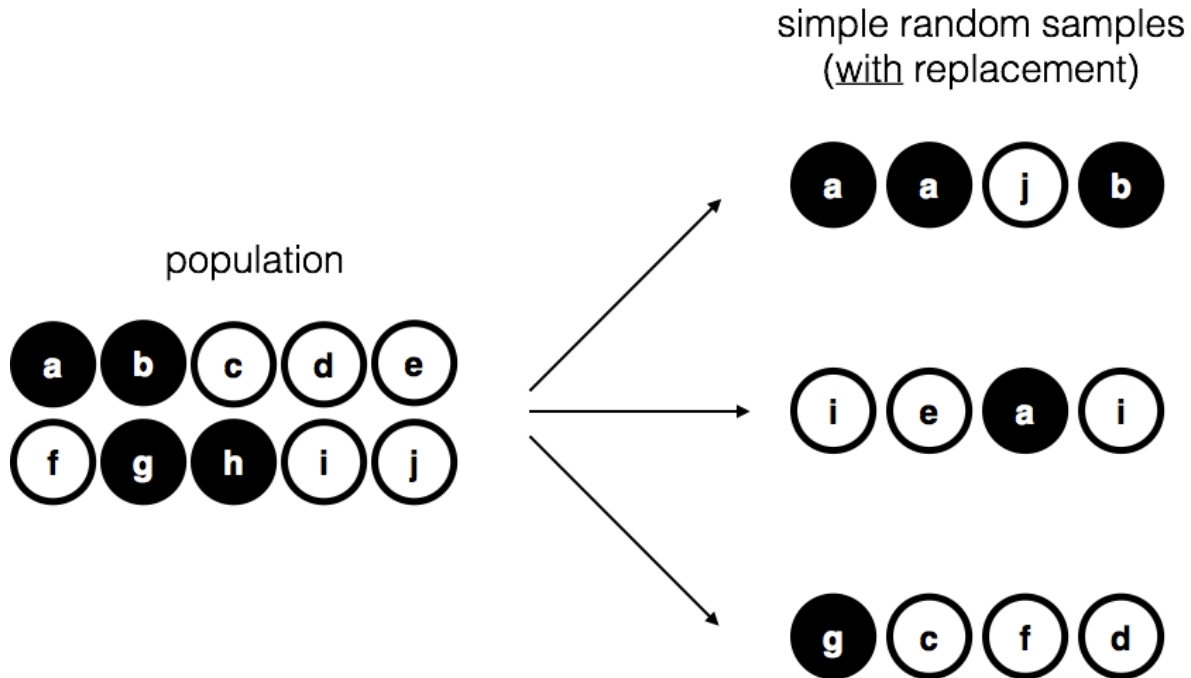


Figure 3.3: Simple random sampling with replacement from a finite population

Most psychology experiments tend to be sampling without replacement, because the same person is not allowed to participate in the experiment twice. However, most statistical theory is based on the assumption that the data arise from a simple random sample **with** replacement. In real life, this very rarely matters. If the population of interest is large (e.g., has more than 10 entities!) the difference between sampling with- and without- replacement is too small to be concerned with. The difference between simple random samples and biased samples, on the other hand, is not such an easy thing to dismiss.

3.1.3 Most samples are not simple random samples

As you can see from looking at the list of possible populations that I showed above, it is almost impossible to obtain a simple random sample from most populations of interest. When I run experiments, I'd consider it a minor miracle if my participants turned out to be a random sampling of the undergraduate psychology students at Minnesota State University, even though this is by far the narrowest population that I might want to generalize to. A thorough discussion of other types of sampling schemes is beyond the scope of this book, but to give you a sense of what's out there I'll list a couple of common strategies:

- **Stratified sampling** is helpful if we can divide the population into different sub-populations (strata). At my institution, the population of students is largely white, so

if I wanted to make sure to ensure that other races are well-represented in my data, I might specifically identify other races on campus and purposely sample from those groups. If a group is particularly rare on campus (such as indigenous groups) I might even **oversample** by getting an overly large proportion of those people to ensure I hear their opinions.

- **Snowball sampling** is useful when I need to find a population that is hidden or hard to access. This can be common in social sciences, because we often want to study populations like this. A good example is if I want to get opinions among transgender people – this population is relatively small, and some transgender people may not want to be identified as such because they may not be out to other people, or they may be worried about their safety. So, I would start by finding a group of transgender people who are willing to share their opinion. At the end of my survey, I ask participants to provide contact details for other transgender people who might want to participate. I contact those people, have them take my survey, and then ask them for more contact details. This process continues – much like a snowball rolling down a hill and growing larger as it collects more snow, my sample will “snowball” into a larger sample with the help of my participants. This gets us a lot of hard-to-obtain data. However, we have to be sensitive about the ethics of this sample (i.e., is it OK for a participant to give me someone’s contact information, or will they feel this violates their privacy?) It also is definitely **not** a random sample, because we tend to know people with whom we have a lot in common, so there might be other groups of transgender people I’m missing out on that might be very different from the sample I end up with.
- **Convenience sampling** is more or less what it sounds like. The samples are chosen in a way that is convenient to the researcher, and not selected at random from the population of interest. Snowball sampling is one type of convenience sampling, but there are many others. A common example in psychology are studies that rely on undergraduate psychology students. These samples are generally non-random in two respects: firstly, reliance on undergraduate psychology students automatically means that your data are restricted to a single sub-population. Secondly, the students usually get to pick which studies they participate in, so the sample is a self selected subset of psychology students not a randomly selected subset. In real life, most studies are convenience samples of one form or another. This is sometimes a severe limitation, but not always.

3.1.4 How much does it matter if you don’t have a simple random sample?

Okay, so real world data collection tends not to involve nice simple random samples. Does that matter? A little thought should make it clear to you that it **can** matter if your data are not a simple random sample: just think about the difference between Figure 3.1 and Figure 3.2. However, it’s not quite as bad as it sounds. Some types of biased samples are entirely unproblematic. For instance, when using a stratified sampling technique you actually **know** what the bias is because you created it deliberately, often to **increase** the effectiveness

of your study, and there are statistical techniques that you can use to adjust for the biases you've introduced (not covered in this book!). So in those situations it's not a problem.

More generally though, it's important to remember that random sampling is a means to an end, not the end in itself. Let's assume you've relied on a convenience sample, and as such you can assume it's biased. A bias in your sampling method is only a problem if it causes you to draw the wrong conclusions. When viewed from that perspective, I'd argue that we don't need the sample to be randomly generated in **every** respect: we only need it to be random with respect to the psychologically-relevant phenomenon of interest.

Suppose I'm doing a study looking at working memory capacity. In study 1, I actually have the ability to sample randomly from all human beings currently alive, with one exception: I can only sample people born on a Monday. In study 2, I am able to sample randomly from the Australian population. I want to generalize my results to the population of all living humans. Which study is better?

The answer, obviously, is study 1. Why? Because we have no reason to think that being "born on a Monday" has any interesting relationship to working memory capacity. In contrast, I can think of several reasons why "being Australian" might matter. Australia is a wealthy, industrialized country with a very well-developed education system. People growing up in that system will have had life experiences much more similar to the experiences of the people who designed the tests for working memory capacity. This shared experience might easily translate into similar beliefs about how to "take a test", a shared assumption about how psychological experimentation works, and so on. These things might actually matter. For instance, "test taking" style might have taught the Australian participants how to direct their attention exclusively on fairly abstract test materials relative to people that haven't grown up in a similar environment; leading to a misleading picture of what working memory capacity is.

There are two points hidden in this discussion. First, when designing your own studies, it's important to think about what population you care about, and try hard to sample in a way that is appropriate to that population. In practice, you're usually forced to put up with a "sample of convenience" (e.g., psychology lecturers sample psychology students because that's the least expensive way to collect data, and our coffers aren't exactly overflowing with gold), but if so you should at least spend some time thinking about what the dangers of this practice might be.

Second, if you're going to criticize someone else's study because they've used a sample of convenience rather than laboriously sampling randomly from the entire human population, at least have the courtesy to offer a specific theory as to **how** this might have distorted the results. Remember, everyone in science is aware of this issue, and does what they can to alleviate it. Merely pointing out that "the study only included people from group X" is entirely unhelpful, and borders on being insulting to the researchers, who are aware of the issue. They just don't happen to be in possession of the infinite supply of time and money required to construct the perfect sample. In short, if you want to offer a responsible critique of the sampling process,

then be **helpful**. Rehashing the blindingly obvious truisms that I've been rambling on about in this section isn't helpful.

3.1.5 Population parameters and sample statistics

Okay. Setting aside the thorny methodological issues associated with obtaining a random sample, let's consider a slightly different issue, which is how that sample relates to the population. The easiest way to talk about this is for us to run a little simulation. Let's say that I'm an all-knowing god-like figure who exists in Theoryland, and therefore I have access to the entire population I want to study. Let's say the population I'm interested in is a tiny slice of people who experience both a button phobia (koumpounophobia) and a fear of otters (lutraphobia). These are both fairly unusual conditions, so let's say that the entire population on the planet of these people is only 10,000 people. And let's say that I want to study this population's IQ. Here in Theoryland, because I'm an all-knowing being, I **know** that the IQ of this population is 100, with a standard deviation of 15. These characteristics of the population are known as **population parameters**. If you use R, it's pretty simple to generate this population using this code:

```
population<- rnorm(10000, mean = 100, sd = 15)
```

Unfortunately for us, we live here in Dataland, where it's not possible for us to access the full population. So, instead, I draw a sample from this population. Let's start with a sample of 25 people – a bit on the small side. When I draw this random sample, I can take a look at the **sample statistics** – that is, things like the mean and standard deviation of that sample. I can also take a look at a histogram that shows me the frequency of scores:

In this sample, I see a mean of 102.09 and a standard deviation of 16.12. You can also see that this histogram is a bit lumpy, but it follows a basic bell curve with most scores in the middle, and a few scores that are on the high or low ends of the IQ scores. And we can see that the means and standard deviations are close to the population parameters, but they are not perfect.

Let's try this again with a really big sample – lets say I get 5,000 people.

Here, my mean is 99.87 and my standard deviation is 15.06, which are closer to the parameter values. You'll also notice my histogram is much smoother and looks even more like a normal distribution. Why did this happen?

This phenomenon is known as the **law of large numbers**. The law of large numbers is a mathematical law that applies to many different sample statistics, but the simplest way to think about it is as a law about averages. The sample mean is the most obvious example of a statistic that relies on averaging (because that's what the mean is... an average), so let's look at that. **When applied to the sample mean, what the law of large numbers states is**

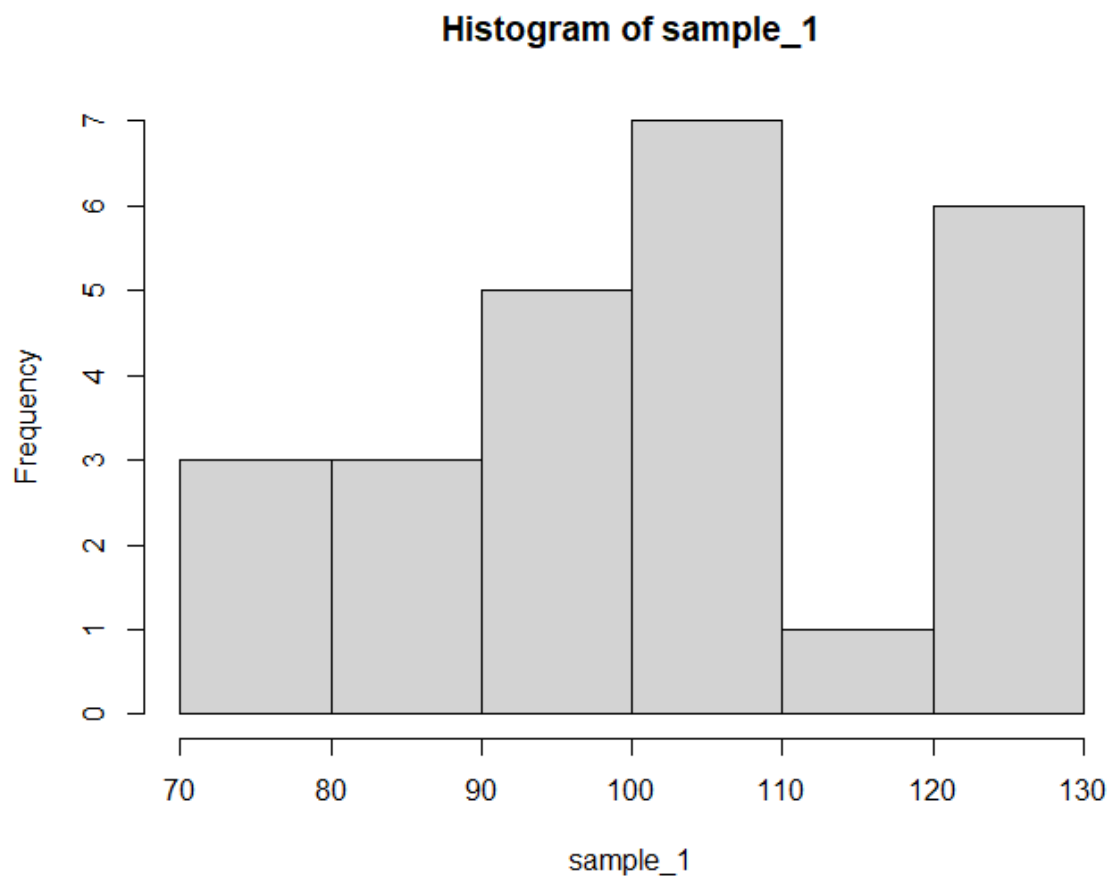


Figure 3.4: A histogram of a random sample ($n=25$) of the population

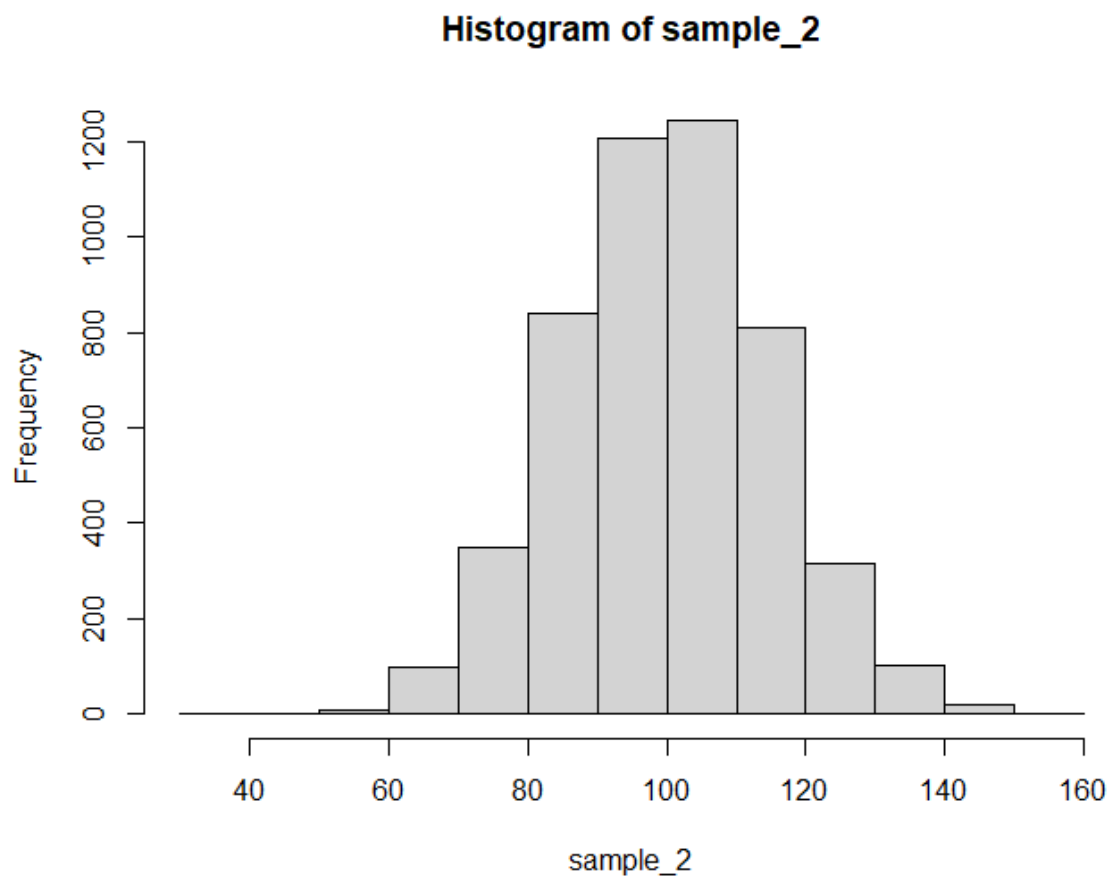


Figure 3.5: A histogram of a random sample ($n=5,000$) of the population

that as the sample gets larger, the sample mean tends to get closer to the true population mean. Or, to say it a little bit more precisely, as the sample size “approaches” infinity (written as $N \rightarrow \infty$) the sample mean approaches the population mean ($\bar{X} \rightarrow \mu$).

The main takeaway here is that a large sample tends to do a better job of approximating parameters compared to a small sample.

3.2 Sampling distributions and the central limit theorem

The law of large numbers is a very powerful tool, but it’s not going to be good enough to answer all our questions. Among other things, all it gives us is a “long run guarantee”. In the long run, if we were somehow able to collect an infinite amount of data, then the law of large numbers guarantees that our sample statistics will be correct. But as John Maynard Keynes famously argued in economics, a long run guarantee is of little use in real life:

[**The] long run is a misleading guide to current affairs. In the long run we are all dead. Economists set themselves too easy, too useless a task, if in tempestuous seasons they can only tell us, that when the storm is long past, the ocean is flat again.** (Keynes 1923)

This is true in social science generally. It is not enough to know that we will **eventually** arrive at the right answer when calculating the sample mean. Knowing that an infinitely large data set will tell me the exact value of the population mean is cold comfort when my **actual** data set has a sample size that is much less than infinity. In real life, then, we must know something about the behavior of the sample mean when it is calculated from a smaller data set!

3.2.1 Sampling distribution of the sample means

So far, we’ve been talking about a sample or two from a population. But imagine again that we are a god-like entity living in Theoryland. We have infinite resources and time, so imagine that we decide to take infinite samples from the population, and for each sample, I calculate the mean of that sample, and I put each sample mean into a data set. That data set that I create provides something we can call **the sampling distribution of the sampling means**. I know that name is long and awkward, but I don’t make the rules around here.

I want to jump into a different example for this topic. Let’s say that I want to give a sample of 20 people a 10-sided die and ask them to roll it 10 times, and I record whatever value they roll. If this die is fair, it’s equally likely to land on each number, so we see a **uniform distribution** – that is, a flat line that indicates there’s a 10% chance of each value being rolled.

So now, let’s say that I decide to take 10,000 samples of 10 people rolling dice. For each sample, I record what value people roll, and I add that to my data set and calculate the mean. Here’s a

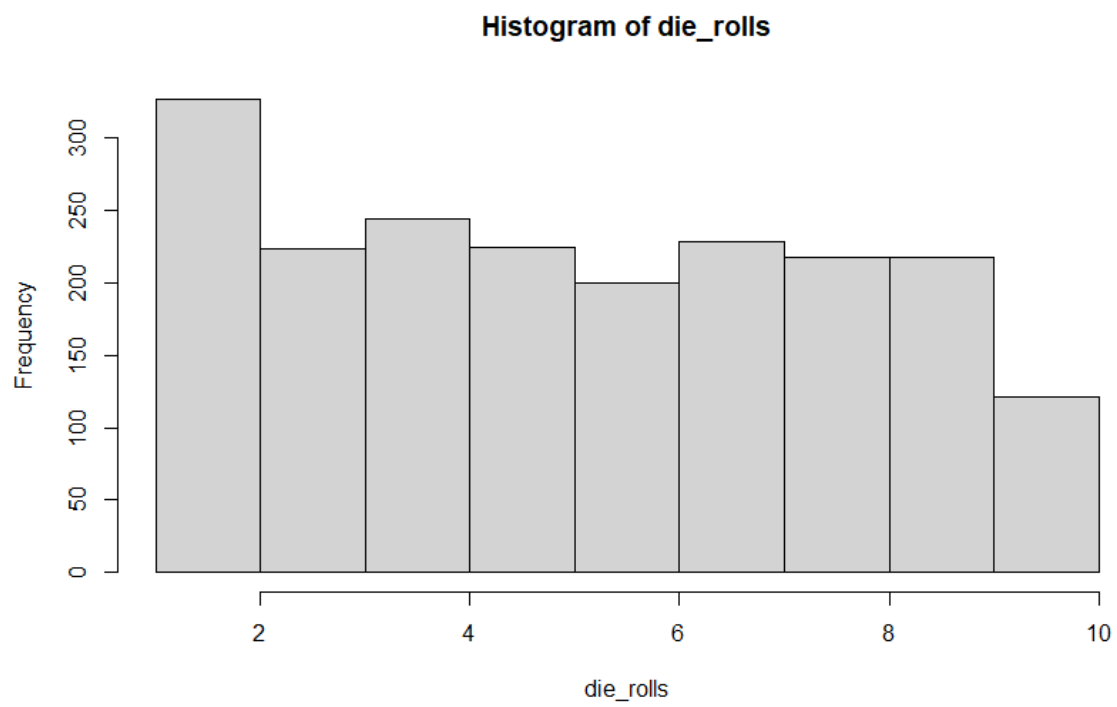


Figure 3.6: A uniform distribution of 2,000 die rolls

little animation that shows how a few of these samples I collect with the mean of each sample indicated by the vertical line:

First question: Are all the sample means the same? The answer to this is “No.” As you can see in the animation, the vertical line moves around a bit – it tends to stay relatively close to the middle values of 5 and 6, but sometimes it is much higher or lower than that.

Second question: If I continued to do this for 10,000 samples and I kept a record of each sample’s mean and put it on a histogram, what would that histogram look like? It’s tempting to say that, because this data comes from a uniform distribution, that the distribution would also be uniform. In fact, this is what we see:

What the heck?! How did that happen? Why do we have a normal curve here? The answer is a fancy little thing called the **central limit theorem**. We’ll talk about that more in a moment, but let me walk through what exactly is happening here.

First of all, let’s think about what the “average” would be if we rolled the dice 10 times and got a perfect uniform distribution where the die turned up each number once. So my data set would be:

1 2 3 4 5 6 7 8 9 10

What’s the mean of this? If we calculate it, we find:

$$(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10)/10 = 5.5$$

So, this tells us the mean of each sample should generally be pretty close to 5.5. That won’t happen every time, but there are a lot of combinations of numbers that would lead to a sample having a mean of 5.5. For example:

1 2 2 4 5 6 7 8 10 10

3 3 4 4 5 5 7 8 8 8

But there’s only one way to get a mean of 1:

1 1 1 1 1 1 1 1 1 1

And there’s only one way to get a mean of 10:

10 10 10 10 10 10 10 10 10 10

So, our distribution of sample means shows that it’s going to be fairly common to get a sample mean around 5.5, and that it’s unusual (but not impossible) to get a mean that’s closer to 1 or to 10. But the main point here is that even though our population distribution is uniform, the sampling distribution of sample means is going to be a normal distribution. OK, cool! But why do you need to care about this? The central limit theorem holds the key to this!



Scan for a helpful animation!

Figure 3.7: Three samples from our die rolling data

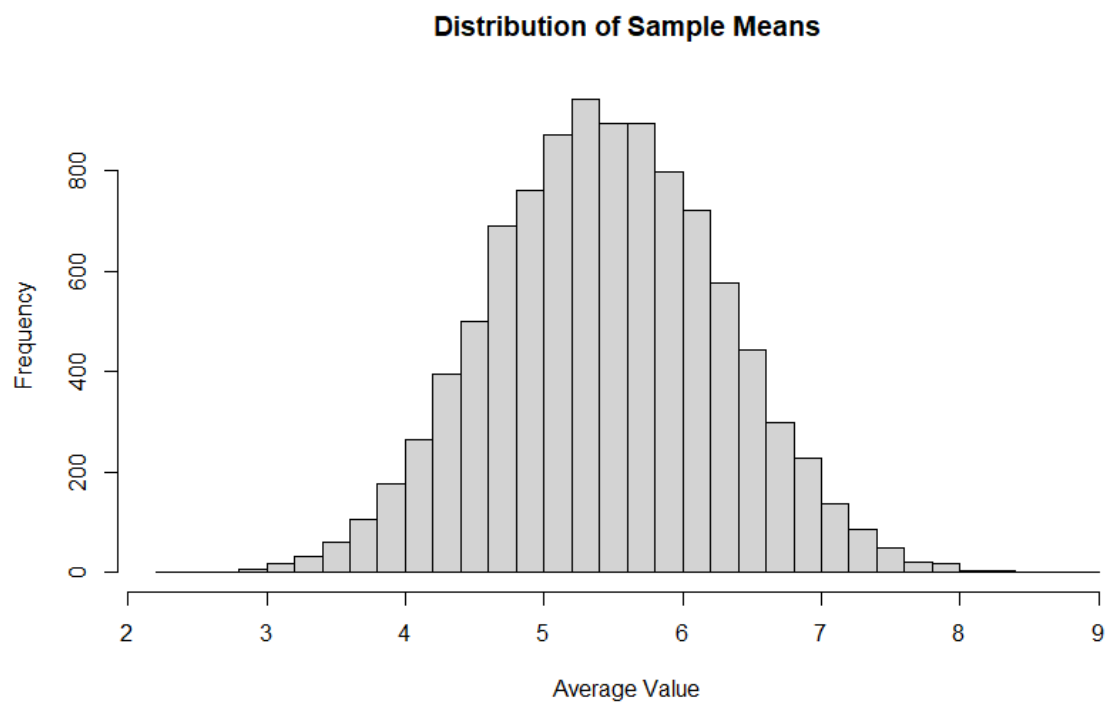


Figure 3.8: A distribution of 10,000 sample means

3.3 The central limit theorem

Now you've seen lots of sampling distributions, and you know what the sampling distribution of the mean is. Here, we'll focus on **how the sampling distribution of the mean changes as a function of sample size**.

Intuitively, you already know part of the answer: if you only have a few observations, the sample mean is likely to be quite inaccurate (you've already seen it bounce around): if you replicate a small experiment and recalculate the mean you'll get a very different answer. In other words, the sampling distribution is quite wide. If you replicate a large experiment and recalculate the sample mean you'll probably get the same answer you got last time, so the sampling distribution will be very narrow.

Let's give ourselves a nice movie to see everything in action. Let's return to the earlier example using IQ scores. `?@fig-sampdistgif` has four panels, each panel represents a different sample size (n), including sample-sizes of 10, 50, 100, and 1000. The red line shows the shape of the normal distribution. The grey bars show a histogram of each of the samples that we take. The red line shows the mean of an individual sample (the middle of the grey bars). As you can see, the red line moves around a lot, especially when the sample size is small (10).

The new bits are the blue bars and the blue lines. The blue bars represent the sampling distribution of the sample mean. For example, in the panel for sample-size 10, we see a bunch of blue bars. This is a histogram of 10 sample means, taken from 10 samples of size 10. In the 50 panel, we see a histogram of 50 sample means, taken from 50 samples of size 50, and so on. The blue line in each panel is the mean of the sample means ("aaagh, it's a mean of means", yes it is).



Scan for helpful animation!

Figure 3.9: Animation of samples (grey histogram shows frequency counts of data in each sample), and the sampling distribution of the mean (histogram of the sampling means for many samples). Each sample is taken from the normal distribution shown in red. The moving red line is the mean of an individual sample. The blue line is the mean of the blue histogram, which represents the sampling distribution of the mean for many samples

What should you notice? Look at how the range of the blue bars shrinks as sample size increases. The sampling distribution of the mean is quite wide when the sample size is 10, it narrows as sample size increases to 50 and 100. When we get to an enormous sample of 1000 participants, the distribution is just one blue bar in the middle. What we are seeing is that the mean of the sampling distribution approaches the mean of the population as sample size increases. This makes some intuitive sense – if I had a sample that was so big that I was only missing a couple of people from the population, you can bet that my sample mean would be almost exactly the population mean.

The sampling distribution of the mean is just another kind of distribution, and we’ve already talked about the mean of that sampling distribution. The sampling distribution also has a standard deviation, which we call the **standard error of the sample mean (SEM)**. You can see in the animation that as our sample sizes get larger, the SEM (the range of the blue bars) gets smaller.

Okay, so that’s one part of the story. However, there’s something we’ve been glossing over a little bit. We’ve seen it already, but it’s worth looking at it one more time. Here’s the thing: **no matter what shape your population distribution is**, as N increases, the sampling distribution of the mean starts to look more like a normal distribution. This is the central limit theorem.

To see the central limit theorem in action, we are going to look at some histograms of sample means from different kinds of distributions. It is very important to recognize that you are looking at distributions of sample means, not distributions of individual samples.

Figure 3.10 shows sampling from a normal distribution. The red line is the normal distribution where each sample is drawn from. The mean for each sample of numbers is computed, and the distribution of sample means is shown by the blue bars. Note that the shape of red line and the blue bars are similar, they both look like a normal distribution.

Here’s one more example with an exponential distribution (shown in red on Figure 3.11) where smaller numbers are more likely to be sampled than larger numbers. Even though way more of the numbers in a given sample will be smaller than larger, according to the sampling distribution of the mean does not look the red line. Instead, the sampling distribution of the mean looks like a bell-shaped normal curve. This is the central limit theorem in action.

On the basis of these figures, it seems like we have evidence for all of the following claims about the sampling distribution of the mean:

- The mean of the sampling distribution is the same as the mean of the population
- The standard deviation of the sampling distribution (i.e., the standard error) gets smaller as the sample size increases
- The shape of the sampling distribution becomes normal as the sample size increases

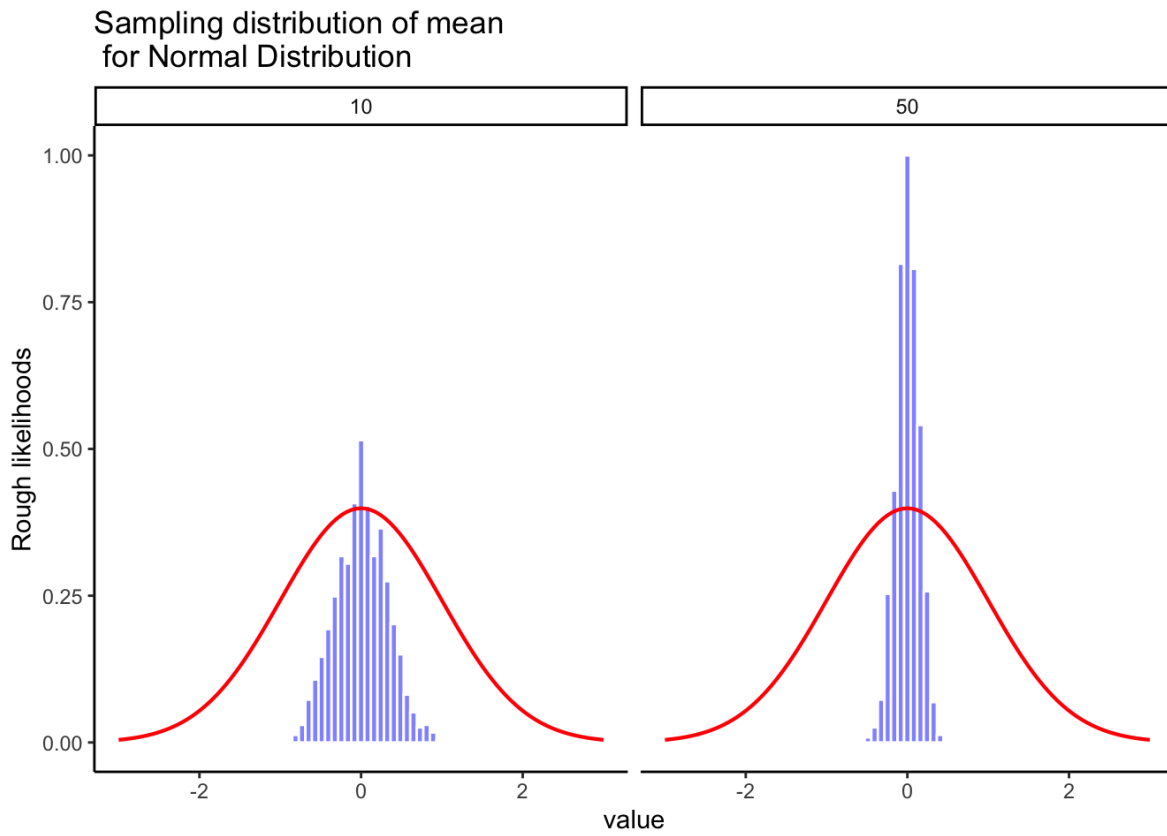


Figure 3.10: Comparison of two normal distributions, and histograms for the sampling distribution of the mean for different samples-sizes. The range of sampling distribution of the mean shrinks as sample-size increases

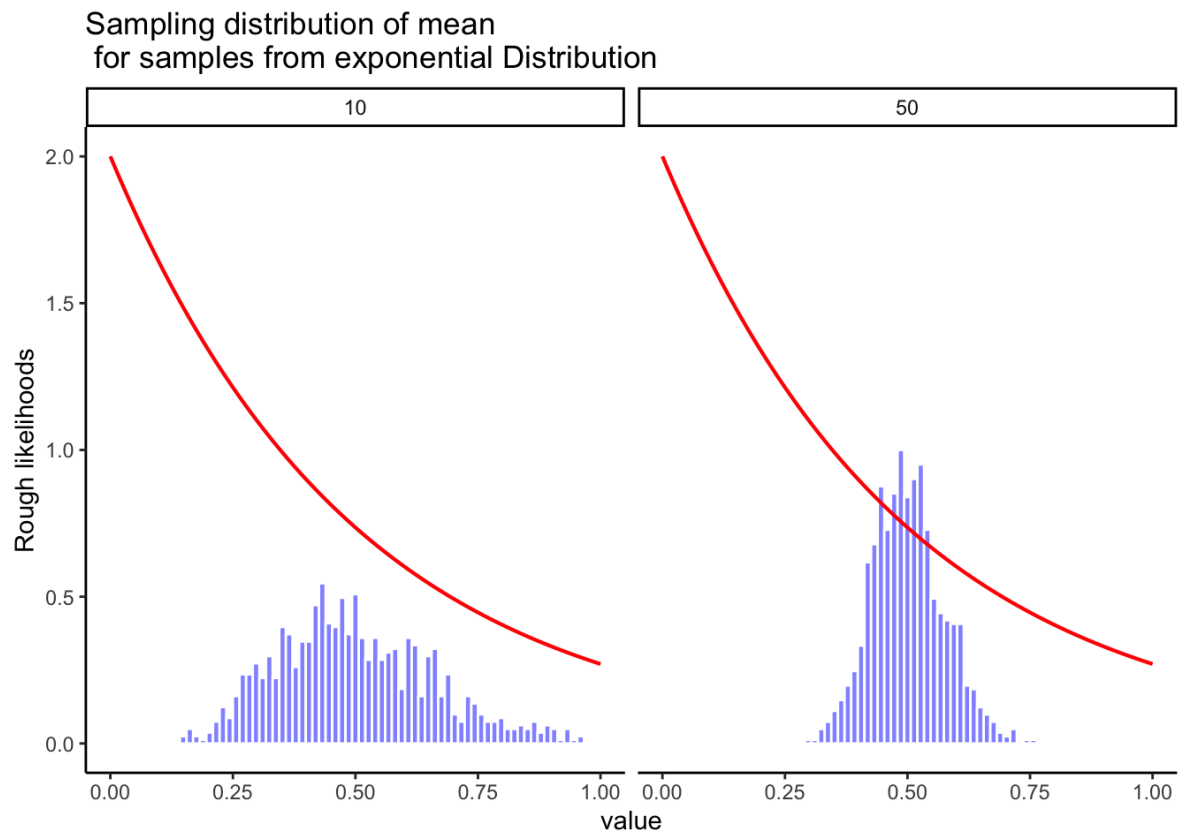


Figure 3.11: Illustration that the shape of the sampling distribution of the mean is normal, even when the samples come from an exponential distribution

As it happens, not only are all of these statements true, there is a very famous theorem in statistics that proves all three of them, known as the **central limit theorem**. Among other things, the central limit theorem tells us that if the population distribution has mean μ and standard deviation σ , then the sampling distribution of the mean also has mean μ , and the standard error of the mean is:

$$S_{\bar{X}} = \frac{\sigma}{\sqrt{N}} \quad (3.1)$$

Because we divide the population standard deviation σ by the square root of the sample size N , the $S_{\bar{X}}$ gets smaller as the sample size increases. It also tells us that the shape of the sampling distribution becomes normal.

This result is useful for all sorts of things. It tells us why large experiments are more reliable than small ones, and because it gives us an explicit formula for the standard error it tells us **how much more reliable** a large experiment is. It tells us why the normal distribution is, well, normal. In real experiments, many of the things that we want to measure are actually averages of lots of different quantities (e.g., arguably, “general” intelligence as measured by IQ is an average of a large number of “specific” skills and abilities), and when that happens, the averaged quantity should follow a normal distribution. Because of this mathematical law, the normal distribution pops up over and over again in real data.

3.4 Models: Ways of Understanding the World

There’s one more concept that fits well here, and that’s the concept of creating a **model**. It sounds fancy, but all it means is that I’m using my data to try and predict and/or understand some sort of variable in the world. If you can measure something, you can build a model around it. For example, if I want to understand why some students procrastinate on their homework and others don’t, I might measure a number of variables (personality, perfectionism, smartphone addiction, physical activity) and some of those variables might help me predict which students are likely to procrastinate.

One of the simplest models we can build is simply using the mean. If I know nothing at all about a student, and I’m asked to make a guess about how much the procrastinate, I might get a sample of participants and see what the average level of procrastination is for them, and use that as my best guess for how this mystery student will behave. As I learn more about procrastination, I might be able to adjust my estimate up or down if I know something about this student. For example, if I know they are highly perfectionistic, I might expect that they will be higher in procrastination; if I know they don’t have a smartphone, I might expect they will be lower in procrastination. Ultimately, if another variable provides a better prediction than just the average score on procrastination, it’s likely a useful variable. Building models is

ultimately the goal of statistics in social sciences, so we'll talk a lot about models and how well data fits a model in future chapters.

4 Understanding Probability

4.1 Learning Objectives

- Understand what a probability is, and how the additive law, the multiplicative law, and conditional probability allow us to understand the likelihood of events occurring given different conditions.
- Understand what a “standard score” is, and know the characteristics of some common standard scores (e.g., z-scores, t-scores)

In order to understand the connection between data and inferences we can make about it, we need to spend a little time talking about probability. Some statistics books go into a lot of depth on this topic, but I actually think there are just a handful of principles that you need to understand in order to grasp how statistics work:

1. Additive Law
2. Multiplicative Law
3. Conditional Probability

If you can get these three concepts down, you’ll have most of what you’ll need for a basic level of statistics. Probability is surprisingly tricky for humans! Our brains are not really built to understand how probability works, and we make a lot of mistakes when we think about the world around us and make quick judgments about what is likely versus unlikely (read any article by Daniel Kahneman and Amos Tversky – these two made a whole career out of finding weird ways we misunderstand probability).

So, in this chapter, we’re going to walk through these laws, then at the end, we’ll connect it back to why this matters for social sciences.

4.2 Basic Probability

Just in case you are a little rusty on the basics of probability, let’s walk through an example. Let’s say that I have a standard deck of cards, which has four suits (hearts, diamonds, clubs and spades) and each suit has 13 cards (2-10, Jack, Queen, King, Ace... let’s assume we’ve put

the two Joker cards aside for this example). When we calculate a probability, all we're doing is taking the likelihood of something happening over all possible events.

Imagine that I want to know the probability that I will draw a Heart out of my deck of cards. We can determine this probability by considering how many Heart cards there are in the deck (13) and how many total cards there are in the deck (52). So, we can indicate the probability of a Heart this way:

$$p(\text{Heart}) = \frac{\text{HeartCards}}{\text{AllCards}} = \frac{13}{52} = .25$$

This tells us a have a 25% chance of drawing a Heart card from my deck.

4.3 Additive Law

Additive law is defined as “Given a set of mutually exclusive events, the probability of the occurrence of one event or another is equal to the sum of their separate probabilities.” When we say “mutually exclusive,” this means that we have a situation where either one event or another can occur, but they can't both occur. In the case of my cards, the suit on the card I draw is mutually exclusive – my card can't be both a Heart and a Diamond, it can only be one of those things. But I might be interested in the probability of drawing a card that is a Heart or a Diamond. Additive law provides the answer for us – we determine the probability of each event happening, and then we add them up:

$$p(\text{Heart or Diamond}) = p(\text{Heart}) + p(\text{Diamond}) = .25 + .25 = .50$$

This tells me I have a 50% chance of randomly drawing a heart or a diamond (this makes sense, as these cards make up half the deck). Notice that when we talk about additive law, we're saying “This OR That” so we can expect that final probability will be larger than the probability of one event on its own. Here's a visualization of what that is the case:

If you want to connect this back to a topic more connected to social science, age might be a good example of this – if we wanted to know the probability of someone being under the age of 18 or over the age of 65, we would add the probabilities for each to determine that final probability.

4.4 Multiplicative Law

Multiplicative law, meanwhile, is defined as “The probability of the joint occurrence of two (or more) events is the product of their individual probabilities.” Put more simply, if I want to know about the probability of two events happening together, I multiply their probabilities together. So, for my card deck example, imagine I want to know the probability that I'll draw a Heart card and the probability that I'll draw a Queen. We know:

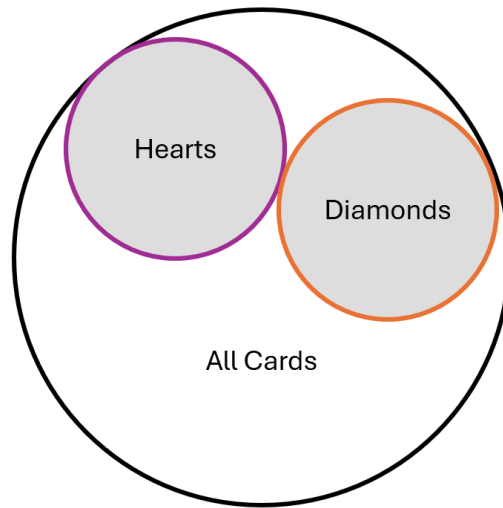


Figure 4.1: A visual representation of additive rule, where the gray area represents the probability of a heart OR a diamond.

$$p(\text{Heart}) = .25$$

$$p(\text{Queen}) = \frac{4}{52} \approx .08$$

So, if I want to know the probability I'll draw a Heart and a Queen, I multiply those probabilities together:

$$p(\text{Heart and Queen}) = .25 \times .08 \approx .02$$

So, I have about a 2% chance of drawing a card that is both a Heart and a Queen. This makes sense; there's only one in the deck, so it would be pretty lucky to draw that specific card! Note here with the multiplicative law, the final value is smaller than the probability of each event happening on its own. This graphic shows why this is the case:

As Figure 4.2 demonstrates here, the intersection between those two events is smaller than each of the circles on their own. After all, we only have four queens, and only one of them is in the heart suit.

4.5 Conditional Probability

Finally, a conditional probability is defined as "the probability that an event will occur given that some other event has occurred." If these two events are unrelated, we can simply use the multiplicative rule to calculate this. This is the case with cards – there's no relationship between a card's suit and its value, they are completely independent of each other. So if

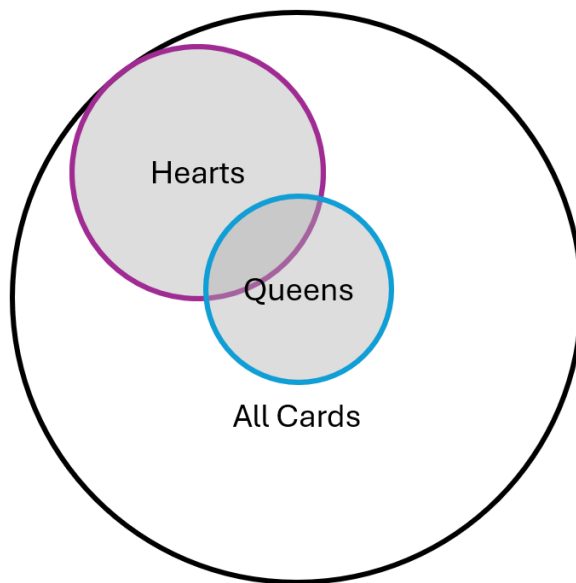


Figure 4.2: A visual representative of the multiplicative rule. The darker intersection between the “Hearts” and “Queens” circles represents the probability of drawing the queen of hearts.

our question is “Given that I’ve drawn a Queen, what is the probability it’s a Heart?” the multiplicative law works just fine. In this case, we already know that there are only 4 Queens in the deck, so we only have about an 8% chance of drawing the Queen. We also know there is only one Queen in each of the four suites, so there’s only a 25% chance the Queen we’ve drawn is a Heart. The multiplicative rule works here too, we can just multiply those probabilities together and see there’s about a 2% chance of drawing that specific card.

Here’s the thing: most of the time in social science, we are hoping that our variables are, in fact, related. So what we’re really doing is seeing whether our data looks like it’s following the multiplicative rule or not. We usually hope that the final results deviate from what we’d get with the multiplicative rule.

Let’s take a simple example to start. Imagine that I teach a class where both Psychology majors and other majors take the class. At the end of the class, I ask students whether they would recommend my class to another student (yes/no). Now, imagine my data looks like this:

Table 4.1: Contingency table of major and recommendation

Recommend the class?	Psychology major	Other major	TOTAL
Yes	25	25	50

Recommend the class?	Psychology major	Other major	TOTAL
No	25	25	50
TOTAL	50	50	100

If I asked the question “Given that a student is a psychology major, what is the probability they would recommend my class?” we would find that the multiplicative rule works perfectly. There’s a 50% chance that a student is a psychology major, and there’s a 50% chance they recommend my class, so:

$$p(Yes|PsychologyMajor) = .50 \times .50 = .25$$

Sure enough, we can see that 25% of my students are in the “Yes/Psychology major” cell. So, this tells me that whether a student recommends my class is unrelated to whether they are a psychology major or not.

Now, let’s imagine that my data looks like this instead:

Table 4.2: Contingency table of major and recommendation

Recommend the class?	Psychology major	Other major	TOTAL
Yes	40	10	50
No	10	40	50
TOTAL	50	50	100

If my data looks like this, we can see that my multiplicative rule is calculated the same, because we have the same numbers in the “total” columns:

$$p(yes|Psychology) = .50 \times .50 = .25$$

But when I look at the data I actually collected, the multiplicative rule did not work well at all! Rather than the 25 students I expected to see, I actually have a lot more students in that cell – 40 students! It appears that psychology students really like my class, but students from other majors kind of hate it. So, what this tells me is that a key premise of my multiplicative law has been violated. The multiplicative law only works when events are independent of one another. If I find that the data doesn’t match up well to what I’d expect with the multiplicative law, that indicates it’s likely the variables are related.

Let’s walk through a slightly more complex example to hammer home the point. Imagine I create a study where I expose children to either non-violent media (a cartoon where characters cooperate) and violent media (a cartoon where characters fight) and then I give them some time on the playground and observe whether the children behave aggressively (pushing or hitting). After I’ve collected my data, I see these results:

Table 4.3: Contingency table of exposure to violent media and aggression

	Aggression	No Aggression	TOTAL
Exposed to Violent Media	90	10	100
Not Exposed to Violent Media	60	40	100
TOTAL	150	50	200

So, I can ask the question “Given that a child is exposed to violent media, what is the probability they will engage in aggression?” If these variables are unrelated, the multiplicative rule should provide a pretty decent estimate of how many children should be in the Exposed/Aggressive cell:

$$p(\text{Exposed}) = \frac{100}{200} = .50$$

$$p(\text{Aggression}) = \frac{150}{200} = .75$$

$$p(\text{Aggression}|\text{Exposed}) = .50 \times .75 = .375$$

So out of 200 kids, I’d expect about 37.5% of them to be in the exposed/aggression box... about 75 kids. However, that’s not what we see here! Instead, we see 90, which is more than we would expect if those variables are unrelated. So, this suggests to me that these variables are, in fact, related. Specifically, children who see violent media are more likely to engage in aggression (and likewise, children who are not exposed to violent media are less likely to behave aggressively).

But how different is “different enough” for us to say that we feel pretty certain there’s a relationship here? If we had 85 kids in that cell, would that be enough for us to say there’s a relationship? What about 80?

This is where hypothesis testing comes in, which we’ll cover in the next chapter.

4.6 Standardized Scores

There is one more topic I want to cover in this chapter while we’re thinking about probability, and that is what’s known as a “standardized score.” This is best illustrated with an example.

Imagine that I have to write a recommendation letter for two students I’ve had in class. One student was in PSYC340 and earned a 281 points in the course. The other student was in PSYC460 and earned a 175 points in the course. On its face, it seems like my PSYC340 student is better. But these courses have a different number of points in the course, and I also know that PSYC460 is very challenging, so students overall tend to have lower scores in the course. How is it possible for me to compare these students in courses that have different standards and different scales of points?

One solution I can use is something called a standardized score. Standardized scores use information about the distribution in order to understand where someone's score is compared to other people. Standardized scores are very useful in statistics, primarily because it "equalizes" scores when they come from different scales. Numbers like 281 and 175 don't really tell us much about how those students did. In order to contextualize those scores, it would help me a lot to know what the average score is in each class, as well as the standard deviation. So, let's add that information:

Table 4.4: Table with student scores, plus class averages and standard deviations

	PSYC340	PSYC460
Student Score	281	175
Class Mean	300	160
Class SD	20	12

Right away, these values give me some insight into how these students did. Even though my PSYC460 student had fewer points, this table tells me they actually scored above average, versus my PSYC340 student was below average in the course. But I'd like a more precise measure of how they compare. One standardized score I could use is something known as a **z-score**. Z-scores pop up in statistics all the time, because they communicate someone's score in terms of how many standard deviations below or above the mean they are. If you have someone's score, you can calculate their z-score using the following formula:

$$z = \frac{X - \bar{X}}{s} \quad (4.1)$$

In plain language, you take someone's score, subtract the mean from it, and divide that value by the standard deviation. The value that results tells you how many standard deviations someone is above (for positive values) or below (for negative values) the mean. Let's do this for each student:

$$z = \frac{281-300}{20} = -0.95$$

$$z = \frac{175-160}{12} = 1.25$$

We can see here that my PSYC460 student performed much better than my PSYC340 student. Let's say that I have another student from the same PSYC450 class with a z-score of +.50. If I want to know how many points they earned in the class, I can just reverse this equation to figure that out:

$$X = .50(s) + \bar{X}$$

$$X = z(12) + 160 = 166$$

Another nifty thing about z-scores is that you can use a z-table to figure out for any given z-score, what percent of people are above or below that score. In the appendix, **z-table-zscores** provides proportions for positive z-scores (negative z-scores take a little extra math). Basically, the area under the curve adds up to 1.00, and the value in the table tells you what percent of values fall below a certain z-score.

As an example, let's say I know someone has a z-score of +2.24, I can find that score on a z-table to figure out what percent of people fall below that score (also known as a **percentile**). If I use the left-hand column to find 2.2, and then the top row to find .04, I see the value listed is .9875. In other words, about 98.75% of people fall below a z-score of 2.24.

If we need a negative z-score, we follow the same procedure, but take the extra step of subtracting the value we find from 1. So, if I need to know what percent of people fall below a z-score of -.37, I find 0.3 in the left column, and .07 in the top row, which gives me the value .6443. I subtract this value from 1 to get .3557, which means about 35.57% of the population is below a z-score of -.37.

You can also figure out the proportion of people who might fall between scores. If I needed to know how many people have a z-score between -.37 and +2.24, I can do that. We already know that 35.57% of people fall below -.37. We also know that 98.75% of people fall below a score of +2.24, which means only 1.25% of people have a higher score than that. And if the area under the curve equals one, we can see that:

$$1 - .3557 - .0125 = .6318$$

So quite a few people – 63.18% – have a score between a z-score of -.37 and +2.24. I realize that right now this seems a little abstract, but z-scores will pop up again in this textbook, so this might be a section you'll want to revisit for a refresher now and then.

There are a few other common standardized scores we use in social sciences, so here's a handy summary of them in case you encounter them:

Table 4.5: Means and SD of common standardized scores

	z-scores	t-scores	IQ
Mean	0	50	100
SD	1	10	15

Standardized scores can provide us a quick shorthand for how someone compares to others. Someone with a t-score of 60 is 1 standard deviation above the mean. Someone who has an IQ of 130 is 2 standard deviations above the mean (which is usually the standard for someone to be considered “gifted”).

5 Hypothesis Testing

5.1 Learning Objectives:

- Understand the idea behind hypothesis testing, and how it bridges the gap between theory and data.
- Know the difference between one- and two-tailed hypotheses (aka directional and non-directional hypotheses) and why it's important to be clear about your hypotheses prior to starting an analysis.
- Understand the possible outcomes of a hypothesis test (correct decision, Type I error, Type II error)
- Know the four ways we can increase the power of a study.
- Understand the difference between the frequentist approach (which uses null hypothesis significance testing) and Bayesian approach (which uses other approaches to defining “importance” of findings).

Hypothesis testing is incredibly important to the work we do, but I also think it is one of the most widely misunderstood aspects of social science... by students, by laypeople, and unfortunately, sometimes by scientists themselves. If you have some experience in research, you know there's a typical cycle research follows:

1. Identify a theory that seems interesting, or observe a phenomenon that seems interesting.
2. Devise a hypothesis about that phenomenon you want to test.
3. Decide how to measure (and/or manipulate) your variables.
4. Conduct your test and collect data.
5. If your data supports the existing theory, add it to the pile of evidence. If the data does not support the existing theory, consider reasons why that might be the case.
6. Start the process over again with a new question.

Solid science is built on devising testable hypotheses based on theories, and then researching and replicating that research to build up a body of evidence. While this all seems simple, it is not! It is easy to get sloppy with this process. Researchers started to raise concerns about this back in the 1950s, but this all came to a head in the 2010s when it was discovered that only around 40% of psychological studies could be replicated (see Baker (2015)). There are several reasons we're in this situation now:

- Journals are not terribly excited about publishing articles that said “We did a study and found that none of our hypotheses worked.” So, studies that show a lack of support for a theory get thrown into a desk drawer and forgotten about.
- Researchers are also not terribly excited to replicate other researchers' studies. Social science and statistics are established on the assumption that we will test theories many times on different samples, but in practice, this doesn't happen as much as it should. If we don't do these replications, we never have an opportunity to refute the findings of a study with a new sample.
- Many researchers are often required to publish their research in order to keep their jobs or their funding. This creates an incentive to game the system so research turns out a significant finding. This can range from questionable methods (p-hacking is one example of this, which I will talk about in more detail later in the next chapter) to outright falsification of data.

In this chapter, I'll spend some time talking through what hypothesis testing is and how it works (or how it *should* work, anyway). But hypothesis testing tends to be a rather black-and-white way of thinking about a world that contains many shades of gray, so I'll also talk about some alternative ways to think about hypotheses and how they relate to theories and results.

5.2 Why do we bother with hypothesis testing?

The best way to make a case for the importance of hypothesis testing comes from an old analogy known as the Texas Sharpshooter:

An army commander is marching his troops through the Texas countryside. They are exhausted and need to rest, and the commander spots a farm on the horizon. They stop at the farm and ask the farmer living there if they can take shelter overnight in his barn. The farmer agrees, so the troops go to the barn and get settled.

The commander walks around the side of the barn and is shocked to see a series of bulls-eyes painted on the side of the barn, with a bullet hole perfectly in the center of each bulls-eye. The commander rushes back to the farmhouse and asks the farmer “Who shot up the side of your barn? I need a sharpshooter for my troops!”

The farmer laughs and says “My son did the shooting, and trust me, you don’t want him.”

“Why not?” asks the commander.

“Because he shoots the barn first, and then draws the bulls-eyes around the bullet holes.”

I mentioned way back in Chapter 1 that humans are not very good at avoiding their biases. Because a lot of social science feels like “common sense,” it is easy to justify any finding after the fact. Imagine that I tell my students that I found a $+0.60$ relationship between how much you like your coworkers and how good your job performance is. They might say “Of course, that makes perfect sense! If you are happy at work, you’ll feel motivated to do a good job.” Now imagine I say that actually, it’s a -0.60 correlation between those variables. My students might say “Of course, that makes perfect sense! If you like your coworkers, you’ll spend time chatting with them instead of doing your work.” The point is, in social science, it’s easy to come up with explanations after the fact!

When we make up a story to fit our data, this is known as **post-hoc theorizing** and it is best to avoid this because it allows our biases to determine the story rather than the data. So, having a clear and testable hypothesis before you collect any data is really how science *should* work.

To have a testable hypothesis, we need to be clear about what our variables are, and what operational definition we are using for each variable. We should also be clear about whether we have an idea about the nature of the relationship. In some cases, there might be a theory that makes the expectation very clear, so we can devise a **one-tailed hypothesis** (also known as a **directional hypothesis**) where we can be specific about what we expect to find:

There will be a *positive* relationship between job satisfaction and job performance.

Children who watch violent media will be *more likely* to behave aggressively than children who do not watch violent media

In some cases, we might not be sure, or there might be competing theories and we want to see which one bears out in a particular situation. In this case, we might have a **two-tailed hypothesis** (also known as a **non-directional hypothesis**) where we think *something* will happen, but we don’t want to specify exactly what we think will happen:

There will be *a relationship* between job satisfaction and job performance.

Children who watch violent media will behave with *a different level* of aggression compared to children who do not watch violent media.

Although we don’t mention it specifically most of the time, we are also establishing what we call a **null hypothesis**. We don’t mention it because it’s incredibly boring! All the null hypothesis says is that nothing is happening at all:

There will be *no relationship* between job satisfaction and job performance.

Children who watch violent media will have *the same level* of aggression compared to children who do not watch violent media.

What it is important here is that you should establish a clear hypothesis before you collect data so that you have a plan for how you will analyze the data and what the results mean for your theory.

5.3 Possible outcomes of a hypothesis test

So, once we have our hypothesis, and we know how we are going to measure our variables, we are ready to collect our data. Once we have our data, we can use it to help us infer what is going on with the population. Ultimately, when we conduct a statistical test, we actually are comparing our findings to what we would expect to find if our null hypothesis is true. In other words, we are asking the question:

What is the probability I would obtain data that looks like this, given the null hypothesis is true?

(Aha! There's one of those conditional probabilities I mentioned in Section 4.5!)

Most of the time, we are hoping for a low probability – in most social sciences, we have set our standard at a 5% chance or lower – this is why you often see $p < .05$ all over the place in articles. If you are curious about why 5% seems to be the magic number, the short version is “We didn’t have computers back then and calculating this probability for every statistical test was a pain.” You can read a longer version of the history of it in (Cowles and Davis 1982). The main point here is that once we think there is less than a 5% chance of us making a mistake, we say that our finding is “statistically significant.” This means that we can say that we “fail to retain the null hypothesis.”

You might be confused as to why we’re so focused on the null hypothesis instead of the other, more interesting hypothesis. It is really confusing! So let me take a moment to emphasize what we are *not* asking, and walk through how all this relates to Dataland and Theoryland.

A question people often incorrectly think we are asking is “What is the probability my hypothesis is true given the data I’ve obtained?” This question seems very similar to the bold question I put above, but these are actually very different questions. So, let me walk through an example to make this difference clear.

Let’s return to the violent media and aggression example. If I am an all-knowing being living in Theoryland, and I have access to all humans past and present, I have access to The Truth about whether exposure to violent media is related to aggression. Conversely, if I am a normal human living in Dataland, I simply don’t have access to all that information; I only have access to my sample that I’ve tested. So, if we think about what I know here in Dataland, and what

is *actually* happening in Theoryland, there are four possible outcomes when I test a hypothesis asking whether there’s a difference in aggression based on whether a child is exposed to violent media:

Table 5.1: Possible outcomes of a hypothesis test

	Theoryland	
Dataland	There <u>is</u> a difference.	There <u>is not</u> a difference.
My data indicates there <u>is</u> a difference.	Correct decision	False Positive (Type I error or α)
My data indicates there <u>is not</u> a difference.	False Negative (Type II error or β)	Correct decision.

Two outcomes are good news. If my data shows there’s a difference, and over in Theoryland, the all-knowing being looks at all the data ever and says “Yep, there is indeed a difference” then I’ve made the right call. Likewise, if my data shows no difference, and the all-knowing being says “Yup, nothing to see here!” then I’ve also made the correct decision.

However, there are also two mistakes I might make. My data might show a difference between groups, so I think that there’s a difference, but the all-knowing being sees that actually, there is no difference if I was able to test all data ever. In that case, I’m saying there’s an effect when there is not. This mistake is known as a false positive or a Type I error. If I do that, I might end up wasting all kinds of time and money studying a topic that actually doesn’t matter.

Alternatively, I might see there is absolutely no difference between groups, but the all-knowing being sees that actually, there are differences between these groups. This is known as a false negative, or a Type II error. If I make this error, I decide these variables aren’t useful, and I move on to other topics when perhaps I should have continued to research this topic.

The problem is that we don’t have any access to Theoryland, so **we will never know whether we have made an error**. We are always at risk of getting a funky sample or measuring a variable incorrectly, so we have to tolerate some level of uncertainty in social science. Most of the time, we are content to risk a 5% chance of making a Type I error. In some cases, perhaps a false positive is really consequential– maybe if finding an effect of violent media on children means we have to gather up all the televisions and videogames and shoot them into the sun, I’d want to make darn sure I have an accurate answer. In that case, we can be more stringent, and maybe only tolerate a 1% chance ($p < .01$) or a .01% chance ($p < .001$) of making a Type I error. But we can never have a perfect zero chance of making an error; errors are just part of social science.

You might be asking why I’m totally ignoring Type II errors. We’ll dig into that in the next section!

5.4 Power

In most social sciences, we don't typically spend much time talking specifically about Type II errors – instead, you might have heard of something known as **power**. Power is our ability to detect a difference if one exists, and it is calculated as $1 - \beta$ (remember that β is our probability of a Type II error). In general, we aim to have at power of at least .80 (or put another way, we'd like our Type II error probability to be under 20%). In other words, when our study has a lot of power, it is more likely we'll find an effect. Ideally, we want power to be as high as possible, and there are several things that can help increase power. Some of them we have some control over, and some of them we don't.

5.4.0.1 Probability of a Type I Error

One way we can increase our chances of finding a significant finding is to increase our chances of making a Type I error. After all, if we're less stringent about how big a difference needs to be for us to consider it present, we're more likely to find something. You can think of α and β like on opposite ends of a see-saw – as one goes up, the other goes down. So, if we choose to set $\alpha = .10$ instead of the traditional $\alpha = .05$, this means we need a smaller difference between groups for us to say that we feel confident there is a difference. However, it means that we are also more at risk of making a mistake about what is important. While we certainly *can* do this, most journals and researchers expect you to justify why you're deviating from the traditional threshold, and saying "Because I want to find a significant effect" is not a compelling argument. So, this is often not a practical solution for us to increase power.

5.4.0.2 The Test We Use

Another way we might be able to increase power is through the design of our research. Some tests, such as an independent t-test, compare two separate groups to one another. This is known as a **between-subject test**. Other tests allow us to test the same participants several times to see if an intervention changes their responses. This is known as a **within-subject test**. Within-subject tests are more powerful for detecting effects. Because we are looking for the change within a person instead of differences between groups, we can disregard some of the differences between individuals in a way we can't if we are doing a between-subject test (we'll dig into the details of this situation once we start talking about repeated t-tests in Section 7.4).

As an example for our violent media study, I've presented it as a between-subject test, where some children watch a violent show and others don't. However, I could also easily set this up as a within-subject test – maybe on separate days I show the same kids different cartoons and observe them. I'd need to be careful to counterbalance my study (some kids are randomly assigned to see the non-violent media first, other kids are randomly assigned to see the violent

media first) but doing this provides me with greater power to detect an effect. So, it seems ideal to always use a within-subject design, right? It doesn't always work out that way!

In my example, I might have a hard time getting access to these children for two separate days, so it might not be realistic for me to use a within-subject design. Sometimes within-subject tests are simply not possible. If I wanted to teach a skill to people using a video versus text, I can't re-test their ability to learn twice. As soon as they've learned the skill, they can't reasonably "unlearn" it for me to test them with the opposite learning condition. So, sometimes a more powerful test is not possible for us to use, so we have limited control over this aspect of power.

5.4.0.3 The Effect Size

Sometimes, particularly in experiments, we may have some control over the size of the effect in our study. In other words, how different is the control condition from the experimental condition? In our violent media example, the control condition is non-violent media (watching characters cooperate) and the experimental condition is violent (watching characters fight). But as experimenters, we have control over just how violent the experimental condition is. If the experimental condition was only a little different from the control group (e.g., the characters get in a verbal argument and push each other) then I'm producing a small effect. For a moderate effect, I could select more aggressive behavior (e.g., a fistfight). Or I could induce a large effect by having my experimental condition by showing the children some extreme violence (a character slices dozens of other characters in half with a sword). But I also have to consider what is realistic – it's not super likely that little kids are watching "Kill Bill" on a regular basis, so a low-to-moderate level of violence is more realistic.

If we are not doing an experiment, we don't have any control over the size of the effect – it is what it is! So, we are not always able to increase effect size to have more power in our study.

5.4.0.4 Sample Size

Usually, the easiest way for us to increase power in a study is to obtain a larger sample size. As we learned in Chapter 3, larger samples mean we have a smaller sampling error, which means our estimates of parameters are more stable and trustworthy. When these estimates are more stable, we can feel confident that a smaller difference is meaningful and real (i.e., not due to chance). So, this is one reason why researchers like to have a decent sample size for their studies – they are more likely to be able to correctly detect differences between groups.

If this is making sense so far, you might have a really important question to ask here: "If having a large sample makes it more likely your result will be significant, can't you just get a huge sample to ensure you get significant results?"

Well...yes, actually that is how it works! When I was learning statistics for the first time, this fact blew my mind. But this is exactly how it works – if I get a sample size of millions of people, I can find a difference of 0.0000000001 to be statistically significant. So, it's worth spending some time here talking about what statistical significance means, what it does not, and some alternative ways of thinking about it.

5.5 Statistical Significance versus Practical Significance

As a reminder from earlier in this chapter, when we find a result is significant, it means we are saying “There is only a 5% chance that we would have data that looks like this given the null hypothesis is true” (assuming we are setting $\alpha = .05$).

What we are not saying is “This effect is important.” If we have a small sample, we might find a very large and important effect isn't statistically significant, and if we have a giant sample, we might find a very tiny effect is significant. So, I want to make sure we separate out the concept of “important” and “significant.”

All we are saying when we say something is “statistically significant” is that we feel pretty certain that our findings are not due to chance; we think that something is going on out there in Theoryland.

One way we might think of importance is something called **practical significance** (sometimes called “clinical significance” in healthcare settings). Practical significance is whether the finding matters to real people in the real world.

Let's return to our violent media example. Let's say that the way I'm measuring aggression is seeing how many times a child hits another child on the playground over the course of a week. Imagine I collect a large sample, and I find that the difference between groups is statistically significant. Great! But once I look at the data, I realize that the difference between kids who do versus do not watch violent media is that they hit their friends .03 fewer times this week. It's a LOT of work to prevent kids from seeing violent media, and all that work means that a kid hits others a tiny bit less? WHO CARES? Even though the result is statistically significant, it doesn't really have much practical effect.

On the flip side, if I do the study with a small sample, I don't have much power in my study so I find in the end that I don't see a statistically significant difference between groups. SAD! But when I examine the averages for my groups, perhaps I learn that the group that watches violent media is hitting other kids 20 times more than the non-violent media group every week. It might not be statistically significant (so, I could just see this big difference due to chance) but dealing with one kid hitting another 20 times during a week is a big deal for the teachers. So, even though this study didn't find statistically significant results, it has practical significance, so maybe I will rethink what kinds of media I show kids because it does seem to have an impact on how they behave, even though there's a chance my findings are just an anomaly and if I conducted my study again, I might not see such a big difference.

Statistical significance is a black-or-white situation. Something either is or is not significant; saying something is “slightly significant” is like saying someone is “slightly pregnant” or “slightly dead.” It doesn’t make sense. Practical significance, meanwhile, is a judgment call, so some people might think something is practically significant and others would not.

If I have a drug that improves symptoms for 40% of people, but makes symptoms worse for 20% of people, is it helpful? If I have a training that improves test performance by 5 points, but it costs \$500 per student, is it worth it? The answer depends on who you’re talking to, and whether it seems like these outcomes are valuable. That’s what practical significance is.

5.6 An Alternative to Null Hypothesis Tests

The way we’ve been talking about hypothesis testing so far is known as “null hypothesis significance testing” (NHST), sometimes known as the “frequentist approach” – we rely on p-values (the probability of a Type I error) to tell us whether something matters. This is currently the most popular way of thinking about results, but it’s not the only way!

The “Bayesian approach” is one alternative that has been gaining support slowly over the years. Rather than having a black-or-white split between “significant” and “not significant,” Bayesian approaches allow for more nuance. For example, instead of talking about statistical significance, instead we would explain to our audience how certain we are of a particular result, and what real-world implications exist for our findings (e.g., “Watching violent media was associated with children hitting other children 5-10 more times per week”).

Another way we can get away from NHST is to report effect sizes. These are often reported as r (essentially a correlation coefficient that can range from -1.00 to +1.00) or using a statistic known as *Cohen’s d* (which essentially describes how many standard deviations different groups are – if the effect size in our study was $d=1.00$, it would mean that the means for the control and experimental group are one standard deviation apart).

Even though I tend to prefer the Bayesian approach because I think it’s more transparent, the downside is that the results are not clean cut. How large of an effect is “important”? Bayesian approaches don’t specify this, so it’s up to reader interpretation. This does make it harder for us to be clear about which results matter and which results don’t matter. I won’t say much throughout the rest of this book about this approach because it’s still not the standard, but I do like students to be aware of it in case they come across Bayesian statistics in the wild.

6 Getting to Know Your Data

6.1 Learning Objectives

- Understand what the assumptions are for parametric tests.
- Diagnose problems with normality (such as skew or kurtosis) in a data set.
- Diagnose problems with heteroscedasticity in a data set.
- Use data visualization to understand the nature of data so you can diagnose and resolve problems
- Apply methods for resolving data that is non-normal and/or heteroscedastic

We've spent a lot of time familiarizing ourselves with the concept of data and how statistics help us bridge the gap between a theoretical population and a concrete sample. We're almost ready to start diving into learning about specific statistical tests, but first, I must emphasize the importance of getting to know a data set and getting a data set ready for analysis. Quite frankly, data cleaning and preparation is much more difficult and time-consuming than running the analyses! Once you know how to run a test, it's quite easy to do so. But if our data is a bunch of garbage, we can't expect to get meaningful results from our analysis, so data cleaning and preparation is a tedious but vital step of the process.

6.2 Assumptions of Parametric Tests

The vast majority of statistics I cover in this text are what are known as **parametric tests** – that is, at least one variable is a continuous variable that we'd expect to be normally distributed in the population. This is generally true in most social sciences – most variables will have the majority of people clustered near the mean, with smaller proportions of people in the very high and very low ends of that variable. So, parametric tests are quite common in social science fields. However, in order to confidently use a parametric test, there are a few **assumptions** (i.e. expectations) we have of our continuous variables.

6.2.1 Assumption 1: One Variable is Interval or Ratio Data

I noted in the previous section that parametric tests have at least one variable that is continuous. As you might recall from our previous chapter, this means that our data needs to be either an interval or ratio data. For the most part, this requirement is straightforward – if none of your variables are continuous, you need to find a non-parametric test to use. One area where some people get into trouble is assuming that ordinal variables are continuous – they are not! You should not treat ordinal variables the same way because with ordinal variables, we don't have precise information about the amount of the variable at hand – we only know the rank order of participants, so it is not appropriate to pretend that ordinal variables are appropriate for parametric tests.

6.2.2 Assumption 2: Errors should be independent

We also want to make sure that my observations are independent of one another. In other words, one participants' score should not affect the score of another participant. If I found out after the fact that, for example, one of my participants was sharing answers on a test or that they were pressuring other participants to answer in a certain way, this is a problem for my data! In some cases, this makes my data unusable, because by allowing participants to influence one another, they've introduced systematic bias into the data, which makes it difficult to distinguish error from effects.

Sometimes, this assumption is impossible to avoid because of something called **dependencies** in data. For example, if I'm collecting data on workers in a manufacturing plant, many employees might share the same supervisor, and perhaps there's something about that supervisor that causes them to answer in a certain way. Or, if I'm measuring the same person multiple times, their responses might share error because some people have certain tendencies in how they respond (e.g., some people stick to the middle of a scale, others are more strict or lenient in their ratings). While this does violate an assumption, if we have information collected about how errors might be related (e.g., we keep a record of employee supervisors or we have an ID variable that allows me to track one respondent's ratings across multiple time periods) we can incorporate those variables into our analyses. One solution to this problem is something known as multi-level modeling, which is far beyond the scope of this book. But suffice it to say that there are ways that we can work with this assumption if we're careful about recording potential dependencies in our data.

6.2.3 Assumption 3: The data is normally distributed

We've talked a little bit about normal distributions, but as reminder, the normal distribution is where our data is symmetrical, with most people clustered close to the mean, and the number of people tapering out as we get further from that mean.

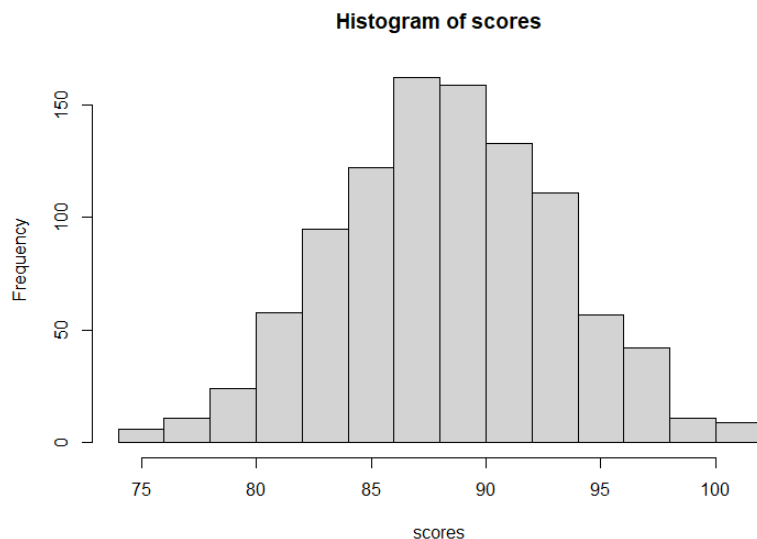


Figure 6.1: A normal distribution.

In small samples, this normal curve can look a little funky, but we typically hope it will somewhat replicate a normal curve. It won't always! Sometimes we might have an **outlier** or an **extreme value** where one data point is very far away from everything else, which will pull the mean away from the center. We might also have a **ceiling effect** or a **floor effect**, where one end of our normal distribution gets squashed. For example, household income often has a **positively skewed distribution**, where the curve is flattened on the left side (because minimum wage produces a floor effect) and the tail might be very long on the right side if we have a few millionaires in our sample.

Exam scores can sometimes produce a **negatively skewed distribution**, where the right side of the distribution is squashed (especially if my test was too easy or students cheat, so too many people get perfect scores) and the left side has a long tail (a few students getting a zero on the test because they missed it).

Sometimes we might also get **kurtosis** in our data, where we don't have many people right in the middle of my distribution, which produces a flat top and thicker shoulder and tails on my distribution (we sometimes call this platykurtic). Or alternatively, maybe I have too many people in the middle of my distribution, producing a sharp peak in the middle, with short tails (also known as leptokurtic).

This assumption we will want to check prior to doing any analyses, as if we violate that assumption, we might want to make some adjustments or use a different type of test. More on this later!

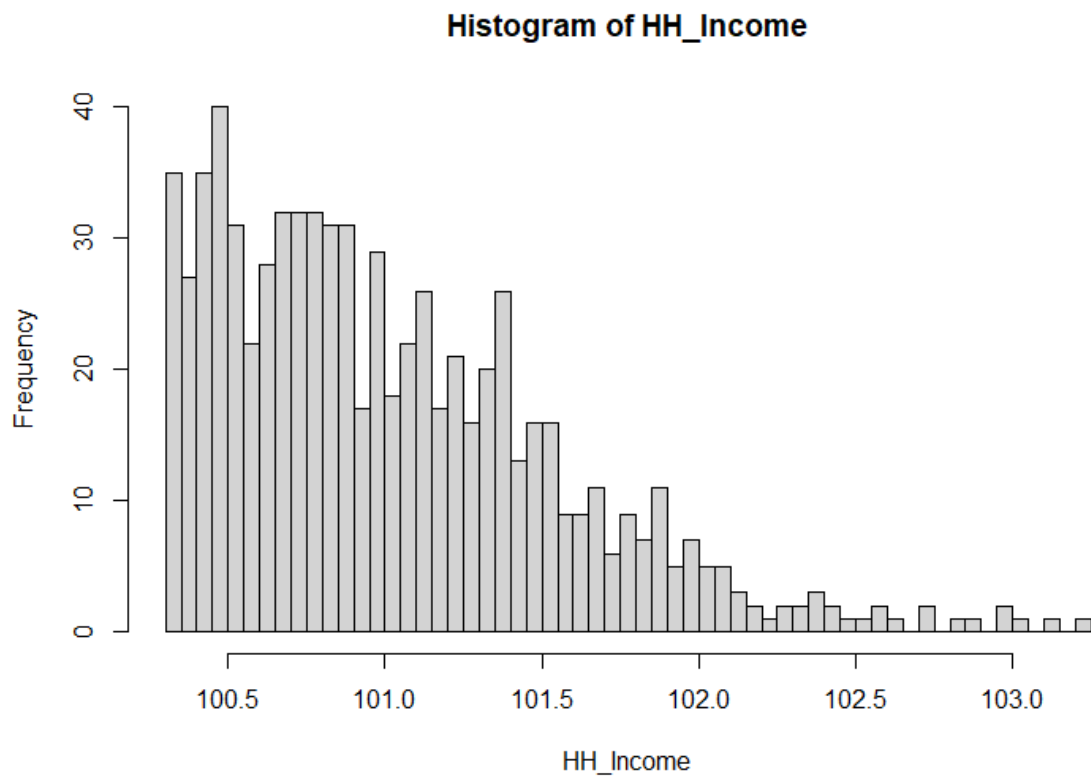


Figure 6.2: A positively skewed distribution.

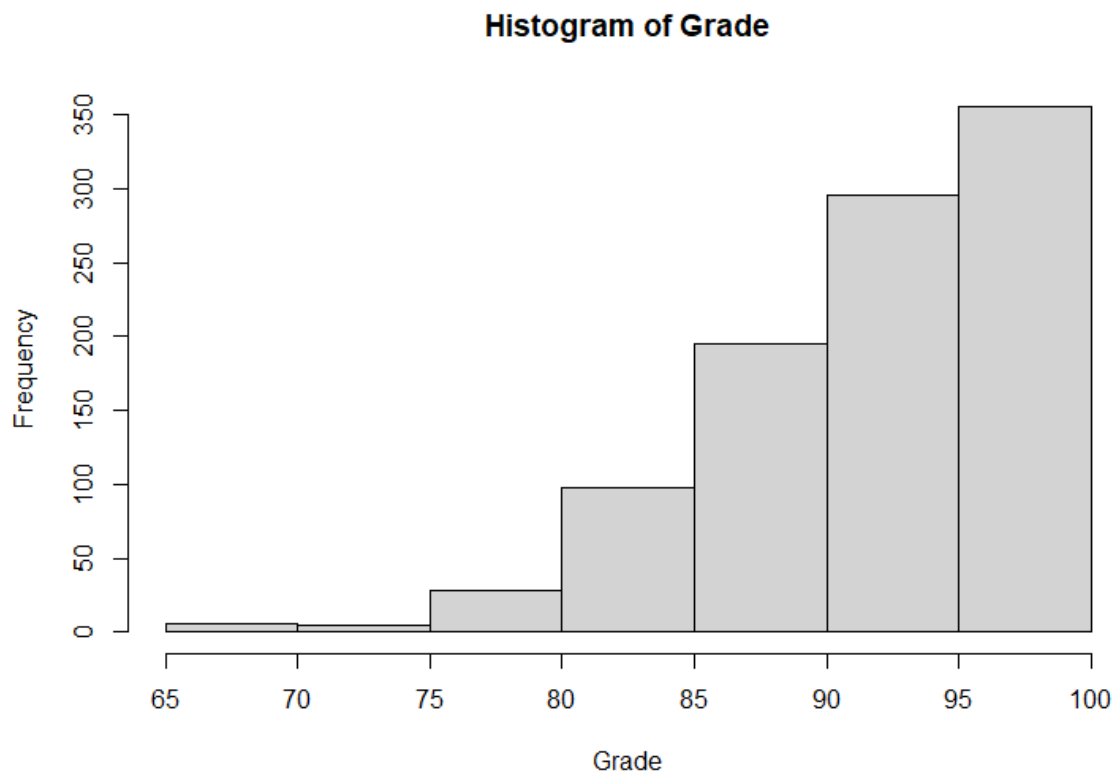


Figure 6.3: A negatively skewed distribution.

6.2.4 Assumption 4: We have homogeneity of variance (homoscedasticity)

Homoscedasticity is a long and fancy name for a relatively simple concept: We want the variance in our sample to be roughly similar at all values of a predictor. Consider the graph below:

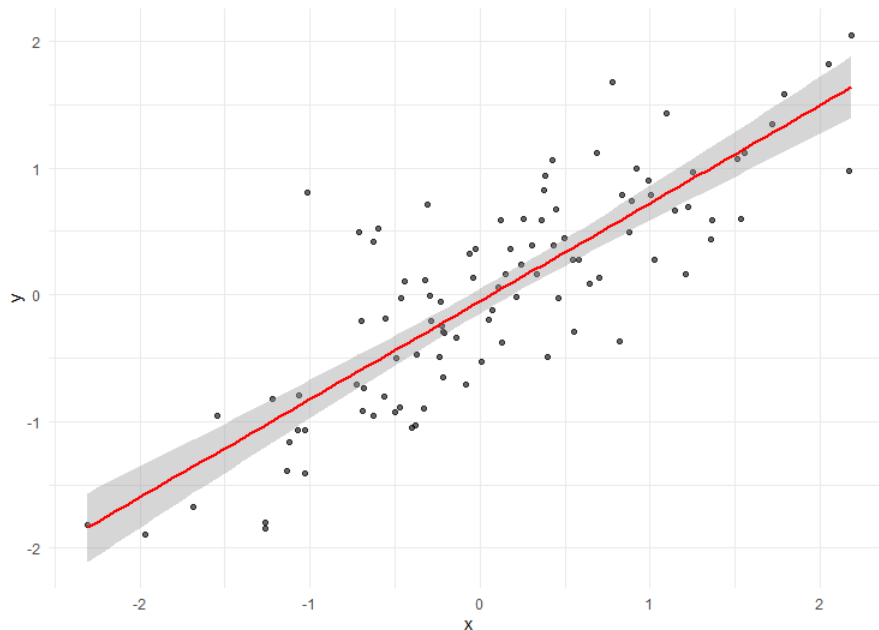


Figure 6.4: Data with homogeneity of variance (homoscedastic)

In Figure 6.4, you can see that the data spreads out similarly from the line as you move from left to right. This is good – it shows homogeneity of variance, or homoscedasticity, which means our assumption is met.

In Figure 6.5, you'll notice that the points sit very close to the line on the left side, and spread out a lot toward the right side, producing a trumpet shape. This indicates that we do not have homogeneity of variance (heteroscedasticity). This is not so good! What this means is that at the low end of our predictor, we're pretty accurate with how people will score on our outcome, but we are less and less accurate as our predictor increases. Heterogeneity can cause some problems with estimates, so if we have heterogeneity, we might need to make some adjustments to our test to get more accurate estimates of parameters.

So, how will we know if our data is violating any assumptions? There are a few different ways you'll want to get to know your data to help you catch these issues and understand your data better.

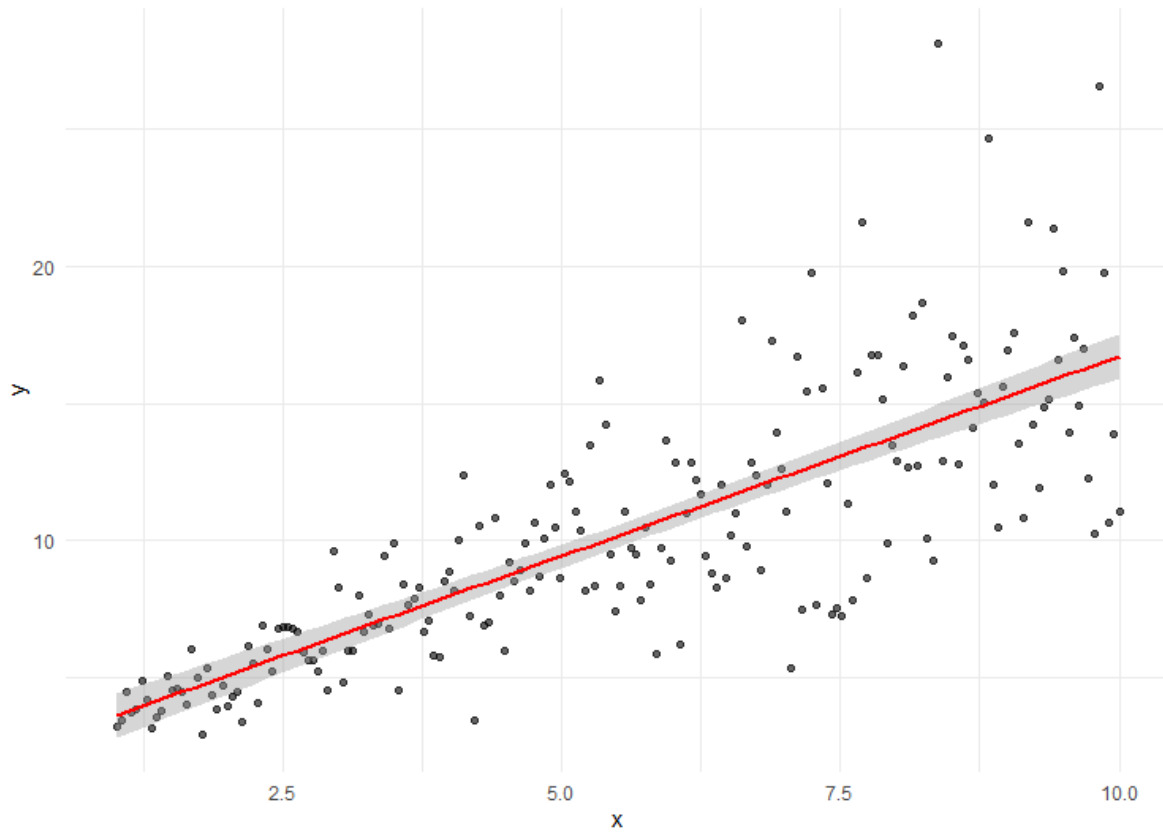


Figure 6.5: Data with heterogeneity of variance (heteroscedastic)

6.3 Data Visualization

One of the simplest ways to check out your data is by visualizing it. This can be helpful for you as you try to understand what your data is trying to tell you. It can also be helpful to an audience if you are trying to explain your data or your findings.

One common visualization of data is the **histogram** – in fact, I’ve been showing you these already! Here’s another just for fun:

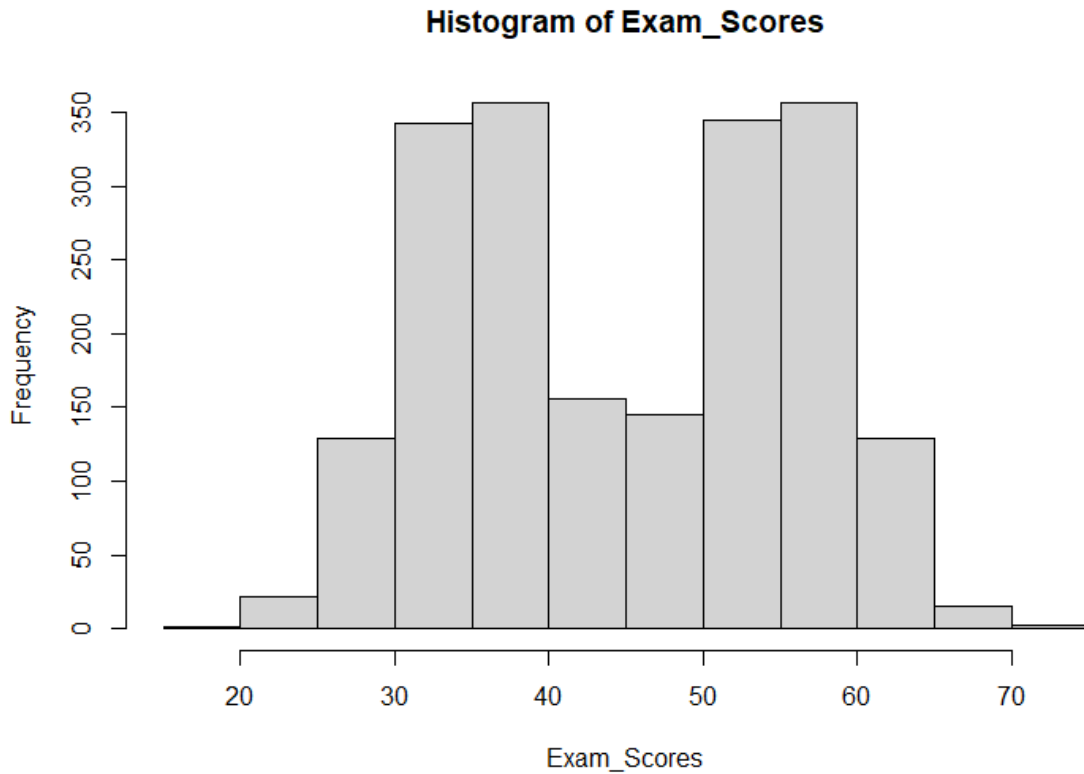


Figure 6.6: A histogram of a bimodal distribution

Right away, we can see some weirdness in this data – it is certainly not normally distributed, as we can see two humps rather than one (also known as a bimodal distribution). So, this is a quick way to see what’s going on with a single variable.

Another way we can examine data is through a **boxplot** (also known as a box and whisker plot). With a box plot, there’s typically a line in the middle denoting the mean. The rest of the scores are broken into 4 equally-sized groups (also known as quartiles). The middle quartiles are noted by the box, and the outer quartiles (the lowest and highest quartiles) are noted by

the whiskers. Some programs also provide a symbol (like an asterisk) to indicate outliers or extreme scores that go way beyond the end of the whisker. This can be helpful for identifying incorrect or unusual data that might need to be addressed:

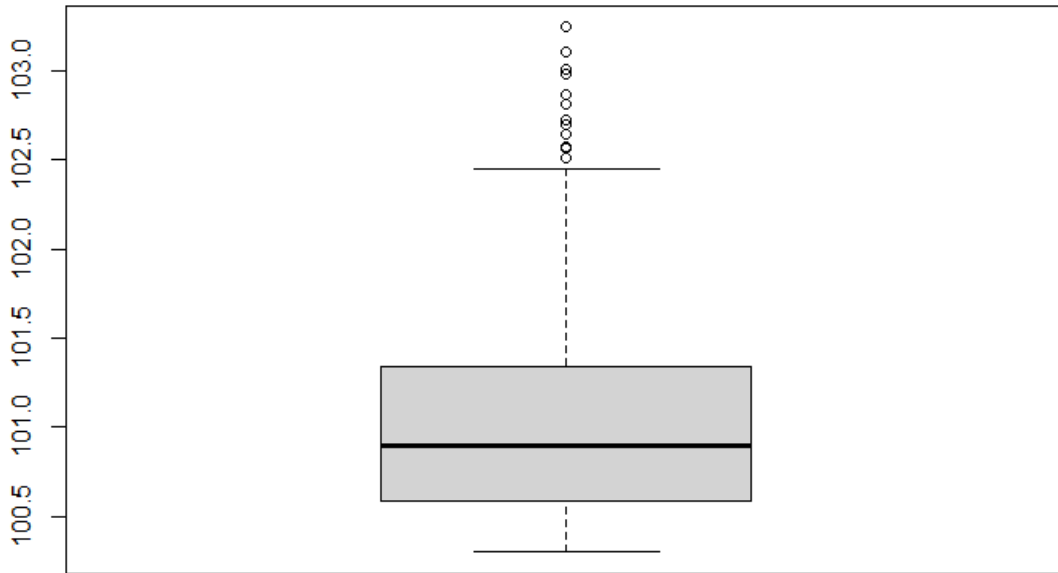


Figure 6.7: A boxplot of positively skewed data.

This example shows we have some skew in our data– the whisker on the bottom end is squashed, and the whisker at the top is quite long, with a trail of dots indicating some outliers.

Sometimes we might want to examine several groups at once. For example, maybe I want to look at the averages across three different groups. One way to do this is through a **bar chart**:

Here, the bar charts provide the average for each group. They also have an error bar – error bars can mean different things, so for this example, I’ve set it up to provide a 95% confidence interval for each mean.

Another similar option is the line chart:

Each point provides the mean for each group, and some programs allow you to also provide an error range for those lines as well. Generally, we use bar charts when we’re trying to show

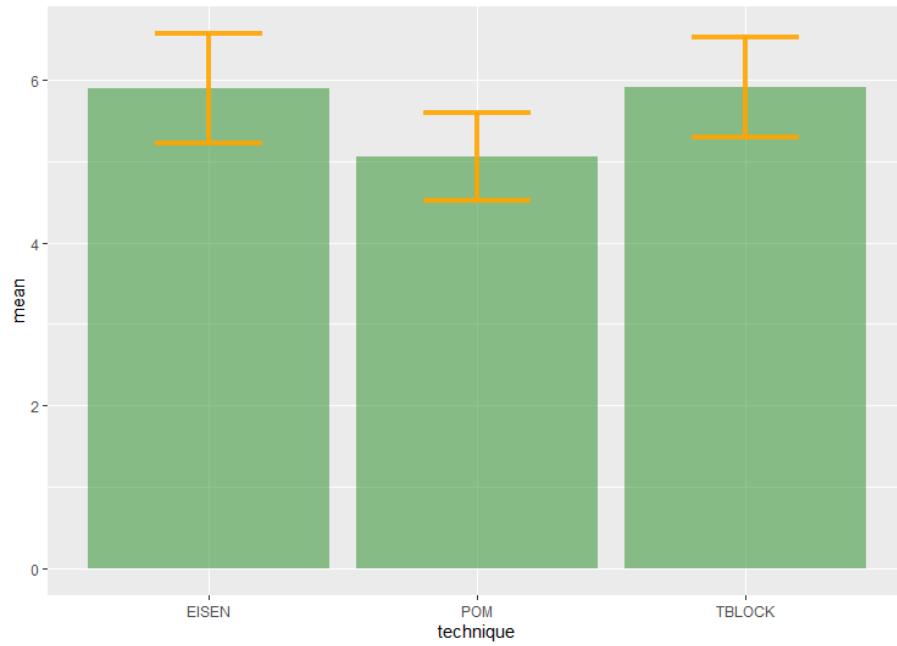


Figure 6.8: A bar chart showing effectiveness of time management technique.

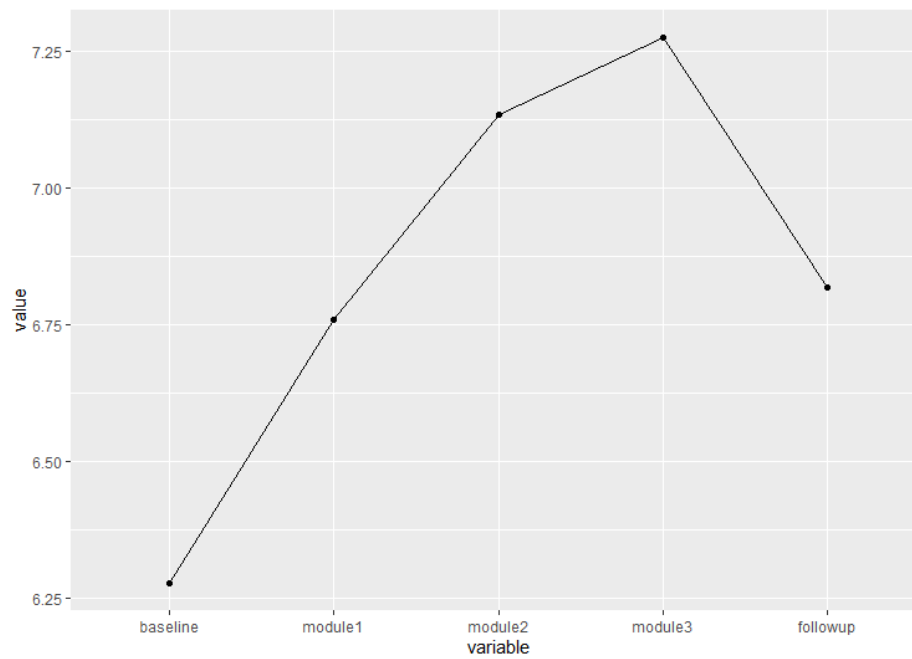


Figure 6.9: A line chart of sleep across multiple time periods.

differences between groups, and line charts if we are charting how something changes across time. This is not a hard and fast rule, so if you prefer one over the other, that's fine.

Finally, if we wanted to see the relationship between two variables, a scatterplot is a helpful option:

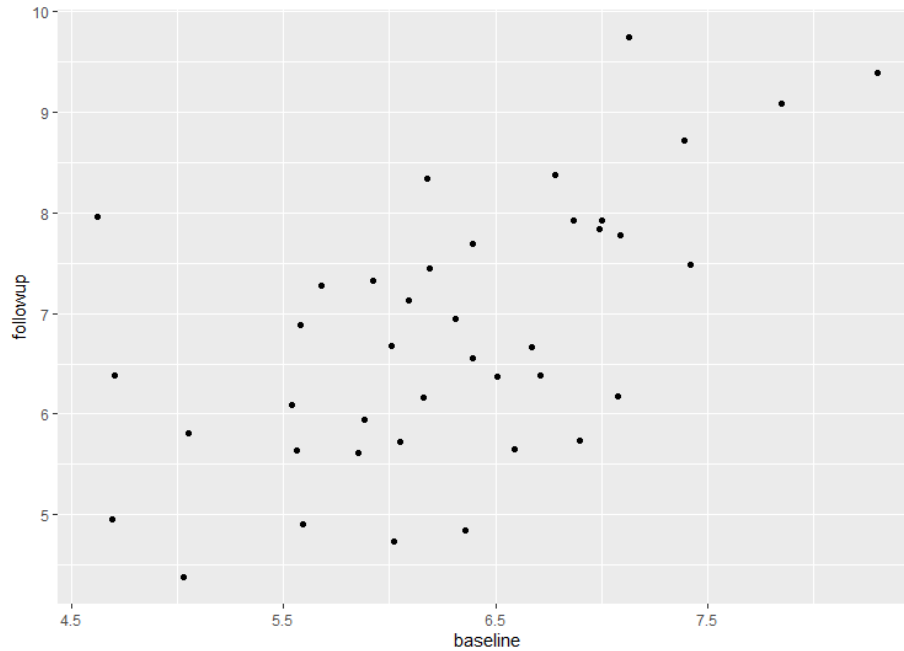


Figure 6.10: A scatterplot of sleep times at a baseline measure and at a follow up measure.

Here, we can see that there seems to be an upward trend, with those who get more sleep at the baseline measure also getting more sleep during the follow up.

I'm being very utilitarian right now, and showing you the plots I generally use to explore my data. If you are presenting to an audience, there are dozens of snazzy ways to present data. Looking up information on “data visualization” will give you plenty of creative ways to show off your data. But for our purposes, these four provide us with a lot of information about our data.

6.4 Resolving Issues

So, what do we do if we are violating an assumption? What do we do if our data looks funky? What you do really depends on a lot of things:

- What does this variable usually look like?

- How was it collected? Could there be problems with how we collected the data?
- Is it possible some data is inaccurate and needs to be corrected?
- Is the data coming from one source/group, or several?
- Does the data need to be manipulated or transformed in some way?
- What tests are we conducting, and are they sensitive to assumption violations?

Let's walk through a few examples where my data looks weird and ways I might choose to approach it.

6.4.1 Example 1: Head size and math score

Imagine that I have collected some data by measuring my participants' head size (circumference in inches) and their score on a 20-point math quiz. Here are my results:

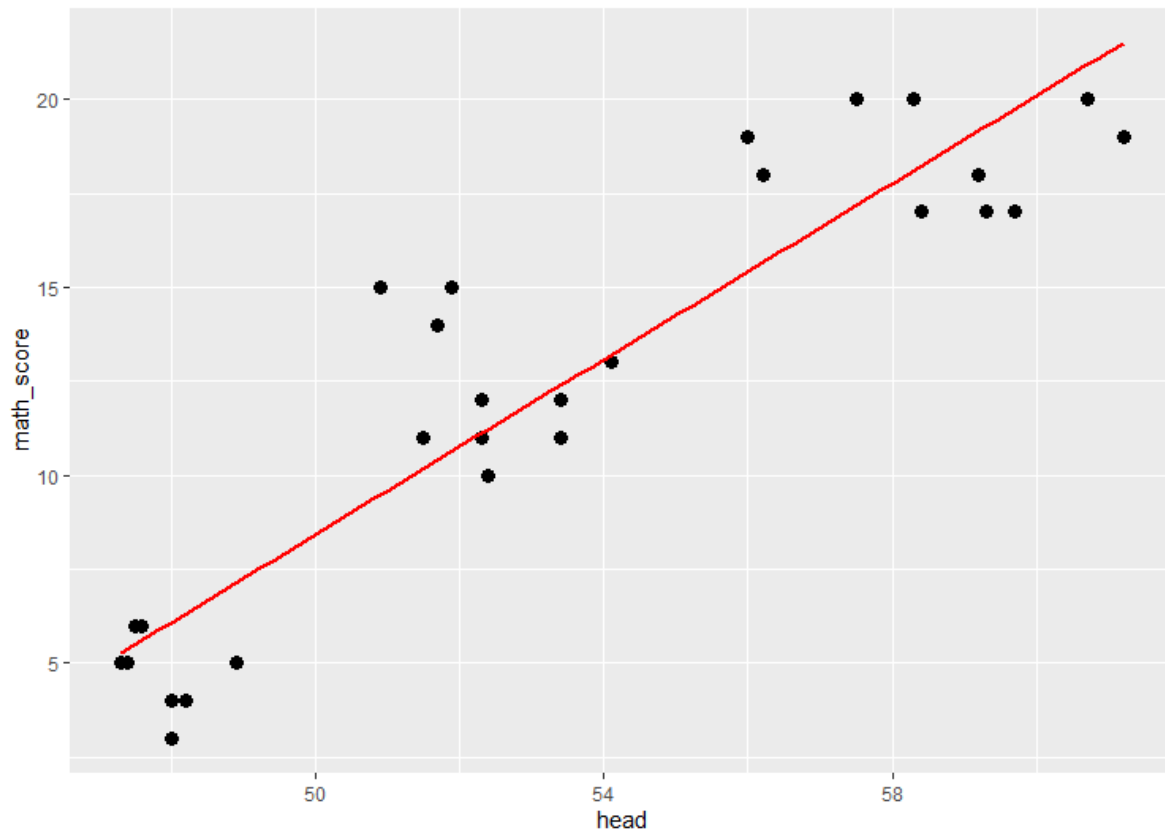


Figure 6.11: A scatterplot of the relationship between head size and math score.

At first glance, this chart seems to be telling me a straightforward story – the bigger your head, the better you are at math, right? Not so fast! There might be some nuance to this data. Perhaps I learn that there are actually three different groups in this data set – toddlers, children, and adults. If I separate out these groups, I get a completely different story...

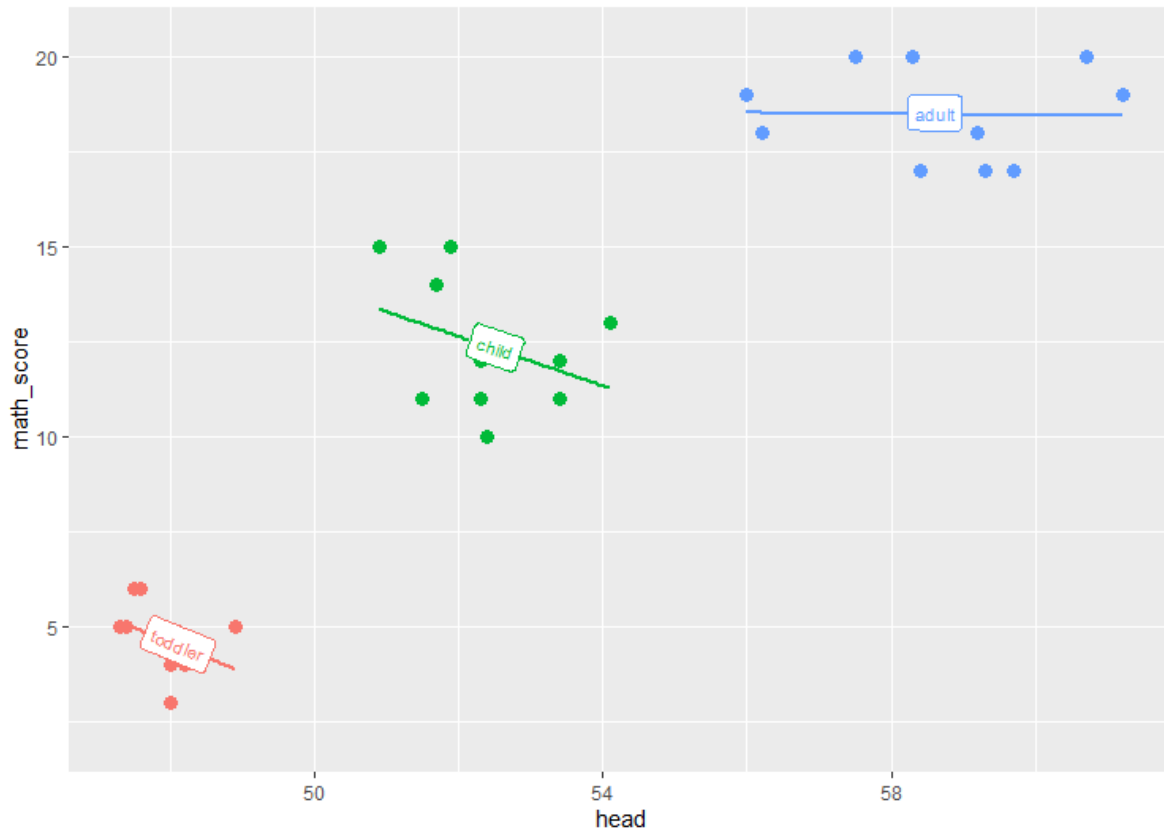


Figure 6.12: A scatterplot with details and the three age groups

Oh, whoops! It turns out that age was a bit of a confounding variable here – when I separate out groups, suddenly what looked like a strong relationship is mostly an artifact of the different groups. Within groups, there’s a weak negative relationship between head size and math score. In fact, Figure 6.12 illustrates the Simpson paradox – when we combine data from different groups, it can obscure the effect going on within a group. So, always be careful about combining different groups!

6.4.2 Example 2: Scuba Diving

For our next example, let’s imagine that I’ve collected data on different sports experiences people have tried (you can find this data set [here](#)). I have this result when I ask people how

many times my participants have gone scuba diving:

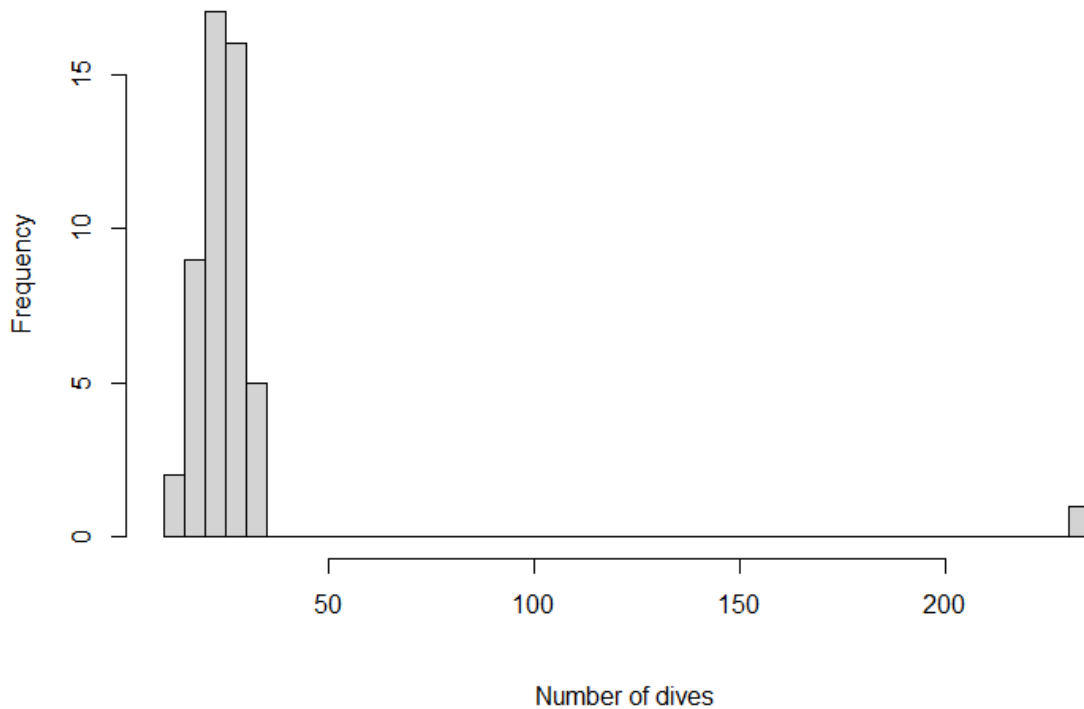


Figure 6.13: Histogram of number of dives.

This is a pretty small sample, so I know that I'm not going to get a great looking normal distribution, but we have a pretty extreme value popping up here! It's possible that this data point is a mistake – maybe this was an online survey, so the 234 might be a typo (perhaps they meant 23, or 24, or 34?) In this case, if I'm not sure, perhaps the safest approach is to simply trim the data by deleting the data point:

That seems to have cleared up the problem. However, let's say that instead, I'm able to get in touch with this person, and they say the number is accurate because they are a scuba instructor! But I still have the problem with this value skewing my data – if I report the mean with this data point it is 28.22, and if I report it without this value, it is 24.02. So, I probably shouldn't leave this value as is. But I have a small data set, and I would like to keep this person in the data set because their other data might prove instructive, but I can't have her value be so extreme! Another option I can use is called **winsorizing**. When we winsorize a data point, we take an extreme value and replace it with a less-extreme value. There are a few different ways you might do this. One option is to find the next highest score (here, that would be 35)

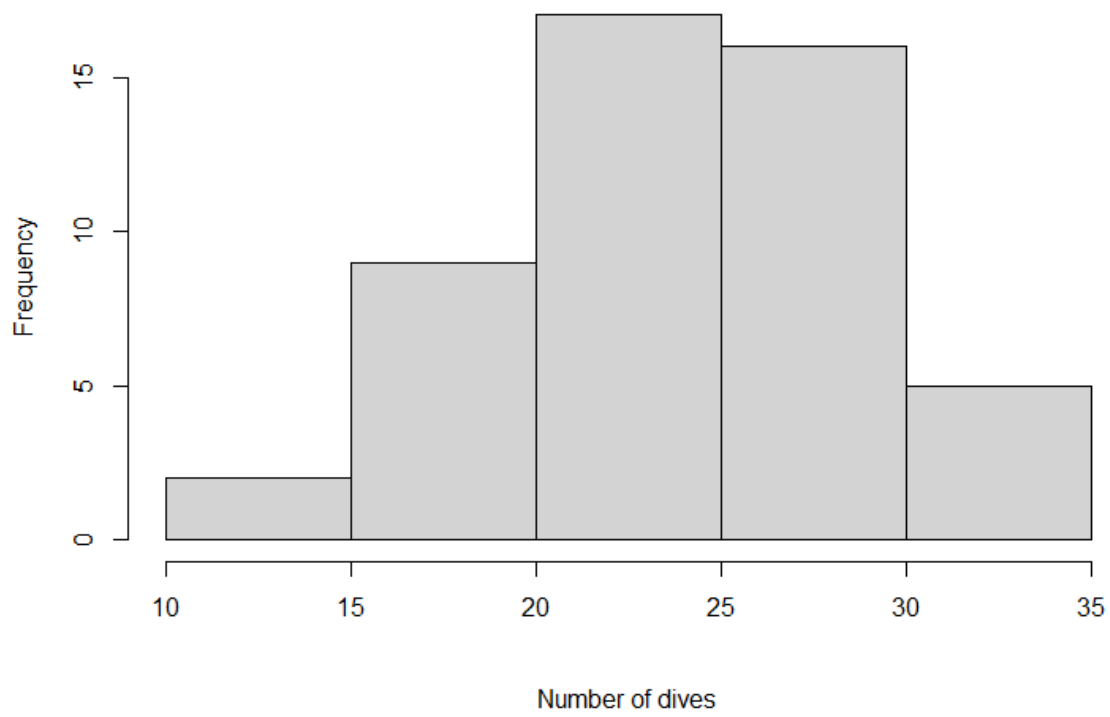


Figure 6.14: A histogram of dives with the outlier removed.

and make the extreme value just higher than that score. So in this case, we might replace the 234 with 36. Another option that is common is to use z-scores and replace the outlier with whatever is a z-score of 3. In this example, the z-score for this scuba instructor is 6.84, which is pretty extreme! So, instead I can take the mean (28.22) and use the data standard deviation (30.10... this is large due to that outlier!) to reassign the value:

$$(3 \times 30.10?) + 28.22 = 121.3$$

So, I could replace that extreme value with 121, which is less extreme. Either ways is fine, but you absolutely must explain in your results that you have winsorized the data and how you have done so.

6.4.3 Example 3: Reaction time

For my third example, let's imagine that I conducted a study on reaction times (data is [here](#)) and I have the following results:

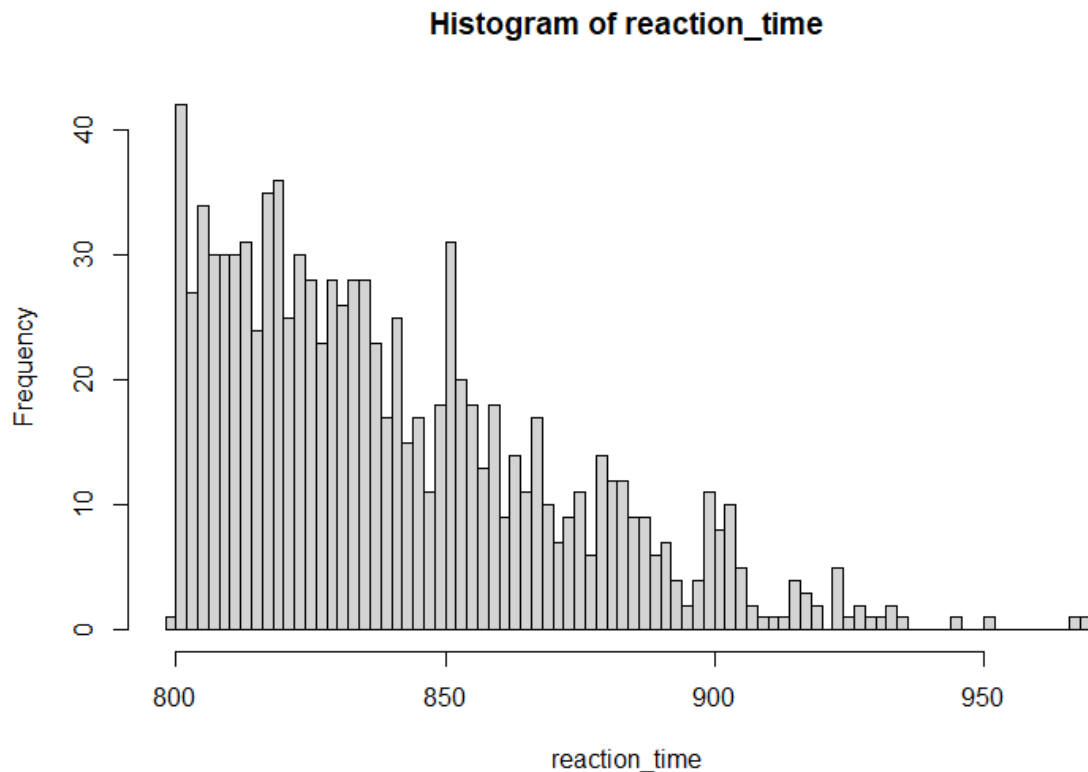


Figure 6.15: Reaction time data.

We can certainly see this data has a lot of positive skew! In some cases, this might be a cause for concern, but reaction time is just one of those variables that tends to be skewed. Humans can only be so fast, and there's almost always a few slowpokes in the data set that lead to a long tail in the positive direction. Because this is such a common issue, almost every article on reaction time applies something called a **transformation** to the data. Transformations are ways we can manipulate the data to rescale it and push it towards more of a normal curve. There are many transformations out there (square root, reciprocal, and reverse hyperbolic to name a few) but the most common one we use on reaction times is a log transformation (that is, I take the natural log of this value). When I do that, here's what my new histogram looks like:

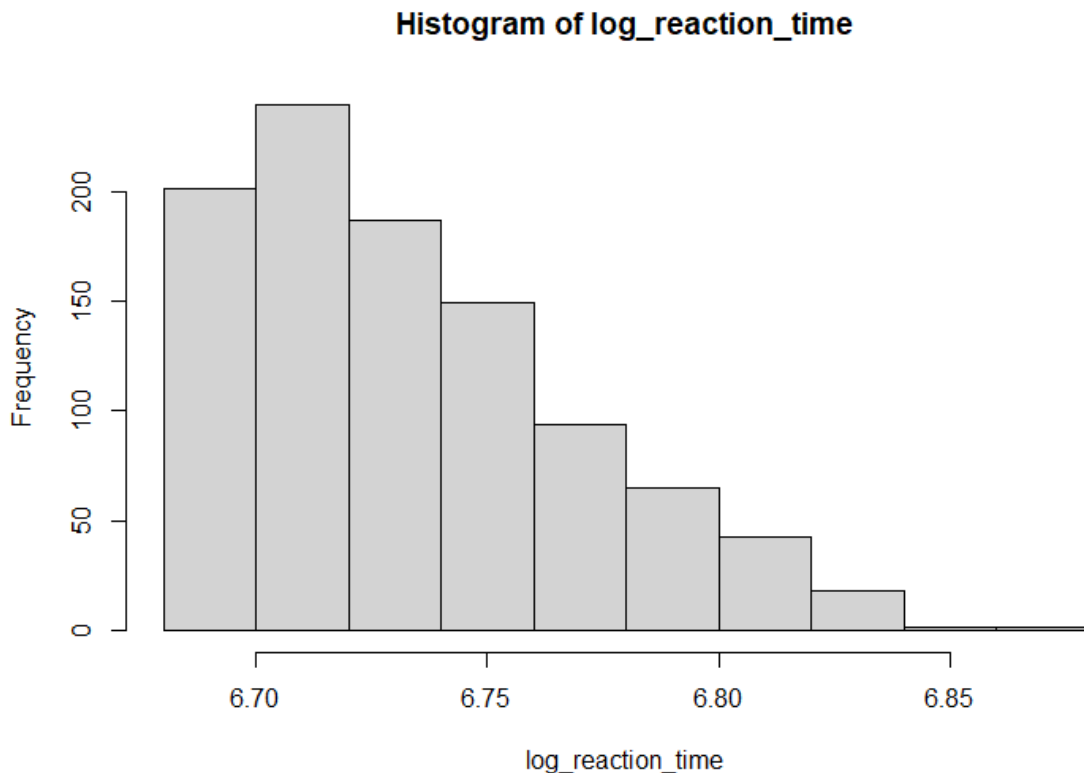


Figure 6.16: Log-transformed reaction time.

As you can see, the log transformation squashes that long tail down a bit to make it more normal. Here's the issue though – when I do this, I can no longer call this variable “reaction time,” it has to be called “log reaction time” to remind my readers about the transformation. So if I want to talk about how sleep correlates with reaction time, I now have to talk about the correlation between “sleep and log reaction time” instead, which makes it harder for people to interpret in practical terms. So, transforming the data is sometimes more trouble than it's

worth. The benefit of applying a transformation is that you get a more accurate estimate of the population and whether your finding is statistically significant, but the downside is that it's harder to interpret and explain.

There are no hard and fast rules for when to use a transformation. With my own data, I usually use the following approach to decide whether to use one:

- Does my data have a lot of skew or kurtosis? If the skew/kurtosis is less than 2, I typically don't worry too much about it. If the skew/kurtosis is greater than 2, I take a look at a histogram to see what's causing the issue. If it's an outlier, I made a decision about how to address the outlier. If it looks like the variable is just skewed in general and I think I need to address it, I would use a Kolmogorov-Smirnoff test to see if it is significant. If so, I might consider using a transformation.
- Does my data have homoscedasticity? I take a look at a graph first (either a scatterplot or some boxplots/histograms) to see if it looks really heteroscedastic. If I'm concerned about it, I apply a Levene's test to see if it is statistically significant.
- Who needs to understand this data? If I'm working with a lay audience, the trade off for accuracy versus clarity is probably not worth a transformation. If I'm working with a scientific audience, or if I'm working with a variable that is often transformed (like salary or reaction time) then I am more likely to apply a transformation.

6.5 Ethics

While having no hard-and-fast rules when it comes to data cleaning gives us a lot of flexibility, it also can sometimes tempt people into bending the rules when it comes to statistics. There are plenty of gray areas out there, but here are a few things that I would advise against if you want to maintain a good code of ethics around data cleaning and analysis.

First, don't falsify data! You'd think this would be obvious, but here in 2026, we've had a spate of fairly high-visibility cases of wholesale fabrication of data, as well as unexplained data manipulation. (I don't want to name names, but if you want the hot gossip, datacolada.org has some fascinating articles about questionable data practices). So, don't make up data! If you remove data points or replace them through winsorizing, be transparent about what you are doing and your rationale for doing so.

Second, establish a clear research plan and stick to it! If we are doing things completely above board, we should:

- Have a specific hypothesis (or several hypotheses) we intend to test with our sample. We should know ahead of time whether our hypothesis is one- or two-tailed.

- We should use a power analysis to establish what sample size we need to detect an effect size of interest. And we should stick to that number – we shouldn’t wrap up a project early once we get the findings we want, and we shouldn’t collect additional data with the goal of “making” something significant just by increasing power in our study.
- If we are using a frequentist approach, this means that the p-value must be under .050 to be significant; technically, .051 is not less than .050, so we shouldn’t treat it as significant. Sharing effect sizes and confidence intervals can help our audience understand our findings with more nuance than we get with just plain null hypothesis significance testing.
- If we are doing multiple tests, we should apply a Bonferroni correction where appropriate to avoid finding something significant based on chance. You will learn more about this in [Section 8.1](#).

Do researchers fudge these details? Yeah, they probably do. Will you someday find yourself in a position where you engage in some not perfectly ethical behaviors? Maybe! But at least you should know when you’re doing something that is not above board. Hopefully, as researchers and journals demand more transparency and accountability, some of this will disappear. But I at least want you to know if you’re doing something you’re not supposed to do!!

7 T-Tests

7.1 Learning Objectives

- Know what a t-test is, and what kinds of research questions it can address.
- Know what aspects of the data influence the size and significance of a t-test.
- Know what a one-sample t-test is, and what questions it can address.
- Know what a between-subject t-test is, and what questions it can address.
- Know what a within-subject t-test is, and what questions it can address.
- Know how to write the results of each type of t-test.

As we start learning about inferential tests, we'll start with a simple question: How do we know that a mean is meaningfully different from another mean?

Take, for example, the (true) story of William Sealy Gosset, who worked for Guinness Breweries (yes, the same Guinness that still sells a very nice stout). Gosset was a chemist and brewer, and as you might imagine, he needed to conduct research to examine what aspects of the brewing process related to tastier beers. The problem was that he was limited to small samples, and he wanted to know whether small samples related well to the population. For example, if Gosset made a tweak to a recipe and found that it improved the tastiness of that beer by 1 point on a 10-point tastiness scale, would that translate to a meaningful improvement in how the population liked the beer? Because he only had access to relatively small samples, he wanted to know more about how big of a sample he would need to detect a difference that would meaningfully translate to the beer-drinking population. He paired up with Karl Pearson (who you'll learn more about later) to devise a test to establish this. Because Guinness had proprietary information they didn't want falling into the hands of their competition, Gosset was required to use a pseudonym, and he wasn't allowed to share that his research was on beer. In the end, Gosset published a paper under the name Student (Student 1908), and this is where the *Student's t-test* comes from.

In this chapter, we'll be talking about three different types of t-test. But, at its most basic, a t-test tells us a ratio between the differences that are systematic differences between two groups, and the differences that are random noise.

7.2 One Sample T-Test

Let's start with the one sample t-test. This test is used when you want to compare the mean of a single sample to that of a population – that is, we want to see if our sample is notably different from the general population. Consider, for example, a study by Clark and Stoner (2008) that investigated whether children who read braille spelled as well as children who read traditional books. In other words, is it reasonable for us to think that braille readers are similar to the general population in terms of spelling skills?

So, let's dive into the math of this question. Here's the formula for the one-sample t-test:

$$t = \frac{\mu - \bar{X}}{S_{\bar{X}}} \quad (7.1)$$

Hmm! This might look a little bit familiar to some equations we've seen before...

$$z = \frac{\bar{X} - \mu}{\sigma}$$

This equation looks a little similar to Equation 4.1, the equation for a z-score. With a z-score, we were trying to compare someone's score to the rest of the sample. This is analagous to a t-test, but instead of looking at an individual and comparing them to the sample, we are looking at a sample and seeing how it compares to the population. I like to think of it as “scaling up” the same idea from one person to a group of people. You'll also notice instead of using the population standard deviation (σ), we are using the standard error of the mean ($S_{\bar{X}}$, see Equation 3.1) . Why would we do this?

In this case, **we don't know the population standard deviation**. That is something that exists only in Theoryland, and we would never actually be able to measure this directly. So, as I mentioned before, we need to estimate this value by using the standard error of the mean.

So, essentially, the t-test tells us the ratio between how big the difference is between two means (the population mean and our sample mean) compared to how much the data points spread out from that mean. Take a look at  below:



Scan for helpful animation!

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This animation illustrates what happens to our t-test and our significance when the standard deviation is the same, but the difference between our means changes. As the difference between means gets larger, the magnitude of our t-test gets larger, and our significance value gets smaller. In other words, as our means get further apart, we can feel more confident that these groups are truly different from one another.

Now take a look at ??? below. Here, the means stay the same, but within each group, the variance (and thus standard deviation) shrinks. As you can see, when standard deviation gets smaller, our t-test value gets larger, and our significance value gets smaller. So, when people within each group are very similar to each other, a small difference between means might be significant, because as the standard deviation shrinks, our distributions stop overlapping as much.



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The point is that the t-test is a ratio of the differences between means and the variance within each group. The less those distributions overlap, the more confident we feel that they are actually different in the population.

Let's get a little more concrete by walking through an example.

In Clark and Stoner (2008) , braille readers had an average score of 99, compared to a population average of 100. We also know from their data that the standard deviation for the sample was 11.70, and the sample size was 23. So, we can plug in these values to get the t-test value itself:

$$S_{\bar{X}} = \frac{11.70}{\sqrt{23}} = 2.44$$

$$t = \frac{100-99}{2.44} = .41$$

This is quite a tiny t-test value! If you were doing your computations by hand, you'd need to compare this value with the critical t-test value on a t-table (?@tbl-ttable) – if our obtained t-test value is lower in magnitude compared to the critical value, we conclude the difference is not significant. Here, our degrees of freedom are 22, and we are doing a one-tailed test with $\alpha = .05$. According to the t-table, our critical t-value is so the critical t-test value is 1.71. Our obtained t-test value of .41 does not clear that bar, so we can declare this test non-significant, and that our sample likely doesn't differ from the population.

(The good news is that we have software that can do these calculations for us, so we don't generally refer to t tables anymore; instead, the software would indicate that the p-value is greater than our usual .05 standard cutoff for declaring something significant).

In other words, if we think of our t-test value as a ratio, we don't see much of a difference between our sample mean and the population mean, and we see quite a bit of variation within our sample; the fact that our sample is one point lower than the population is very likely just due to random chance. (We can't be 100% certain about this, though! It is certainly possible that we had a weird sample just by chance, so we might be making a Type II error... but we would never know if that was the case.)

So ultimately, the authors reasonably concluded that braille readers seem to be quite similar to other students in terms of their spelling skills, which seems like good news. Of course, it's a small sample, so we would want to keep in mind whether this sample was representative, and whether the size of the difference is practically significant.

So, in conclusion, if you have a sample that you need to compare to some average in the population that you know already because you're using some sort of standardized measure, the one-sample t-test is one possible approach to answering that question. It's not the type of question that pops up a lot in the field, but it's a good baby step into t-tests.

7.3 Between-Subject T-Test

A much more common test you'll encounter is the between-subject t-test. Here, we have means from two separate samples, and we want to compare those means to see if they are significantly different.

As an example, let's stick with spelling as a topic. Imagine that we have two randomly assigned groups of children who are learning how to spell. In one group (Group A), we show them only the correct spelling of a word. In the second group, we show them both the correct spelling, plus some common mistakes students make when they spell the word (Group B). We want to know whether these different instruction techniques lead to different scores on a spelling test.

In this case, the students are in completely different groups, and we're testing their spelling scores only once. So, let's say we obtain the following data (which you can also find in the file [spelling between.csv](#)):

Table 7.1: Table with 10 scores for Group A and 15 scores for Group B

Group	Score
A	113
A	112
A	110
A	109
A	109
A	108
A	106
A	105
A	103
A	111
B	112
B	110
B	108
B	107
B	105
B	104
B	105
B	104
B	104
B	104
B	104
B	103
B	107
B	105

Group	Score
B	104

You can see that within each group, there is some variance – we have some good and not-so-good spellers in each group. The question, then, is whether we can see a meaningful difference related to which instruction method they used. At a basic level, this question is quite similar to the one-sample t-test, but we’re replacing the population mean with a sample mean.

However, there’s another issue here – you may notice that we don’t have perfectly balanced samples (there are a few more students in Group A) and the variances within each group are not perfectly equal. So, we need to make some adjustments to our t-test formula to make it work for this situation:

$$t = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\left(\frac{s_p^2}{n_A} + \frac{s_p^2}{n_B}\right)}}$$

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2}$$

Let’s break down what’s going on here. The numerator is not too different in this case from what we saw in Equation 7.1 – again, we are subtracting one mean from the other to see how different they are... that’s pretty straightforward. Also, it’s pretty arbitrary which group we put in as “Group A” versus “Group B,” so whether the t-test value is positive or negative doesn’t particularly matter.

But there’s a lot going on in the denominator! So, let’s break this down piece by piece. It actually is still a standard deviation, but we’ve needed to make an adjustment to it, because our two groups have different numbers of participants, and different standard deviations. So, we essentially need to weight each contribution the groups make to the standard deviation by multiplying those weights by their respective degrees of freedom (n-1). That is what the pooled variance (s_p^2) is doing. Then, we take that pooled variance and pop that value over the number of people in each group, and add those two values together. But remember that with a t-test, we need a standard deviation, not the variance, in the denominator. So, we have to take the square root of that final value.

Long story short, what this equation does is similar to what we saw with the one-sample t-test – the difference between means is in the numerator, and a standard deviation is in the denominator. This t-test value is what determines whether the difference between groups is significant.

So, as an illustration, let’s try applying this to our made up data set on spelling.

$$\bar{X}_A = 108.60$$

$$\bar{X}_B = 105.73$$

$$s_A^2 = 10.04$$

$$s_B^2 = 6.64$$

$$n_A = 10$$

$$n_B = 15$$

Using these values, we first calculate our pooled variance:

$$s_p^2 = \frac{(10-1)10.04 + (15-1)6.64}{10+15-2} = 7.97$$

And then we apply that value to our t-test equation:

$$t = \frac{108.60 - 105.73}{\sqrt{\left(\frac{7.97}{10} + \frac{7.97}{15}\right)}} = 2.49$$

Again, if you are using statistical software, the program should provide you with a significance value. But just for clarity's sake, you can also use `?@tbl-ttable` to find the critical t value; a two-tailed test with $\alpha = .05$ and 23 degrees of freedom gives us a critical t of 2.07. Because our obtained t-test value is larger than this, we can conclude this is a significant difference. In other words, our results indicate that the group that saw the incorrect spelling did markedly worse than the other group, possibly because when it came time to take the test, the students had a hard time remembering which was the correct versus incorrect spelling. So, we have an answer to our question about what technique works best for students.

7.3.1 What to do if your assumption of normality and/or homoscedasticity are violated

The good news is that t-tests are typically quite **robust** against violations of this type. However, if you have concerns about the normality or homoscedasticity of your data, you can apply what's called a **Welch's correction**. This correction doesn't use pooled variance, but keeps variances separated in the denominator. It also makes an adjustment to the degrees of freedom for the test – specifically, it reduces the degrees of freedom to account for the increased uncertainty of the test. Ultimately, the correction raises the critical t-test value, making it slightly less likely that you'll obtain a significant result. Most of the time, the results won't differ by much, and in fact, some social scientists recommend using the Welch test even if you don't have issues of normality or heteroscedasticity Delacre et al. (2019). If you do use a Welch correction, you should report it in your write up, and adjust the degrees of freedom and the t-test result accordingly.

7.3.2 Calculating an Effect Size for Between-Subject T-Tests

As mentioned in Section 5.6, effect sizes can be helpful to convey the practical significance of the results, especially in cases where your sample size is small. To calculate effect size as a **Cohen's d**, we can use the following equation:

$$d = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{s_p^2}} = \frac{108.6 - 105.73}{\sqrt{7.97}} = 1.01$$

This suggests that there is about one standard deviation that separate the group means – that’s a fairly large effect, so it shouldn’t be surprising we found a significant difference with such a small sample size.

7.3.3 Writing the Results of a Between Subject T-Test

When sharing the results of a between-subject t-test, you want to convey several pieces of information to your audience:

- What type of t-test you are doing;
- What the average scores are for each group and their accompanying standard deviations;
- The t-test score, its degrees of freedom, and its significance;
- In some cases, the effect size
- If you used a Welch correction, you should indicate that.

So, for this example, here’s an example of how I might write up the results:

In condition A, students scored an average of 108.60 ($SD=3.17$) on the spelling test, while they scored an average of 105.73 ($SD=2.58$) in condition B. A two-tailed between-subjects t-test indicated a significant difference, $t(23)=2.49$, $p=.021$. This represents a fairly large effect size ($d=1.01$).

As a writer, you have lots of flexibility in how you might write this aside from using the correct notation for your test, so here’s another way someone might write this:

Results of a two-tailed independent t-test indicated that students did better in condition A ($M=108.60$, $SD=10.04$) than they did in condition B ($M=105.73$, $SD=3.17$); this difference was significant and represented a large effect size, $t(9)=2.79$, $p=.021$, $d=1.01$.

If I applied a Welch correction, I could simply say “a Welch test” instead of “a t-test”. You will also find that you usually have decimals in your degrees of freedom, so make sure to include those in your notation reporting the Welch test. For example:

A Welch t-test indicated a significant difference between groups, $t(16.61)=2.38$, $p=.029$

Results can also be conveyed through graphics – a bar graph is typically the easiest way to communicate t-test results graphically.

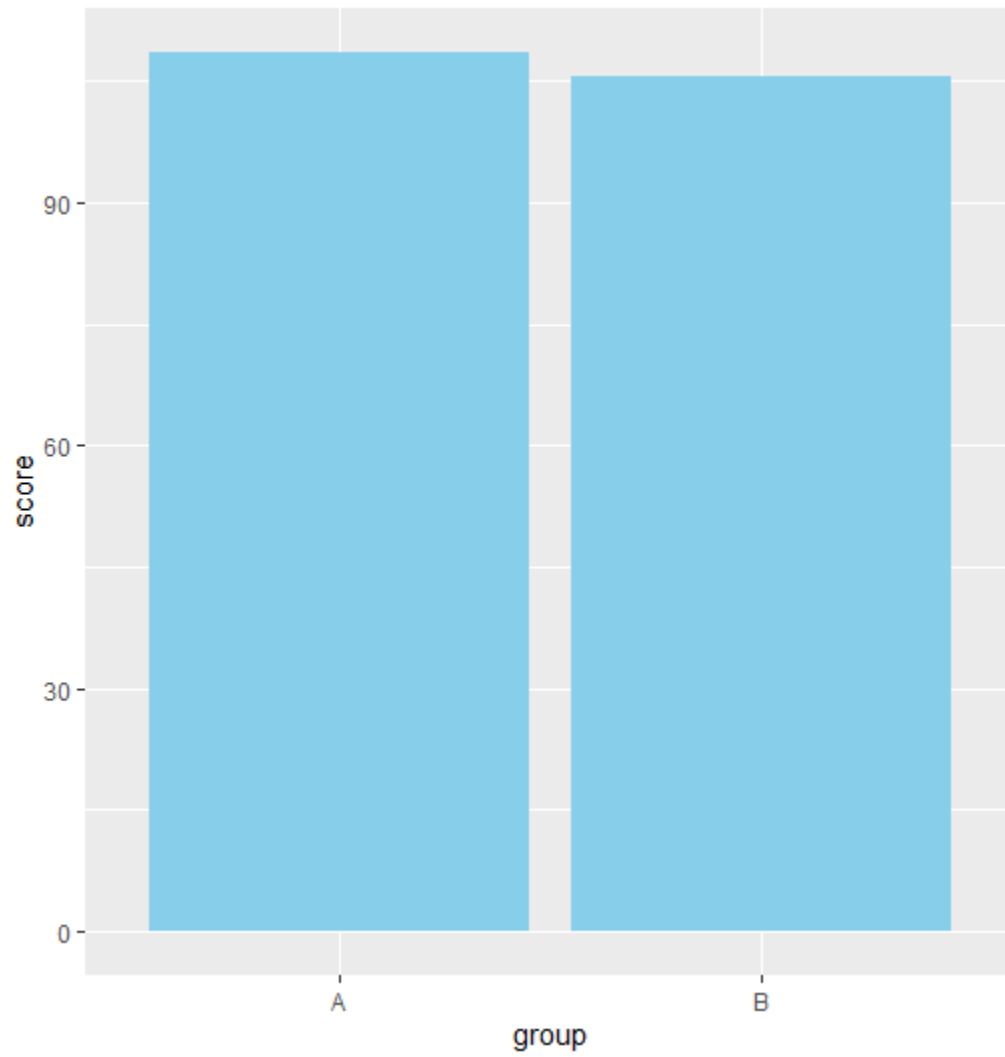


Figure 7.1: A bar chart of group means.

7.4 Within-Subject T-Test

The final type of t-test you'll encounter is the within-subject t-test. This test has a lot of different names, depending on the nature of the research you're doing. In many cases, this t-test is used if you are testing participants twice, like with a pre-test and a post-test to see if there's been improvement after a treatment or intervention; this is often called a repeated measures t-test or a dependent t-test. In some research contexts, it might not be feasible to assess people with both a pre- and post-test, so sometimes participants are matched on a characteristic, and are given the two different conditions and assessed on their outcome. Because they were matched ahead of time, they likely are similar and so they can be treated the same in terms of how they respond to two different treatments. This is known as a matched-sample t-test, but mathematically, all of these different t-tests are the same. The main difference here is that the two scores share some relationship, so we can't treat this as a between subject sample, where we assume the scores between groups are independent of one another.

We can take our example from the last section and re-design it as a within subject t-test (the data set is [here](#)). So, in this case, we might teach our students some words using only the correct spelling, and have them take a spelling test on those words (Condition A). We then teach them some different words using the other technique of showing them both the correct and the incorrect spelling (Condition B). There are some methodological problems with doing it this way, but let's not worry too much about that for the sake of this illustration.

So now, our data might look like this:

Table 7.2: Table with data from 10 people who took a spelling test twice.

Participant ID	Condition A	Condition B	Difference
1	113	110	3
2	112	108	4
3	111	107	4
4	110	105	5
5	109	104	5
6	109	100	9
7	108	102	6
8	106	108	-2
9	105	101	4
10	103	101	2

The way we think about comparing scores is a little different here. Now, we are interested in differences between scores for each person. You'll notice in the table there is an extra column that subtracts the score for Condition B from Condition A; this is known as a **difference score**. If there was no effect at all, we'd expect the difference score to be zero, and as the difference

score changes in magnitude, we have more confidence that our conditions are different. Here, we can see that participant 6 stayed the same, and participant 8 did better in Condition B, but for the most part, we see that scores for Condition A tend to be higher than Condition B. Of course, we need to statistically test these findings to be sure! Here's the equation for a within-subject t-test:

$$t = \frac{\bar{D}-0}{\frac{S_D}{\sqrt{n}}}$$

Where $\bar{D}$ is the average difference score, and S_D is the standard deviation of those difference scores. We find that the average difference score is 4.0, and the standard deviation of differences is 2.83, so:

$$t = \frac{4.00-0}{\frac{2.83}{\sqrt{10}}} = 4.47$$

The critical t-score for this test is 2.26, so the difference is clearly significant in this case; students see a significant drop in scores if they use a technique that shows them mistakes than if they don't see mistakes.

There are a couple additional points worth noting here. First, imagine that we had one unusually poor speller in the group. Let's say that participant #10 has a score of 3 in Condition A, and a score of 1 in Condition B – so, they are about 100 points lower than most students. However, because we are interested in the **change** in scores, the value in the difference column wouldn't change, nor would the standard deviation of difference scores. We essentially throw away the variance between participants in this equation. We don't really care if participants are different from one another, we just care whether their scores change in a systematic way. This also means that this test is more powerful than a between-subjects t-test. If we take these same scores and treat them as if they are in a between-subject test: `::{\#tbl-betweenspell}`

Table 7.3: The same data but as a between-subject test `::`

Group	Score
A	113
A	112
A	111
A	110
A	109
A	109
A	108
A	106
A	105
A	103
B	110
B	108
B	107

Group	Score
B	105
B	104
B	100
B	102
B	108
B	101
B	101

We would find a t-score of about 3.00. This value is significant, but not nearly as large in magnitude as the 3.97 we found when we treated the data as within-subject. So, you don't need as many participants or as much data with a within-subject test as you do with a between-subject test to detect a difference. There are many reasons why it might be nonsensical or impossible to use a within-subject design for your research, but if it is possible, it is worth considering, especially if you are limited on resources or participants.

7.4.1 What to do if your assumptions are violated

The good news here is for within-subject t-tests, we don't need to worry about heteroscedasticity (I'll explain more on this in Section 9.1). As for normality, we don't worry about normality in the measures themselves, we worry about normality in the difference between those values. So, you would need to calculate the difference scores and look at whether that was normal. This test is fairly robust, so minor issues with normality are not a concern, but if you have an outlier, you might want to explore alternatives, such as a Yuen's test (which is not available on all software) or a non-parametric test such as a Wilcoxon Signed-Rank test (Park 2020).

7.4.2 Calculating an Effect Size for Within-Subject T-Tests

There are several ways of writing up a within-subject t-test, so if you are curious, you can check out Kline (2004) for a review. The most straightforward test is known as **Cohen's d**, which is where you take the difference over the standard deviation. So, in this case:

$$d = \frac{\bar{D}}{s_D} = \frac{4.00}{2.83} = 1.41$$

As we learned in Section 5.6, this is quite a large effect. Cohen's d is one of the most popular measures of effect size, but it is not considered the most accurate, so you can read Kline (2004) to learn about the Hedge's correction, which often shows up in some statistical software as an alternative to Cohen's d.

7.4.3 Writing the Results of a Within-Subject T-Test

When sharing the results of a within-subject t-test, you want to convey several pieces of information to your audience:

- What type of t-test you are doing;
- What the average scores are for each testing period/condition and their accompanying standard deviations;
- The t-test score, its degrees of freedom, and its significance;
- In some cases, the effect size

So, for this example, here's an example of how I might write up the results:

In condition A, students scored an average of 108.60 ($SD=3.17$) on the spelling test, while they scored an average of 104.60 ($SD=3.53$) in condition B. A two-tailed within-subjects t-test indicated a significant difference, $t(9)=4.47$, $p=.001$. This represents a fairly large effect size ($d=1.41$).

Just as for the one-sample and the independent t-test, you have some flexibility over how you write your results, as long as you get the statistical notation right. Bar graphs and line graphs tend to be the most common way to convey the results of this test graphically.

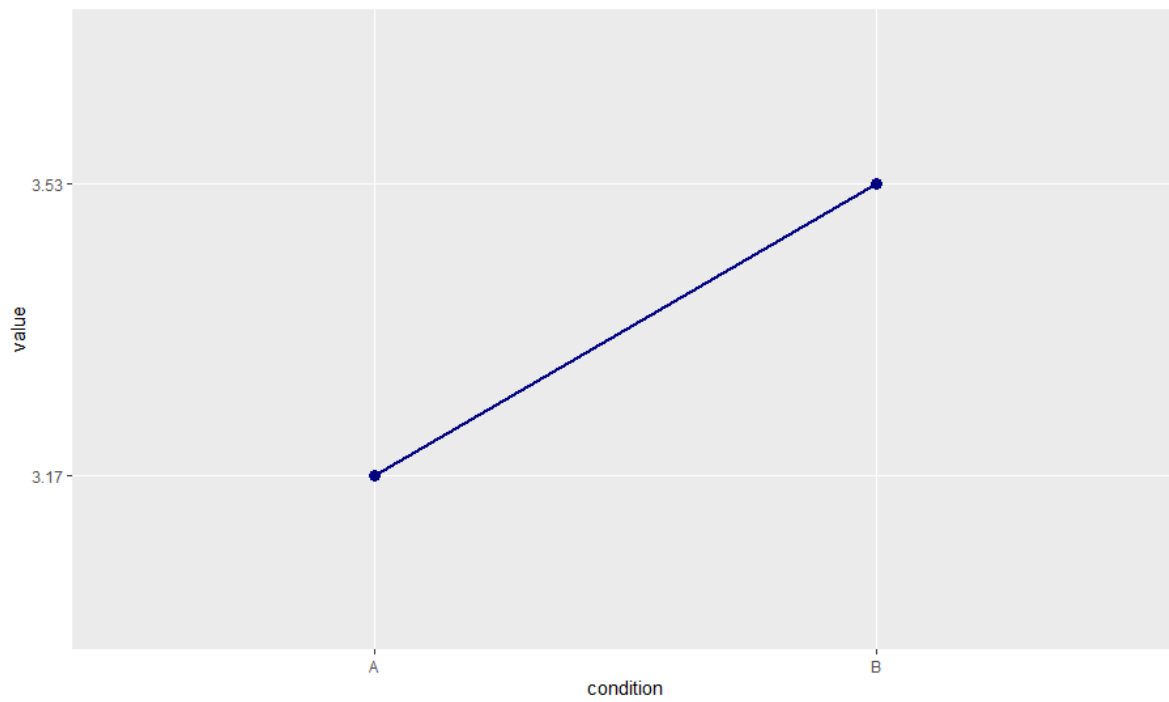


Figure 7.2: A line chart of the data

8 Between-Subject ANOVA

8.1 Learning Objectives

- Know what an ANOVA is, and what kinds of research questions it can address.
- Know what aspects of the data influence the size and significance of an ANOVA.
- The difference between a contrast and a post-hoc test, and what they tell you about your data.
- How to write the results of an ANOVA and its related contrast/post-hoc test.

Much of the time, we're going to end up with a more complex design in our research where we need to test the differences between 3 or more groups. For example, imagine we were curious about what makes someone feel "satisfied" after a meal, and we establish four groups: A Control group who receives nothing after their meal; a Dessert group who receive a slice of pie after their meal; an Aperitif group, who receive a cocktail after dinner, and a Buttermint group who get a small butter mint after their meal. The simulated data for this study can be found [here](#).

Now, we certainly could simply do 6 separate t-tests between pairs of groups (A/B; A/C; A/D; B/C; B/D; C/D). But one problem we run into here is what we call **familywise error**. When we do multiple tests, we compound the likelihood of finding a Type I error. Specifically, if we do three t-tests, our likelihood of a Type I error is:

$$\text{Familywise Error} = 1 - (1 - \alpha_p)^c$$

Where α_p is the alpha level set for each comparison, and c is the number of comparisons you're making. So, if we set each t-test at the normal $\alpha = .05$ for four comparisons, suddenly our error rate has ballooned to a 14% chance of a Type I error. Not acceptable!

So, if we want to hold our error rate steady at .05 total, we have two options.

The first is to do what we call a **Bonferroni correction**. Here, we simply reset our alpha level for each test to a value that will ensure the total error rate remains at 5%. The Bonferroni correction is pretty straightforward:

$$\frac{\alpha}{\text{number of comparisons}} = \alpha_p \quad (8.1)$$

For my example:

$$\frac{.05}{6} = .008\bar{3}$$

Sure enough, if you plug that value into the familywise error equation above, you'll see the overall error rate is about .05, just where we would want it.

The other option we have at our disposal is the analysis of variance, or the ANOVA. An ANOVA is useful because it's what we call an **omnibus test**; it tests across all groups for detectable differences without identifying exactly where those differences are. If our ANOVA is not significant, then we are done with our test and we can conclude that there are no differences among groups. If we get a significant result, we typically need to do more investigation, through either a contrast or a post-hoc test, to determine where the differences lie.

Bonferroni corrections come in very handy because sometimes you might need to do multiple tests. But for this example, most of the time, researchers would use an ANOVA to examine differences among groups.

8.2 Between-Subject ANOVA

The between-subject ANOVA is analogous to the independent t-test; that is, we have 3 or more groups that experience different conditions, and we see if their mean scores differ from the other groups.

At a theoretical level, it shares a lot with the math for an independent t-test. Once again, we are comparing the amount of differences we see between groups to the amount of variance we see within groups. Just like before, this is a ratio of differences we can explain with our research to differences we can't explain with our research.

So, let's use my example research question and a simulated data set to explore how the equation works to answer our questions.

Table 8.1: Scores for 10 participants in each of the 4 conditions, plus their group means and standard deviations.

Scores from the Control Group	Scores from the Dessert Group	Scores from the Aperitif Group	Scores from the Buttermint Group
1	2	1	4
2	3	1	3
3	4	1	4
3	3	2	5
4	4	1	5
3	3	2	5
4	2	3	4

3	3	2	5
2	4	1	5
2	4	1	5
Means for each group:			
2.7	3.2	1.5	4.5
SD for each group:			
0.948683	0.788811	0.707107	0.707107

Let's talk through what an ANOVA is conceptually before tackling the equations. Below, you can see a boxplot of the data we have. You can see black dashed lines on each box, which indicates each group mean. You can also see a thicker red line that goes all the way across the graph; that is the **grand mean**, which is the average of all participants lumped together.

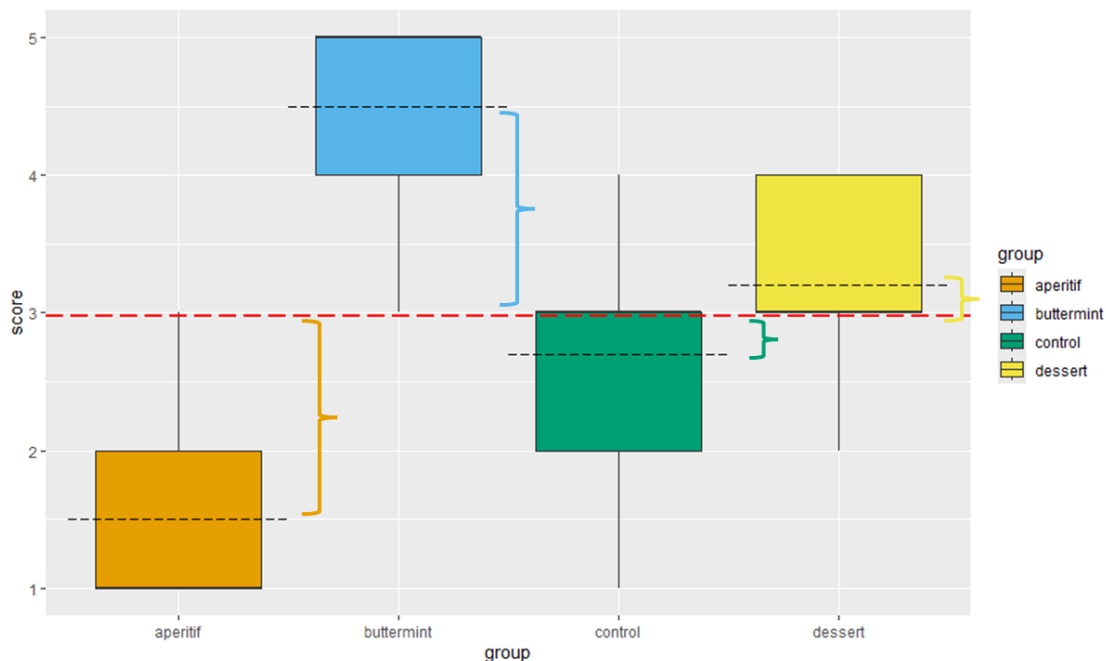


Figure 8.1: A box plot of group means and the grand mean.

If we were to take each data point and compare it to the grand mean, that represents all the variance in the data set. Once we know how much variance there is overall, we can start to **partial** the variance – we start splitting up that variance into differences we can explain and differences we can't explain. In my study, I am hoping that the different after dinner treat affects satisfaction with a meal, so I'm hoping that the means of each group are different. Figure 8.1 also has some colored brackets that highlight the gap between each group mean and

the grand mean. These differences – the differences between groups – is what I am actually interested in this data. Arguably, these differences are explained by the condition.

However, even within groups, we see there's some variation. If we take a look at how each person differs from their group's mean (highlighted by the brackets in Figure 8.2) , that is variance we can't explain. People within each group are different, and we have no idea why! But that's the nature of social science.

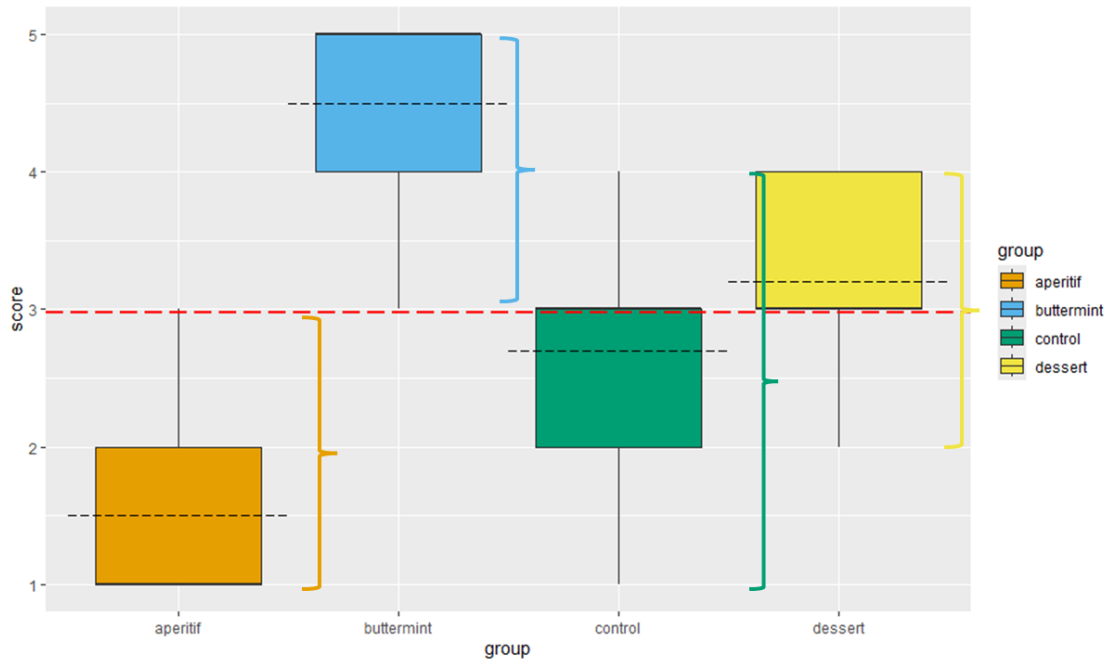


Figure 8.2: A box plot of group means with differences within groups highlighted by brackets

So, we now have split our total variance into the two parts we need:

- Differences between groups, which are differences we believe can be explained by our predictor.
- Differences within groups, which are differences we can't explain because people are random.

Ultimately, with an ANOVA, we are hoping that the proportion of variance we can explain is larger than the variance we can't explain. Now that we've established this conceptual basis, let's look at the math that is going on behind these concepts. Here's the equation for sum of squares total, which is all the variance in our data:

$$SS_{Total} = \sum_{i=1}^N (x_i - \bar{x}_{grand})^2$$

Let's take this piece by piece. Just as I mentioned earlier, we want to take each individual score, and subtract the grand mean from it. We need to square that value (otherwise, it would just be zero). We start with our first subject, and we keep doing this until we have added up values from every participant in our data set.

Next, we want to establish the variance we can explain based on group membership, which is the sum of squares model:

$$SS_{Model} = \sum_{g=1}^k n_g (x_g - \bar{x}_{grand})^2$$

For this equation, we take each group mean, subtract the grand mean from it, square it (because again, if we don't it would just be zero). We then multiply that value by the number of people in each group (mathematically, we are basically acting as though everyone in each group got the group mean, and we're summing that up). We do this starting with the first group, and we keep going until we've added these numbers for all groups.

The final equation establishes how much variance exists within each group, which we might call the sum of squares residual. The formula is a little more gnarly:

$$SS_{Resid.} = \sum_{g=1}^k \sum_{i=1}^n (x_{ig} - \bar{x}_g)^2$$

Essentially, here we are taking each individual's score and subtracting their group mean. We square it, and sum it up. We do this for each person within each group. The good news here is that as you've learned, "residual" just means "leftover stuff." So, another way we can get this is:

$$SS_{Resid.} = SS_{Total} - SS_{Model}$$

Again, SS_T is all the variance in our data, and SS_M is the variance we can explain based on group membership. The only variance left over is variance not explained by group membership!

I'll spare you the calculations here, but if you wanted to do this by hand, you'd get the following results:

$$SS_{Total} = 68.98$$

$$SS_{Model} = 22.70$$

$$SS_{Resid.} = 68.98 - 22.70 = 46.28$$

One issue with sum of squares is that their magnitude is based on the number of people or the number of groups. So, we need to account for that to make these proportions comparable. Normally, we could just do this by dividing by the number of groups or the number of people in each group. But these estimates are slightly smaller in the sample compared to the population we want to extrapolate our results to, so we have to subtract 1 from the number of groups or the number of people to account for that bias. Good news, degrees of freedom are right there with that adjustment you need! The degrees of freedom for SS_T is the number of total participants minus 1, the degrees of freedom for SS_M is the number of groups minus 1, and the SS_R is the difference between the two:

$$df_T = N - 1 = 40 - 1 = 39$$

$$df_M = k - 1 = 4 - 1 = 3$$

$$df_R = df_T - df_M = 36$$

We can now divide the sums of squares by their related degrees of freedom to get what we call **mean squares**. We don't really need to worry about this for the total, we only need it for the model and residual.

$$MS_M = \frac{46.28}{3} = 15.43$$

$$MS_R = \frac{22.70}{36} = .63$$

Finally, we obtain the ratio of what we know, MS_M , to what we don't know, MS_R .

$$F = \frac{15.43}{.63} = 24.46$$

The critical F value for an ANOVA with 3 and 36 degrees of freedom is 2.87 (see [?@tbl-f-table](#)), so we can be reasonably confident there are true differences among the groups. Remember that our conclusion is quite limited, though. The significant F-test indicates that at least one group is different from at least one other group. So, because this finding is significant, we need to do more investigation. If it's not significant, we're done and we can conclude that our data suggests no differences among the groups.

8.3 Contrasts

In some cases, we may only be interested in particular comparisons among our groups. For example, let's say that I have two differences I'm particularly interested in. First, I want to know if the control group is different from the 3 other groups. Second, I want to know if there's a difference between food-related after dinner treats (Dessert versus Buttermint). Third, I want to know if the Control group is different from the Aperitif group. If these are my questions, there are a lot of comparisons I don't need to do (for example, I'm not interested in whether an aperitif is different from a buttermint, so I don't need to do that test).

In those cases, we can preserve power by doing a contrast – that is, we select specific comparisons we want to investigate and test only those comparisons.

In most software programs, setting up contrasts can be a little fiddly, as it requires some dummy coding. Essentially, when setting up a contrast, you need to follow three basic rules:

1. Groups that you want to keep together need to share the same number. If you want to ignore a group, give it a 0.
2. The sum of all your codes should equal 0
3. Once you separate a group out, you can't put it back together with other groups.

This takes a little bit of practice, but once you get your head around these rules, it's not too difficult. Let's take the two contrasts I set up, and walk through them as an example.

The first contrast I want to compare Control to Aperitif, Buttermint, and Dessert. So, I need to give Control a number, and Aperitif/Buttermint/Dessert a different number. Because I am comparing 3 groups to 1 group, I need to think about how I might ensure I end up with a sum of zero. Here's one possible solution:

Table 8.2: Dummy codes for the first contrast

Group	Contrast 1 Code
Aperitif	1
Buttermint	1
Control	-3
Dessert	1
Sum:	0

We've met all three rules. The Control group has a different value than the other groups, and the values sum to zero. The third rule is moot in this case because it's my first contrast, so this is the first time I am separating out a group. Now let's move to my next contrast, where I want to examine whether Dessert or Buttermints are better for satisfying someone after a meal. Here's how I might set that up:

Table 8.3: Dummy codes for the second contrast

Group	Contrast 2 Code
Aperitif	0
Buttermint	1
Contrast	0
Dessert	-1
Sum:	0

Again, I've met all three rules. Buttermint and Dessert have different values, and they sum to zero. Because I separated out the Contrast group earlier, I can't add it back into this model. Looks good!

However, when I try to turn my attention to the final contrast, Control group versus Aperitif group, I run into a problem— I'm violating the third rule. Because I separated the Contrast group out before, it doesn't make sense to compare it to Aperitif... we sort of already did that. So, sometimes you might need to think through your comparisons or draw them out.

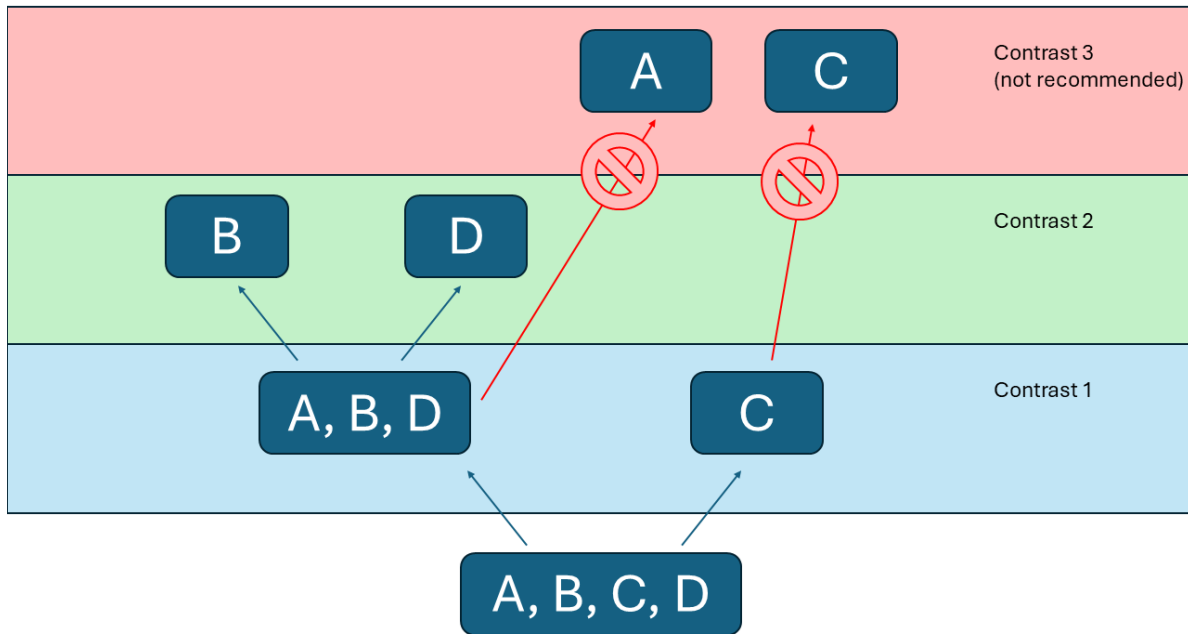


Figure 8.3: An illustration of the contrasts from the example.

Think of it like a tree– the branches should spread out; they shouldn’t reconnect after they’ve split apart. If you do find that you need to make some of those types of comparisons, it might be better to do a post-hoc test, which we’ll discuss in the next section.

When I conduct the first two contrasts, I see they are significant. In other words, I find that my Control group is significantly less satisfied than the other groups. I also find that the Dessert group is more satisfied than the Buttermint group. And I’m done!

8.4 Post-Hoc Tests

Sometimes we don’t need to be quite so precise in our comparisons. Maybe we just want to know which after dinner treat is the most satisfying, and we’d like to see the comparison among all groups rather than selected groups. In that case, the better tool is the post-hoc test. There are a ton of different post-hoc tests out there. Some are a bit outdated and were more useful when we didn’t have statistical software to do the calculation for us. Some might be more conservative and others might be more liberal. If you would like a rundown of most post-hoc tests, you can check out (Toothaker 1993) for an overview.

In general, I tend to use the following post-hoc tests:

- I use the Bonferroni post-hoc when I want to be conservative about Type I errors.

- If the sample sizes are very different in each group, I use the Hochberg's GT2
- If I see big differences in variance in my groups, or if I believe the variance in the two populations is different, I use the Games-Howell procedure.
- If my group sizes are roughly equal and don't demonstrate heteroscedasticity, I use the REGWQ (Ryan's procedure).

In the example, my groups are perfectly balanced with 10 participants each, and I don't see much heteroscedasticity, so in this case a REGWQ is what I'd most likely use.

I usually suggest that students not get too hung up on which post-hoc test to use; most of the time it won't make a huge difference. The one thing that you should not do is run all the post-hoc tests and choose the one that gives you the results you want. That is taking advantage of chance, which we try not to do in statistics.

In this case, all my recommended post-hoc tests indicate the same outcome— the Control group is significantly lower than all other groups, and the Dessert group is significantly higher than all groups. Buttermint and Aperitif are not significantly different from one another.

8.4.1 What to do if your assumption of normality and/or homoscedasticity are violated

Much like with t-tests, ANOVA tests are relatively robust against violations of assumptions. And much like t-tests, problems with this can often be addressed using the **Welch test**. The Welch test adjusts the degrees of freedom and how variances are pooled, so if you do use the Welch test, you'll want to note that you did so, and you can expect to report some decimals in your degrees of freedom.

8.4.2 Calculating an Effect Size for Between-Subject T-Tests

The most common estimates of effect size are η^2 and $\hat{\omega}^2$. In general, $\hat{\omega}^2$ is slightly more complex but also more accurate. Here's the equation:

$$\hat{\omega}^2 = \frac{SS_M - (df_M)MS_R}{SS_T + MS_R}$$

Basically, this is the proportion of variance explained by the model over the total variance, with an adjustment to address the fact that we're trying to translate results from a sample to a population. In the case of our data, $\hat{\omega}^2 = .64$, so a very substantial proportion of differences among participants can be attributed to the condition they were in.

8.4.3 Writing the Results of a Between-Subject ANOVA

When sharing the results of a between-subject ANOVA, you want to convey several pieces of information to your audience:

- What type of ANOVA you are doing;
- What the average scores are for each group and their accompanying standard deviations;
- The F-test value, its degrees of freedom, and its significance;
- In some cases, the effect size
- If you used a Welch correction, you should indicate that.
- If you find a significant omnibus test, you should conduct a contrast or post-hoc test and explain your results. Depending on the software you use and which post-hoc you choose, you may not have access to t-test results for the individual post-hoc tests. Provide as much information as you can, with the p-value being the bare minimum.

So, for this example, here's an example of how I might write up the results if I did contrasts:

To test the hypothesis, I conducted a one-way, between-subjects ANOVA with planned contrasts. The ANOVA was significant, $F(3, 36)=24.46, p<.001$. The first contrast, which compared the control group to the other groups, found that the control group was significantly lower than the other groups in terms of satisfaction, $t(36)=6.78, p<.001$. The second contrast, which examined whether there was a difference between participants who had a buttermint from those who had dessert, found that people who had dessert were significantly more satisfied, $t(36)=3.66, p<.001$

Alternatively, for a post-hoc, there can often be a lot of comparisons, so writing this efficiently helps streamline information for your reader. For my example, most differences are significant, so I most likely would call attention to the comparison that was not significant. You'll want to adjust your approach based on your findings, but in scientific writing we try to be thorough but brief:

To test the hypothesis, I conducted a one-way, between-subjects ANOVA. The ANOVA was significant, $F(3, 36)=24.46, p<.001$. A Tukey's post-hoc indicated all groups were significantly different at $p<.05$, with the exception of the aperitif and buttermint groups, which were not different, $p=.50$.

A table or graph is also helpful for conveying group means:

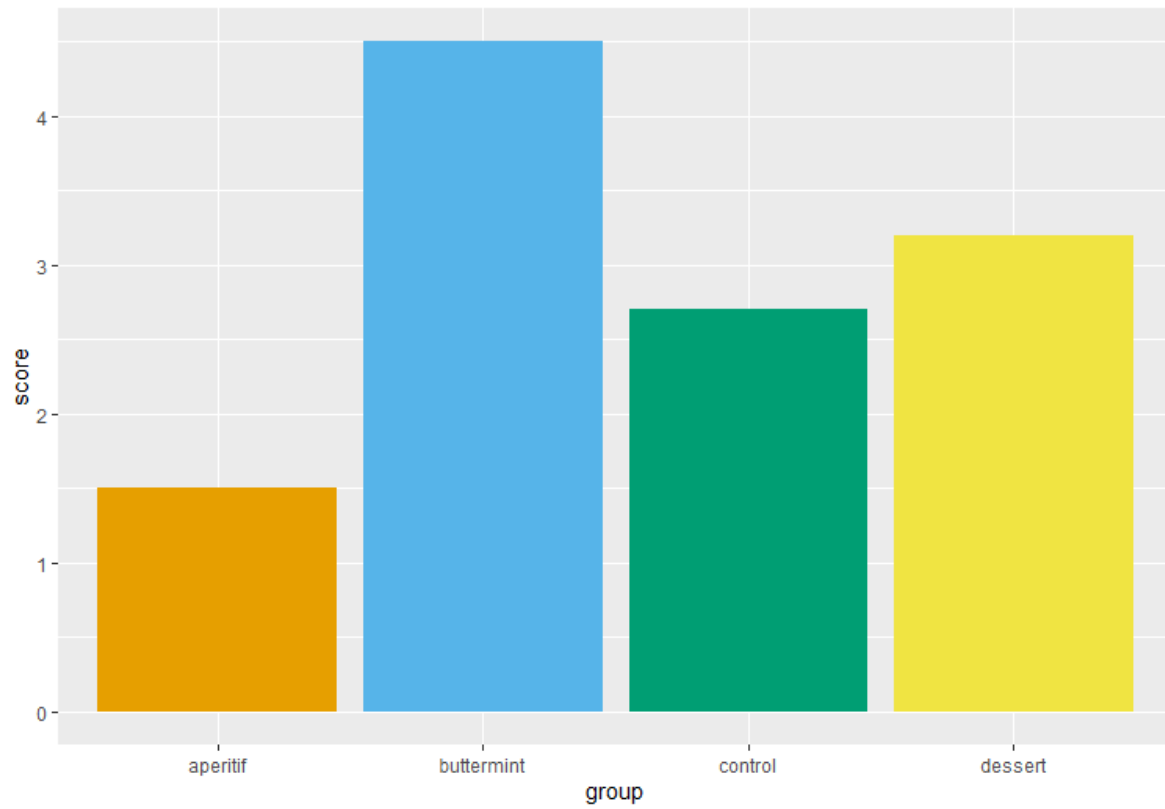


Figure 8.4: A bar chart of means for each condition

9 Within-Subject ANOVA

9.1 Learning Objectives

- Know what a Within-Subject (Repeated) ANOVA is, and what kinds of research questions it can address.
- Using an ANOVA approach to analyzing the data
- Using a linear approach to analyzing the data
- Using contrasts and post-hoc tests to describe differences in the data
- Writing a within-subject ANOVA

If you will recall, I dropped a little note in Chapter ?? that indicated that with all of these “difference” models we’re talking about, you can also conceptualize them as linear models. For the most part, it won’t make a big difference between using difference versus linear approaches, but that is not the case for within-subject ANOVAs. So, first, we’ll talk about when we might use a repeated ANOVA, and then we’ll talk about how these two approaches differ, and how using one approach versus the other will affect your findings.

9.2 The Basics of Within-Subjects ANOVAs

Just as I noted in Section 8.1 that between subject ANOVAs are analogous to independent t-tests, I’ll note here that within-subject ANOVAs (also known as “repeated ANOVAs”) are analogous to dependent t-tests. Instead of re-testing our sample once, we retest them 2 or more times, which means we will have a minimum of 3 scores for each participant who completes our study.

And just as I noted in the dependent t-test, we are not too interested in differences between participants; instead, we are interested in how each participant changes across the different testing sessions. This is going to be hugely helpful for interventions and treatments, because our participants or patients might be wildly different in, for example, the symptoms they experience, but if we can improve the experience for all our participants, it’s not necessarily a concern if some participants don’t improve quite as much as someone who had a more severe issue.

Let’s walk through an example together to understand what this might look like. Imagine that we are working in a sleep clinic, and one problem a lot of our clients have is that they don’t have great “sleep hygiene” – that is, we know there are a lot of ways to improve your sleep habits (having bedtime rituals, using your bedroom only for sleeping, avoiding screens for at least an hour before bed) and we know our clients are not great about following these habits. So, we design a training program with several modules:

- Module 1: Making your bedroom a sleep sanctuary
- Module 2: Developing effective bedtime rituals
- Module 3: Reducing screen use in the evening

Imagine that we get a baseline of how many hours per night, on average, our participants get before they begin our intervention, after each module, and then a 3-month follow up to see if the improvements are sticking. After our study, we might end up with a data set that looks like the Sleep Study data set I’ve created [here](#).

Table 9.1: Example data from the Sleep Study data set

ID	baseline	module1	module2	module3	followup
1	6.02	6.46	6.21	6.43	4.73
2	7.00	7.70	7.71	8.20	7.93
3	5.58	6.39	6.92	8.26	6.89
...	...	...	...	...	...

You can see in this data that people are very diverse – some are getting a lot of sleep in general, and others are not. However, our goal is to help everyone in our sample get more sleep overall, so this data can help us learn about what modules improve their sleep, and whether they are able to maintain these improvements over time. So, we need to learn more about how changes between each measurement look across all our participants.

9.3 The ANOVA Approach to Within-Subject Data

One option for analysis is to basically extend a repeated t-test into a repeated ANOVA, where we simply calculate the differences between each score and establish if those differences are large enough and consistent enough for us to feel confident that a change is occurring. Here, this would look very similar to what we discussed in Section 7.4 – we would calculate the difference between each measurement and use that to establish whether any time period produced a large enough change to be significant. However, if we take this approach, we have a new assumption that we need to consider: sphericity

9.3.1 The Assumption of Sphericity

As we estimate the differences between each measurement period, we can also estimate the variance of these difference scores. For example, from the data provided in Table 9.2, we can see some (but not all) of the comparisons between measures in the table below:

Table 9.2: Difference scores in Sleep Study data

ID	baseline - module 1	baseline - module 2	baseline - module 3	baseline - followup
1	-.44	-.19	-.41	1.29
2	-.7	-.71	-1.2	-.93
3	-.81	-1.34	-2.68	-1.31
...	...	...	...	...
Variance of Difference Scores:	.05	.32	-2.75	-2.16

We would have these comparisons across all measurements, and establish the variance of these difference scores. The assumption of sphericity indicates that these variances between difference scores should be roughly equal for all comparisons (that is, the average scores can be very different across each measure, but the variance in how much they change should be similar across all measures). We can see here, that doesn't seem to be the case! So, in order to establish whether this is the case, we can conduct a test known as the Mauchly's Test. The details of how this test is calculated is not particularly important; suffice it to say that if this test is significant, we need to make some adjustments to our results.

There are typically two corrections people use if there are issues with sphericity– the Greenhouse Geisser correction (which is a bit conservative) and the Huynh-Feldt correction. If you don't use a correction when sphericity is violated, just know that you won't have quite as much power, and it can lead to some funky results for your post-hoc tests. Pituch and Stephens (2016) recommends taking the average of the Greenhouse-Geisser and Huynh-Feldt values; Field (2009) astutely makes the argument that if there is a difference between the estimates (as in, the Greenhouse-Geisser yields a non-significant result while the Huynh-Feldt results in a significant result) that it's a good idea to look at effect sizes to understand whether the result is important from a practical standpoint. I tend to take that approach.

In the case of this data, we see that there is a significant Mauchly's test, and both the Greenhouse-Geisser and the Huynh-Feldt lead to significant findings, so the Greenhouse-Geisser correction is a good choice here. We obtain the following results:

Table 9.3: Results of a Greenhouse-Geisser corrected repeated ANOVA

Source	SS	df	MS	F	Sig.
Variable	23.97	1.65	14.56	19.57	<.001
Error	47.77	64.20	.74		

So, these results indicate we have a significant ANOVA. Just as with other types of ANOVAs, a significant result simply indicates there is a significant difference between at least two measurements, but we'd need to use a contrast or a post-hoc to learn more about the nature of these differences. Just as with a one-way ANOVA, contrasts are best if we have specific comparisons we'd like to make, while a post-hoc is best if we want to compare all groups to one another. Post-hocs are generally the same as what we saw with a one-way ANOVA, but with contrasts, there are some common contrasts that are worth discussing.

9.3.2 Contrasts

While you certainly can program custom contrasts as you did with a one-way ANOVA, many programs allow you to use some pre-programmed contrasts with your data.

The first, and probably most straightforward, is what we generally call a simple contrast. With a simple contrast, we designate a particular measurement as our comparison group (usually this is the first or the last measurement in your data set, but it doesn't have to be). In our example, perhaps we are curious about whether all time periods demonstrate a difference from our baseline measure. If that's our question, a simple contrast makes a lot of sense:

If you run this, you'll find these results:

Table 9.4: Results of a simple contrast

Comparison	F	Sig.
Baseline - Module 1	171.24	<.001
Baseline - Module 2	93.20	<.001
Baseline - Module 3	43.63	<.001
Baseline - Followup	10.20	.003

It can also help to look at a graph to understand the direction of the differences:

Ultimately, we see sleep improves during this program and stays higher than what it was at the baseline. Thus, we might conclude this training is helpful for improving sleep.

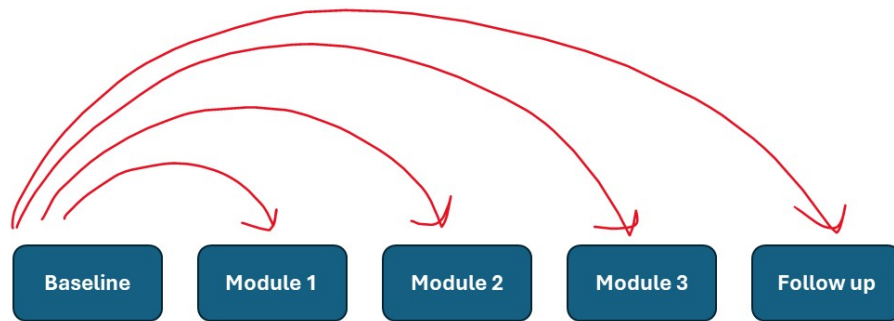


Figure 9.1: Simple contrast

Another common option is what's known as a repeated contrast. This is where you compare each adjacent measure. If our question is whether people keep improving over time, a repeated contrast is a good option:

If we chose to use this approach, we'd find the following results:

Table 9.5: Results of a repeated Contrast

Comparison	F	Sig.
Baseline - Module 1	171.24	<.001
Module 1 - Module 2	24.92	<.001
Module 2 - Module 3	1.55	.22
Module 3 - Followup	28.55	<.001

Here, we have a slightly different conclusion, because our question is different. In this case, we see that sleep improves after module 1 and module 2, but doesn't significantly improve after people participate in module 3. So, perhaps students have learned what they need to learn, and there's no need for module 3. We also see a significant drop between module 3 and the followup. It doesn't go back to baseline, but it does seem to line up with where students were after module 1, so clearly some of the improvements are not sticking around for the long-term.

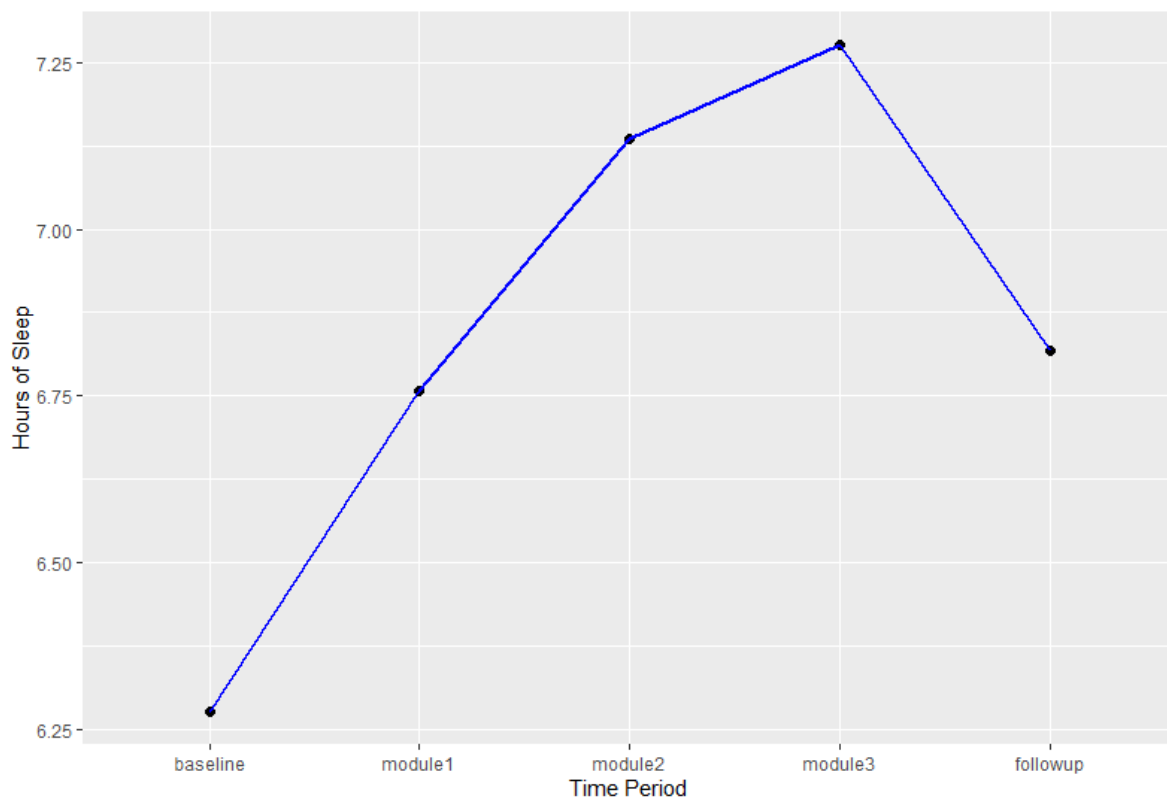


Figure 9.2: Average Sleep Times

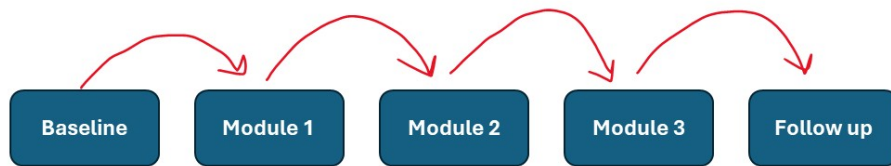


Figure 9.3: Repeated contrast

Another option that exists is the Helmert contrast. Here, you would compare each testing period to the mean of all periods that come before it:

This doesn't make as much sense for our example, but you can imagine if we were conducting research where each measurement built on the ones before it (like if we were gradually increasing the dosage of a drug, or if we expected people to build on a skill over time) then the Helmert is a good option.

9.4 The Linear Model Approach to Repeated Measures

Throughout this book, you are going to learn that sometimes we might have several tests that do the same thing. This is the first example of this (and unfortunately, the most complicated). If you are taking a class, your instructor will hopefully tell you whether you need to read this section. I am going to warn those of you who feel nervous about math that it's going to get a little bananas in this section. I'm sharing this with you to make two main points:

- Linear models are often more accurate and appropriate, but sometimes they are a bigger pain in the neck than they are worth, depending on the statistical program you are using (e.g., linear models are annoying in SPSS, but are pretty easy in RStudio)

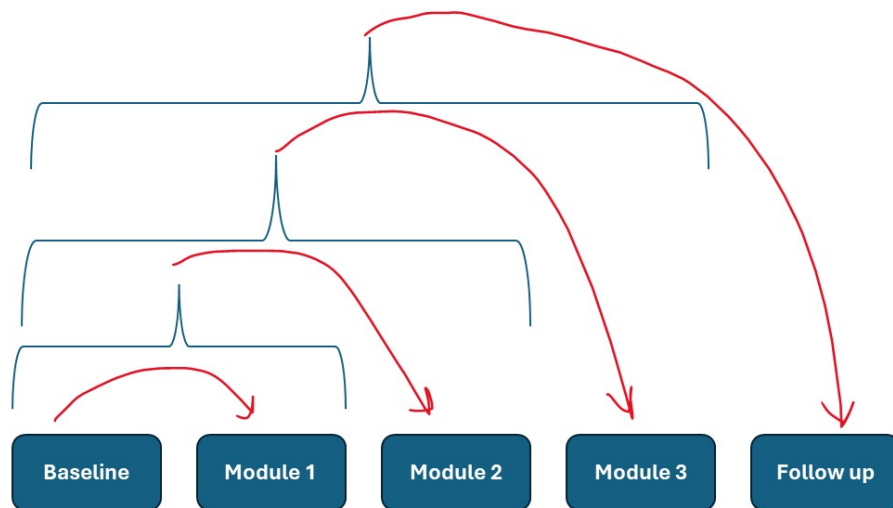


Figure 9.4: A Helmert contrast (the braces indicate means of multiple groups)

- Sometimes (but not always) your results will be very different between an ANOVA and a linear model depending on what is going on in your data.

So here's the deal:

Although historically we have treated repeated measures as ANOVAs, sometimes that is not exactly accurate. Part of what's going on here is that we are, in some ways, violating the assumption of independence. One way to think about this is that we need to consider that humans are different from each other, and all of us have some idiosyncratic ways that we respond to items.

For example, maybe Alan is taking medications that make sleep difficult, and maybe Benoit is an athlete who feels drowsy even after getting 8 hours of sleep. What this means is that we will see some correlations among the scores that come from an individual; Alan will tend to have less sleep overall compared to Benoit. When that's the case, we are dealing with something known as a multi-level model. Basically, we can think of each individual's response (Level 1) as nested within that individual (Level 2). So, there are differences between each measure people provide, but there are also perhaps some meaningful differences between participants that we ought to consider (versus using an ANOVA, where those differences are basically tossed aside and ignored).

The main point I'm making here is this: an ANOVA is primarily focused on differences within an individual, and ignores differences between individuals. Meanwhile, a linear model allows us

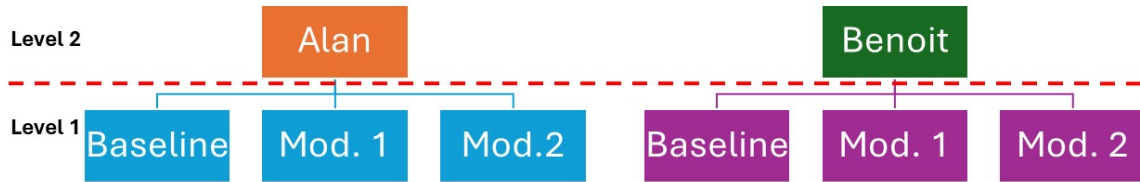


Figure 9.5: An illustration of nested data

to look at changes within an individual plus differences between individuals. A linear model would allow us to answer more questions, such as:

- Are there variables that lead to some people getting more sleep than others in general? (For example, maybe athletes get more sleep in general).
- Are there variables that can explain how much a person is likely to change? (For example, maybe students who have busy schedules don't improve much because they simply don't have enough time for sleep, regardless of their sleep hygiene).

The math that goes into this is much too complex for our purposes, but if you are curious, I recommend taking a look at a book covering multi-level modeling, such as Luke (2026). Anyway, what matters here for our purposes are two key points:

- Sometimes the results for an ANOVA versus a linear model can be quite different!
- If you use a linear approach, it does not have sphericity as an assumption. So all that stuff about Greenhouse-Geisser and Huynh-Feldt? You get to skip that and just report an uncorrected F-test result!

In our case, if we run a linear model on the sleep data set, we end up with results that look like this:

Table 9.6: Results table for linear model analysis

	df1	df2	F	Sig.
Sleep Hours	4	156	19.57	<.001

And here's a repeated contrast for comparison with our results in the last section:

Table 9.7: Repeated contrast for sleep data

Comparison	T-ratio	Sig.
Baseline - Module 1	3.88	<.001
Module 1 - Module 2	3.05	.01
Module 2 - Module 3	1.13	1.00
Module 3 - Followup	-3.70	.001

As we compare our results here with those from the previous section, we can see that the degrees of freedom are quite different, and we have some slightly different estimates for our contrast, but ultimately, we come to the same conclusion.

So, which approach should you use? Linear approaches are a more accurate way to model the data, but sometimes it can be more trouble than it's worth if the difference is minor. It also matters which statistical program you are using – for example, it's easier to run ANOVAs in SPSS, while it's easier to run linear models in RStudio. For most students in an introductory stats class, the real answer is that you should do whatever approach your instructor recommends, and you can learn more about these two approaches in an advanced class later on if you want. :)

9.4.1 What to do if your assumption of normality and/or homoscedasticity are violated

I have some good news for you! We aren't too concerned about normality or homoscedasticity in the variables themselves, we are more worried about this in the difference scores (sphericity). As I mentioned, if you are using an ANOVA, you can apply the Greenhouse-Geisser or Huynh-Feldt corrections if sphericity is a concern. If you are using a linear model, you don't need to concern yourself with sphericity at all.

9.4.2 Calculating an Effect Size for Within-Subject T-Tests

Even more good news – the effect size is calculated the same way as for a between-subject t-test – we use $\hat{\omega}^2$:

$$\hat{\omega}^2 = \frac{SS_M - (df_M)MS_R}{SS_T + MS_R}$$

9.4.3 Writing the Results of a Within-Subject ANOVA

When sharing the results of a between-subject ANOVA, you want to convey several pieces of information to your audience:

- You should indicate that you are conducting a within-subject ANOVA or a repeated ANOVA (they mean the same thing). It is also helpful to explain whether you are using an ANOVA or a linear model approach.
- What the means and standard deviations are for each testing period.
- The F-test value, its degrees of freedom, and its significance;
- In some cases, the effect size
- If you find a significant omnibus test, you should conduct a contrast or post-hoc test and explain your results. Depending on the software you use and which post-hoc you choose, you may not have access to t-test results for the individual post-hoc tests. Provide as much information as you can, with the p-value being the bare minimum.

Here are a couple of examples of how we might write our results

To examine the effect of our intervention on sleep, I conducted a repeated ANOVA with a Greenhouse-Geisser correction ($\epsilon = .41$), which was significant, $F(1.65, 64.20)=19.57, p<.001$. A repeated contrast indicated that there were significant improvements in sleep from the baseline to after the first module ($p<.001$) and between the first module and the second module ($p<.001$). The improvements tapered off between the second module and the third module ($p=.22$), and there was a significant decrease in sleep between module 3 and the follow up ($p<.001$).

A linear model ANOVA revealed a significant effect, $F(4, 156)=19.57, p<.001$. As shown in Figure 6.9, participants experienced an increase in sleep after each module, but a decrease in sleep at the follow up. However, a simple contrast indicates that all time periods demonstrate significantly more sleep compared to the baseline, suggesting that the treatment produces lasting improvements in sleep among participants.

10 Factorial Designs

10.1 Learning Objectives

- Know what a factorial is and what types of data it requires.
- Know the assumptions of a factorial ANOVA.
- Know how to interpret main effects and interactions in a factorial design.
- How to write up the results of a factorial ANOVA.

Now that we have the basics of an ANOVA down, we can build to a more complex design – the factorial ANOVA. In this technique, we are now adding another independent variable to see how two (or more) variables influence a particular outcome.

10.2 A Two Way ANOVA

As an example, let's say that we want to examine several different time management techniques for college students: Pomodoro technique, Eisenhower matrix, and Time blocking. However, perhaps we also believe that which technique works might depend on whether the student is majoring in a STEM field versus a Humanities field. We get a sample of students in both STEM and Humanities, and assign them to the three different techniques. You can find the time management file [here](#).

This is a factorial design. We can also describe it as a “two-way” ANOVA, because we have two independent variables. We might also see it described as a 2x3 ANOVA (we have one variable, or “factor,” with two levels, and a second variable with three levels).

So, you may be asking why we don't just do two separate ANOVAs for each variable? One reason is, again, **family-wise error**. But a more important issue is something known as an **interaction**. In research like this, we sometimes find that the influence of one independent variable is affected by the other independent variable. I'll share more about what interactions look like in a moment.

The good news is that, mathematically, a factorial ANOVA is very similar to the one-way ANOVA we've worked with so far. So, just as before, we need to understand how much total variance is in our model, and we do this by comparing each data point to the grand mean:

$$SS_{Total} = \sum_{i=1}^N (x_i - \bar{x}_{grand})^2$$

And just as before, we can also partial some of the variance into the sum of squares model, which describes how much variance is explained by group membership. But we have to remember that there are now two independent variables, so some of the variance in this model is explained by the technique the student uses, and some of it is explained by their major, so we now have two equations we need to use:

$$SS_{tech} = \sum_{g=1}^k n_{tech}(x_{tech} - \bar{x}_{grand})^2$$

So, for the sum of squares attributed to technique, we take the average for each group and compare it to the grand mean. Likewise:

$$SS_{maj} = \sum_{g=1}^k n_{maj}(x_{maj} - \bar{x}_{grand})^2$$

For the majors, we take the average of each of our two major groups and compare it to the grand mean. So far, this looks a lot like what we've done so far. But we have one additional piece of model variance we need to attend to:

$$SS_{cell} = \sum_{g=1}^k n_{cell}(x_{cell} - \bar{x}_{grand})^2$$

In this data, we have 6 different groups (STEM/Pomodoro; Humanities/Pomodoro; STEM/Eisenhower...) and each of these groups have their own mean. These means tell us a little something about differences that aren't completely explained by someone's group membership on both variables... there seems to be something extra here. That little extra something is the interaction. Mathematically, any variance from these cells that isn't attributed to our independent variables is the interaction effect:

$$SS_{tech \times maj} = SS_{cell} - SS_{tech} - SS_{maj}$$

If you look at Figure 10.1, you'll see that what technique works best depends upon whether a student is a STEM major or a Humanities major. That's what interactions are all about – sometimes the combinations between our independent variables yield interesting results like this. So, the factorial ANOVA allows us to look at these interactions in a way two one-way ANOVAs don't.

Finally, as you know, our data will contain some differences that we simply can't explain, so we also have a residual sum of squares, calculated similarly to how we do it in a one-way ANOVA:

$$SS_{Resid.} = SS_{Total} - SS_{cells}$$

If you recall from Chapter 7, we then go through the task of calculating degrees of freedom and mean squares. However, with a one-way ANOVA, we had only one F ratio we were interested in. Now, we have three: the **main effect** of technique, the **main effect** of major, and the **interaction** between technique and major. These three effects can occur in any combination – maybe they are all significant, maybe we have main effects and no interaction, or maybe we have no main effects but we have an interaction.

In this particular data set, we have the following results:

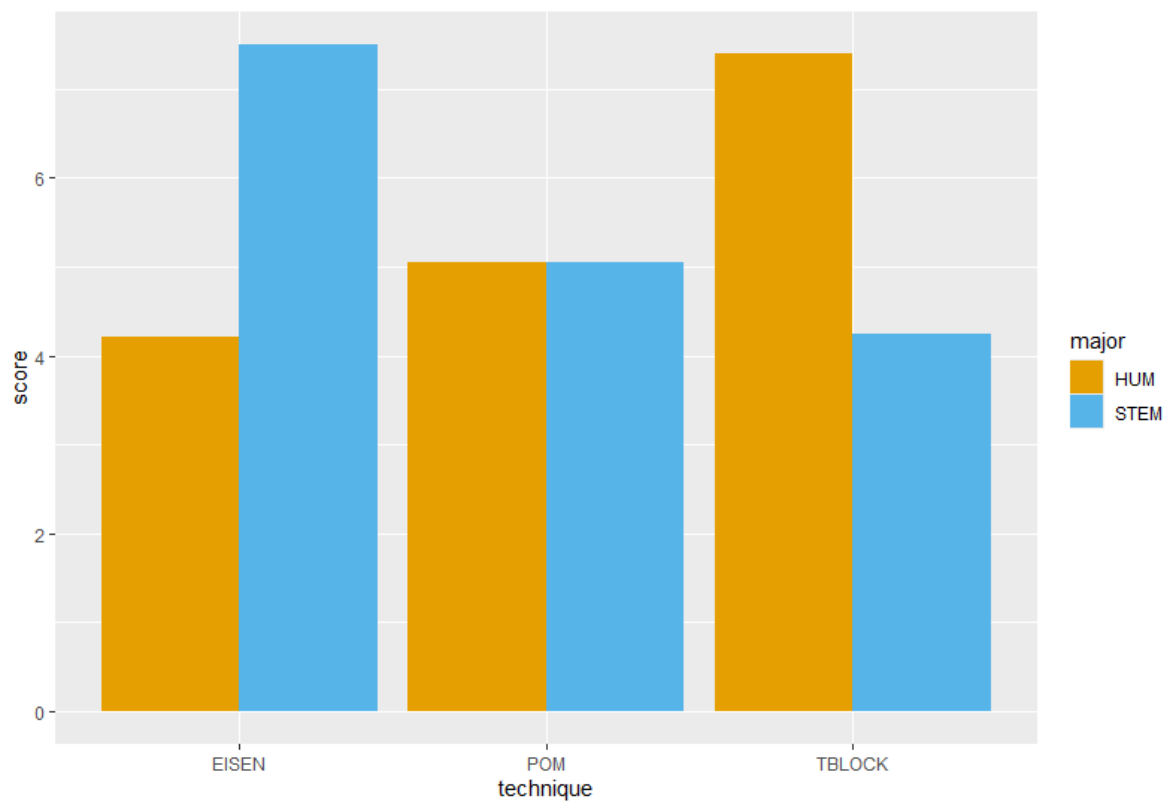


Figure 10.1: A figure of the two-way ANOVA

Table 10.1: F-test results. These results were generated using R; results from SPSS are slightly different.

Variable	df	SS	MS	F-value	Sig.
Major	1	.08	.075	.039	.84
Technique	2	18.98	9.49	4.98	.008
Major x Technique	2	209.95	104.97	55.12	<.001
Residuals	114	217.12	1.91		

This means we have a main effect of technique, no main effect of major, and an interaction. In fact, Figure 10.1 demonstrates nicely what is going on – on average, different majors perform roughly the same. We also see some differences in how techniques work. But perhaps most importantly, we see the type of technique that’s most effective depends on what major a student is earning – if a student is getting a STEM degree, the Eisenhower matrix is most helpful for time management, while if a student is getting a Humanities degree, Time blocking is the most helpful.

As you might imagine, a factorial ANOVA is similar to a one-way ANOVA in that it only tells us if one group is significantly different from another group. For this result, we’d want to conduct a contrast or a post-hoc test on the techniques to see which groups are significantly higher than which. We didn’t see a main effect of major, but even if we had, there are only two groups, so we know where the difference is and there’s no need for a post-hoc test.

As for interactions, there isn’t exactly a way to easily test where the big differences are in an interactions. Instead, if you are curious about whether there are significant differences, you can conduct a **simple effect analysis**. This is where you take a “slice” of the data on one variable, and examine what’s going on. For example, we could ask “For STEM majors, what technique is the best?” Alternatively, you could take cell means and compare them with a t-test. For example, we could compare the STEM/Pomodoro group to the Humanities/Eisenhower group. Keep in mind that you should use a Bonferroni correction for each comparison, so this technique is best for answering specific questions about the data – to compare every group, we’d need 15 t-tests, which is quite a lot. So, using t-tests for examining interactions is most appropriate when you have specific comparisons you’d like to probe.

10.3 A Brief Note on N-Way ANOVAs

Two-way ANOVAs are not the limit for you! You can add more independent variables... but they get much harder to understand, and you need a large sample size (i.e., a lot of power) to find significant results.

Just to share one example of what this might look like, imagine that I am looking at customer reactions to store clerks. I set up a study where I have two genders of shoppers (men and

women), two genders of store clerks (men and women), and two different types of stores the shoppers go to (a clothing store and a computer store).

Because we have three independent variables (customer gender, clerk gender, store type) this is a three-way ANOVA (and specifically, we could call it a $2 \times 2 \times 2$ ANOVA (three variables, with two levels each).

This now means our results are also going to be more complex. We will ultimately have:

- A main effect of customer gender
- A main effect of clerk gender
- A main effect of store type
- A two-way interaction between customer and clerk genders
- A two-way interaction between customer and store type
- A two-way interaction between clerk gender and store type
- A three-way interaction between clerk gender, customer gender, and store type

It gets complicated fast! So, you don't see a lot of research that goes beyond a three-way ANOVA, but it certainly is possible to create more complex models. One difficulty with a three-way ANOVA like this is that it's difficult to clearly convey the results. Really, we'd need a 3D graph to be able to put all three predictors into one graph, but those tend to be hard to understand. So instead, we'd probably need to break our charts into two 2D graphs.

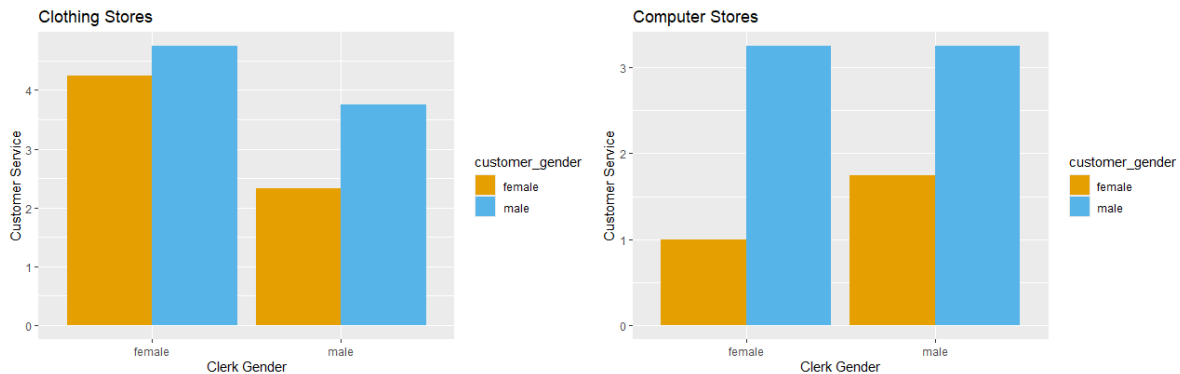


Figure 10.2: A graph of a 2x2x2 ANOVA result

Likewise, most people find two-way interactions a little complex to talk about; three-way interactions even are more difficult, so sometimes these analyses get complex enough that it can be hard to translate the practical implications to other people. Imagine trying to explain in words all the things going on in Figure 10.2. It's not impossible, but it's worth nothing here that adding even one additional independent variable makes the results much more complex to interpret in practical terms.

10.3.1 What to do if your assumption of normality and/or homoscedasticity are violated

Most programs don't allow for easy correction of this problem, but if you have concerns about these issues, we can again apply a **Welch test**, just like we did for a one-way between-subject ANOVA. For the most part, though, ANOVAs are generally robust so most of the time, violations of these assumptions are not a major concern.

10.3.2 Calculating an Effect Size for Between-Subject T-Tests

I bet you know where I'm going this this... if you need an effect size, the best option is $\hat{\omega}^2$. You can calculate this for each main effect and any interactions. Here's the equation once again:

$$\hat{\omega}^2 = \frac{SS_M - (df_M)MS_R}{SS_T + MS_R}$$

10.4 Writing the Results of a Factorial ANOVA

Because you are conveying so much information when you do a factorial design, I strongly recommend you use a figure (or at the very least, a table) to help your audience understand the data. Just reporting a main effect or interaction as significant doesn't help your audience understand how these effects look; a figure is the most efficient way of communicating this, as it provides a visual summary of all the group means very quickly. If you opt not to use a figure, you are going to need to report a lot of means and standard deviations!

When sharing the results of a between-subject ANOVA, you want to convey several pieces of information to your audience:

- How many independent variables you are using in the model, and how many levels each factor has. This can be efficiently conveyed by using the notation I've used in this chapter (e.g., a 2×3 ANOVA) but you can also describe the nature of your variables in more detail.
- What the average scores are for each group and their accompanying standard deviations;
- For all main effects and interactions, you must provide the F-test value, its degrees of freedom, and its significance;
- Any simple effects, contrasts, or post-hoc tests you use to further explain significant findings in your omnibus test.

Here are some examples of how we might explain the results of the analyses we've discussed in the chapter:

According to a 2 x 3 ANOVA, there is a main effect of technique, $F(2,114)=4.98$, $p=.001$, such that the Pomodoro technique is significantly less helpful than Time blocking or Eisenhower matrix. There is no significant main effect of major, $F(1,114)=.04$, $p=.84$. There is a significant interaction, $F(2,114)=55.12$, $p<.001$. As demonstrated by Figure 10.1, Time block is more helpful for Humanities majors, whereas Eisenhower matrix is more helpful for STEM majors.

According to a 2 x 2 x 2 ANOVA, there was a significant main effect for store type, $F(1, 22)=23.40$, $p<.001$, with clothing stores having higher customer service. There was also a significant main effect for customer gender, $F(1, 22)=18.91$, $p<.001$, with male customers receiving better customer service. There was no main effect for clerk gender, $F(2, 22)=3.54$, $p=.07$. I also found a significant interaction between store type and clerk gender, $F(1, 22)=8.43$, $p=.008$. Female clerks tended to demonstrate better customer service in clothing stores than they did in computer stores, while male clerks demonstrated better customer service in computer scores than they did in clothing stores. There was no interaction between clerk and customer gender, $F(1, 22)=.02$, $p=.89$. There was also no significant three-way interaction, $F(1, 22)=1.86$, $p=.18$. Results are presented in Figure 10.2.

11 Non-Parametric Tests

11.1 Learning Objectives

- Know what a chi-square goodness of fit test is, and what types of research questions it can address.
- Know what a chi-square test of independence is, and what types of research questions it can address.
- Know how to calculate effect sizes for chi-square tests.
- Know how to write up the results of a chi-square test.

All the statistical tests that we've covered so far are parametric tests – that is, they are intended to predict a continuous outcome of some sort. I do find that students sometimes need to work with categorical data, so I think it's worth the time to dip a toe into some non-parametric tests, namely the chi-square goodness of fit, and the chi-square test of independence.

11.2 The Chi Square Goodness of Fit Index

The chi-square goodness of fit index (GFI) pops up in a variety of places (for example, if you ever go on to do structural equation modeling, this is sometimes an indication of how well a model fits the data). For the purposes of this text, though, the chi-square GFI is helpful in examining whether participants are falling into categories in any predictable way.

For example, imagine that we want to learn about people's preference for transportation among the options of car, bus, bike or walk. If there's no strong preference for one of these options, we'd expect to see people equally distributed among the options. So, we will compare the proportion of people who report each preference with our null hypothesis of equal numbers of people in each group. Imagine we collect data from 100 people, and here's what we find:

Table 11.1: Table of preferred transportation

	Bike	Car	Bus	Walk	Total
Expected	25	25	25	25	100
Obtained	28	32	10	30	100

I should mention briefly that the null hypothesis does not have to assume that all groups are equal; for example, left-handers make up about 10% of the population, so if we wanted to see if a certain group deviates from this value (for example, American Presidents) we can set up our expected value to reflect the population proportions.

Table 11.2: Expected and obtained frequency of left-handed American Presidents (for the curious, the result is not significant!)

	Left-Handed	Right-Handed	Total
Expected	5	40	45
Obtained	7	38	45

Returning to our transportation example, we see that there are some deviations from what we expected to see here – cars are more popular, and buses are less popular than our baseline expectation. The chi-square test will tell us whether these deviations are large enough that we can feel like they are not due to chance. The calculation is relatively straightforward:

$$\chi^2 = \sum_{i=1}^n \frac{(\text{obtained}_{ij} - \text{expected}_{ij})^2}{\text{expected}_{ij}}$$

Essentially, we take the difference between the expected and obtained values in each transportation type, square that value, and divide that value by the expected number. We do that for each group and sum up these values. In this data, we find the chi-square value to be 12.32. The degrees of freedom for this are the number of groups minus 1 (so, 3 in this case) and the critical value for the chi-square test for 3 degrees of freedom is 7.82 (you can see Table 15.4 for reference), so our finding is significant. The way we interpret this finding is a bit different from what you're used to if you've been thinking about t-tests and ANOVAs. All this test tells us is that people are not distributing themselves equally among different transportation modes; we have to refer back to the table to interpret it by identifying where those deviations are. In this example, we can see that car is most preferred, and bus is least preferred. We also see that walking and biking are a bit above what we expected as well. However, there's no post-hoc test or contrast needed for a chi-square; we can just report the table and point out where the biggest deviations are.

11.3 Chi-Square Test of Independence

A second type of useful non-parametric test is the chi-square test of independence (TOI). This differs from a chi-square GFI in that it requires two variables, and we're looking to see whether those two variables are related to each other. In fact, you've already seen a version of this in Section 4.5, when I talked about hypothesis testing using a hypothetical example of children watching violent media.

Here, let's walk through another slightly more complex example. Mays et al. (2014) did a study looking at how parental smoking behavior relates to youth smoking. I'm simplifying the data a bit, but generally they found three categories of parental smoking behavior (current smoker, former smoker, non-smoker) and four categories of youth smoking behaviors (never-smoker, early regular smoker, early experimental smoking, late experimental smoking). Here's how the data looks:

Table 11.3: Table of youth and parental smoking behaviors; expected values, rounded to the nearest whole number, are provided in parentheses.

	Never-smoker	Early Regular	Early Experimental	Late Experimental	TOTAL
Non-smoker	16 (32)	16 (6)	38 (25)	37 (44)	107
Former smoker	36 (38)	3 (8)	28 (29)	59 (51)	126
Current smoker	69 (51)	5 (10)	26 (38)	67 (68)	167
TOTAL	121	24	92	163	400

Conveniently, the chi-square TOI uses the same equation as the chi-square GFI:

$$\chi^2 = \sum_{i=1}^n \frac{(\text{obtained}_{ij} - \text{expected}_{ij})^2}{\text{expected}_{ij}}$$

But the expected values are not as straightforward as they were before, because we don't have nice, equal groups. As we learned in section Section 4.4, though, the multiplicative rule comes through for us here. We use the total rows/columns to help us calculate the probability someone is in a particular category, then use the multiplicative rule to calculate what each cell total should be **if the two variables are independent of one another**. If the obtained number deviates a lot from that expected value, we can conclude those variables are likely related.

Let's take, for example, youth who are never-smokers and who had a non-smoker parent. The probability of being a never smoker is $\frac{121}{400} = .30$ and the probability of having a non-smoker as a parent is $\frac{107}{400} = .27$. So, according to the multiplicative rule, there is a $.30 \times .27 = .08$ chance of someone being in the never-smoker, non-smoker cell. In a group of 400 people, that would be $.08 \times 400 = 32$ people. When we compare this expected value of 32 to the obtained value of 16, we can see that there might indeed be a relationship. But, we'll want to make sure so we conduct our chi-square test. The degrees of freedom for this are the number of rows, minus one, times the number of columns, minus one:

$$(3 - 1) \times (4 - 1) = 6$$

Ultimately, we end up with a value of 48.20 for our chi-square value, which is significant (see Table 15.4). Again, there is no need for post-hoc tests, we simply need to examine the table and interpret where participants are notably above or below the expected value. If we examine residual values, we see that the non-smoker/never-smoker category is much lower than expected, as is the current smoker/early experimenter group. Meanwhile, the current smoker/never-smoker group and the non-smoker/early experimental groups have more than we would have expected. So there is some indication that having a current smoker as a parent is associated with their child avoiding smoking, and that having a non-smoking parent might be associated with some early experimentation with smoking. This was a correlational study, so we can't reasonably argue a causal link here, just an association between these categories.

11.4 Chi Square Assumptions

Non-parametric test have some different assumptions from the ones we're used to seeing with parametric tests. Specifically, chi-square assumptions are:

1. Independence: Chi-squares assume that each participant inhabits only one cell; we cannot use this test with a dependent/repeated measures design.
2. Expected values should be at least 5 participants.

For Mays et al. (2014) study, both of these conditions are satisfied. Youth can only be in one category, and when they discussed parental behaviors, they had only one parent fill out the form. Although we can see that some of the obtained values in the table are under 5, none of the *expected* values were less than 5. In the event that you do have data that violates this assumption, the Fisher's Exact Test can be used as an alternative test that is more forgiving of small expected values.

If you violate the assumption of independence, you need to find a more appropriate test to analyze your data.

11.5 Calculating the Effect Size of a Chi-Square Test

For a chi-square GFI, you can use a Cohen's w , which is a straightforward calculation:

$$w = \sqrt{\frac{\chi^2}{N}}$$

So, for our example, we can use the formula to see we have a fairly moderate-sized effect:

$$w = \sqrt{\frac{12.32}{100}} = .35$$

For a chi-square TOI, the effect size is known as ϕ , which we will cover in section Section 12.3.1.

11.6 Writing the Results of a Chi-Square Test

When sharing the results of a chi-square test, you want to convey the following information to your audience:

- What type of chi-square test you are doing;
- What the obtained values are (often best conveyed in a table, as I have done in this chapter)
- The chi-square test value, its degrees of freedom, and its significance;
- In some cases, the effect size

So, for the study, I might provide a table of results and say something like:

A chi-square test of independence indicates a significant relationship between youth and parental smoking behaviors, $\chi^2(6) = 48.20$, $p < .001$, $\phi = .$ As can be noted from Table 11.3, youth who have non-smoking parents are much more likely to do some early experimentation with smoking, whereas youth who have a parent who currently smokes are more likely to completely avoid smoking.

You will have the best sense of what particular cells hold the most interest for your audience, so use your best judgement for how to explain the relationship. It's not always particularly straightforward!

12 Correlations

12.1 Learning Objectives

- Know what a Pearson correlation is, and what types of data it requires.
- What phi and Cramer's V correlations are, and what types of data they require.
- What a point-biserial is, and what type of data it requires.
- What Spearman's rho and Kendall's tau are, and what types of data they require.
- How to write up the results of a correlation.

12.2 Pearson Correlation Coefficient

Perhaps one of the most prevalent statistics in most social sciences is the Pearson correlation. This is prevalent enough that most introductory social science courses at least provide the very basics of what a correlation is. The general introduction into correlations tells us that correlations have three important pieces of information they tell us:

1. Correlations can range from -1.00 to +1.00
2. The strength of the relationship between variables is indicated by the magnitude of the number. The closer it gets to +/- 1.00, the stronger the relationship. A correlation of zero indicates no relationship.
3. The direction of the relationship is indicated by whether the value is positive or negative. Positive relationships indicate the variables move in the same direction together (when one goes up, the other goes up). Negative relationships indicate an inverse relationship between variables (when one goes up, the other goes down).

This is a great place to start, but it's worth discussing a few more nuances of the statistic. First, one point worth mentioning is that the Pearson correlation assumes a linear relationship. So it's worth it to check a scatterplot to make sure the relationship looks roughly linear, because if it's something else (like a curve or a doughnut shape) the results might indicate there is no relationship between variables when there actually is... it's just not a linear relationship.

Secondly, if we were to plot a line that best fit our correlation, the Pearson correlation coefficient doesn't provide us information about what that line would look like – that is information that we get from a regression, which is covered in section [Section 13.1](#).

Let's walk through an example. Let's say I'm doing a study on meditation (data can be found [here](#)), and I want to see whether the number of minutes someone meditates throughout the week is associated with their happiness, their stress levels, and how much they like meditating. I collect data from 20 participants, and it looks like this:

Table 12.1: Meditation data

ID	Meditation	Happiness	Stress	Liking
1	111	53	42	41
2	97	39	50	46
3	123	41	55	75
4	130	58	36	60
5	99	70	50	61
6	83	65	59	32
7	99	67	46	48
8	89	54	52	50
9	80	55	39	44
10	115	54	51	52
11	76	57	46	42
12	42	33	71	73
13	73	63	49	44
14	118	59	63	45
15	121	69	43	44
16	82	53	38	36
17	91	50	54	48
18	97	68	27	49
19	53	41	64	62
20	167	71	43	57

First, let's dig into the correlational analysis between minutes spent meditating and levels of happiness. Here's a scatterplot of the relationship, which shows that it appears to be linear, so a Pearson correlation is a good option for us to use.

The first step to understanding the correlation is understanding something we call **covariance**. Covariance is a way of identifying if two variables “move together” – we see some of this in our scatterplot – people who tend to meditate more also tend to have a higher level of happiness. It's not a perfect relationship; even though participants #5 and #7 are meditating the same amount, they have different happiness scores. But as scores move higher on minutes of meditation, happiness also trends up. Mathematically, here's what that looks like:

$$cov_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

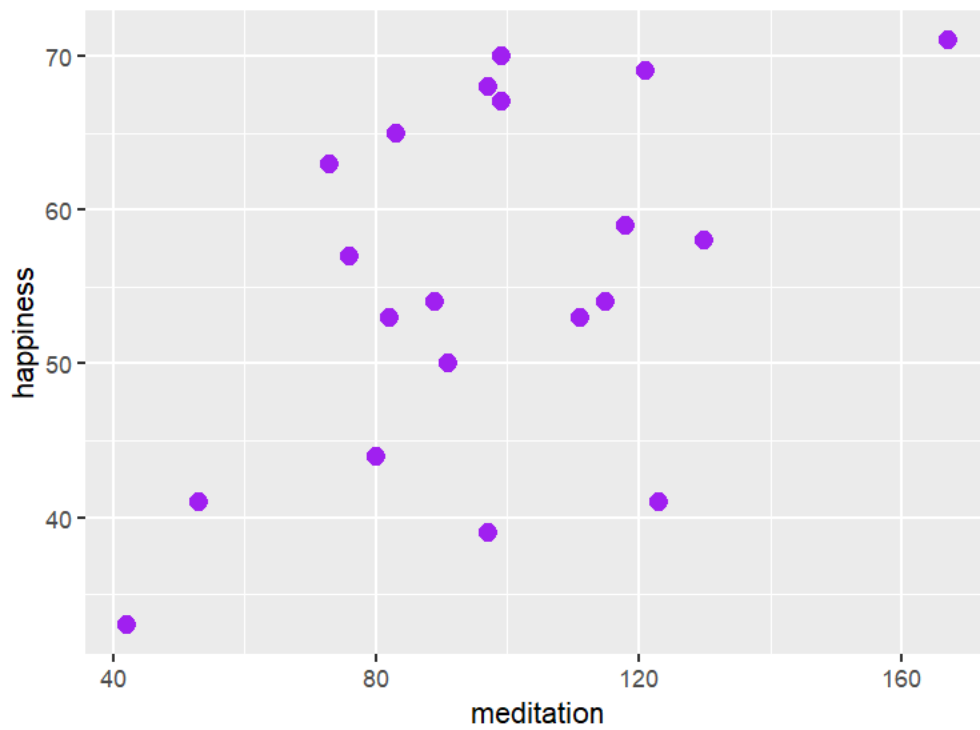


Figure 12.1: A scatterplot of meditation and happiness

What covariance does is it takes each individual's score on meditation, and subtracts the mean of meditation time (which is 97.30 in these data). It then takes each individual's score on happiness and subtracts the mean of happiness (55.45 in these data). It then multiplies those two values together. Once that's done, you sum those values up, and divide that total by $N-1$ (to adjust for the fact that our sample will slightly underestimate what's going on in the population).

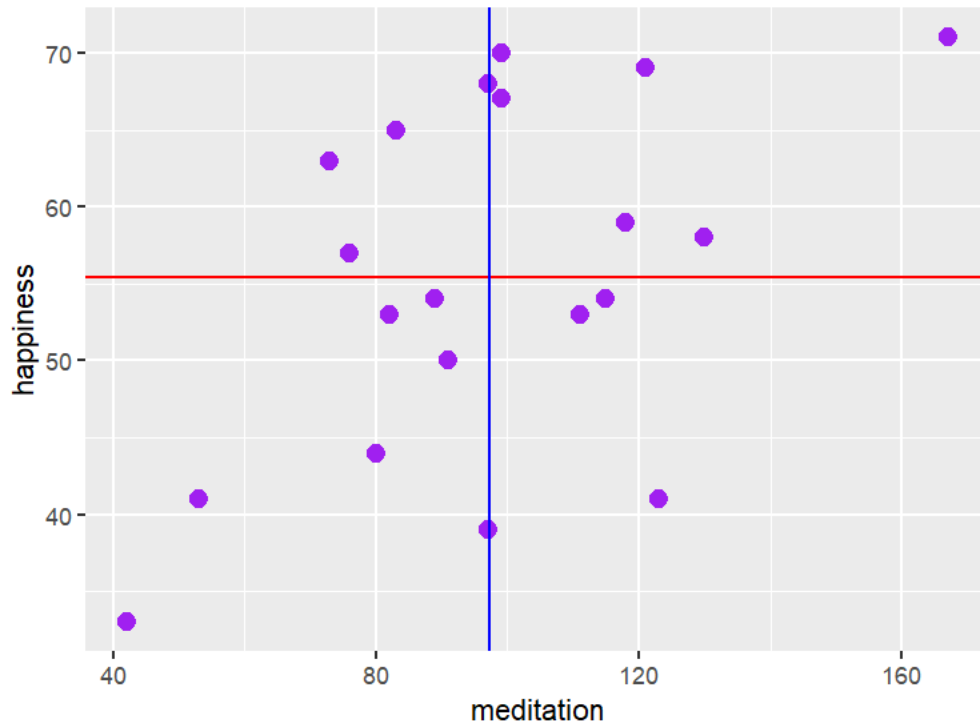


Figure 12.2: Scatterplot with lines at the mean for meditation and happiness.

Let's take a moment here and note that the majority of data tends to clump up in the upper right and lower left quadrants of this graph. This means that the $(x_i - \bar{x})(y_i - \bar{y})$ part of the equation will usually be positive (in the upper right quadrant, we are multiplying two positive values together, and in the lower left quadrant, we are multiplying two negative values together, which equals a positive number). Put a pin in this point, as we'll return to it during our second example.

Covariance is a very important concept, and one that you'll hear more throughout the rest of your statistical journey. But, there is a shortcoming of covariance: when we have a scale that is quite large as we do here, the covariance is also going to be large. As we've learned with other statistics, it's helpful if we can figure out a way to standardize this value. And as you've seen before, standard deviation is one way we can do this. So, the equation for a Pearson correlation is:

$$r = \frac{\text{cov}(xy)}{s_x s_y}$$

This ensures that this value can never be less than -1.00 or higher than +1.00, which helps make it more interpretable among data sets that might have smaller scales (like if we are using a 1 to 5 Likert scale or a GPA).

I'll spare you the tedious math here, and tell you that the calculations indicate $r=.49$, which is a moderately large correlation in most social sciences. Significance is determined in most statistical programs as a t-test:

$$t_r = \frac{r\sqrt{N-2}}{\sqrt{1-r^2}}$$

In the case of our data, this results in a t value of 2.38. Degrees of freedom are N-2 here, so our critical t value is 2.10, so we can conclude that our correlation is significant.

Because correlations are so important, it's worth briefly walking through two more examples (and we'll discuss them again in section [13.1](#) when we review regression).

Let's take a look at the scatterplot for minutes of meditation and stress:

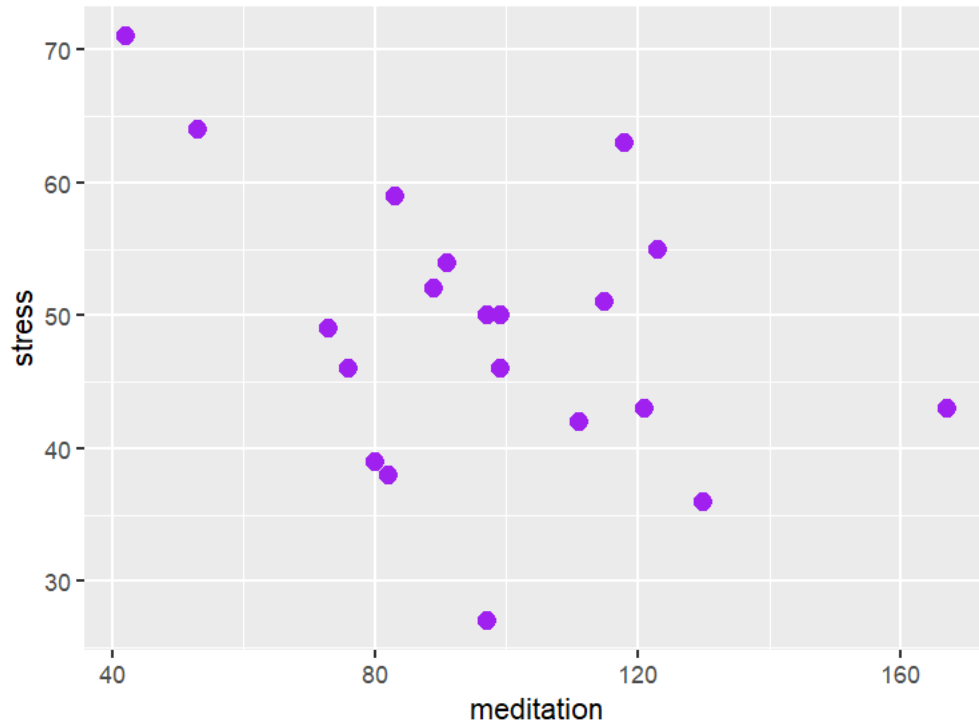


Figure 12.3: Scatterplot of meditation and stress

Right away, we can see that it looks very different from the previous example, because the data points are gathered in the top left and bottom right quadrants. If you think back to our formula for covariance:

$$cov_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{N-1}$$

you can see that for most data points in this correlation, if they are above the mean in meditation time, they are below the mean on stress, and vice versa. This means that the covariance will be negative (because a positive number times a negative number leads to a negative product), which in turn, means the correlation will be negative ($r=-.41$, to be exact!).

Meanwhile, if we look at the scatterplot between meditation time and liking meditation:

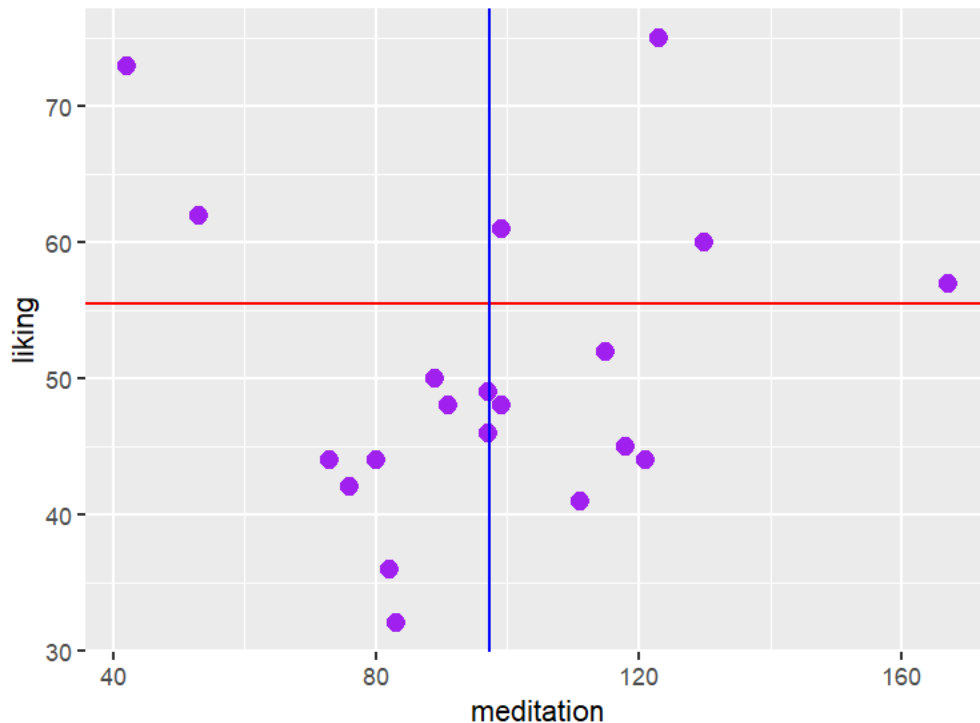


Figure 12.4: A scatterplot of meditation and liking

you will see that the data points are distributed pretty equally across all four quadrants. This means that when we sum it up, some of the positive and negative values will cancel each other out, which means our correlation will be pretty small ($r=.04$ in this case).

I would be remiss if I didn't say what all statistics teachers say: Correlation is not causation. Correlations can't be used in experiments, so based on these results, we can't say that meditation is causing higher happiness and lower stress, they are just related. It might be the other way around (happy people and people with low stress are more likely to meditate) or there could be another causal factor (people with money have the time to meditate, and they are happier and

less stressed). So, be careful in how you talk about correlations so that you're not implying a causal link.

12.3 Other Correlations

The vast majority of the time, when someone says “correlation” they are talking about a Pearson correlation, but there are some other useful correlations that are worth knowing about. Whereas a Pearson correlation is intended for two continuous variables, we sometimes might have different types of variables where we need a correlation.

12.3.1 Phi and Cramer's V

To start, I want to briefly address a correlation that I already hinted at in section [Section 11.3](#) – the phi correlation. The phi is useful when we have two categorical variables, each with two categories apiece. For example, imagine we want to learn about the relationship between dog ownership and exercise:

Table 12.2: Frequency of dog ownership and daily walks

	Takes a daily walk	Does not take a daily walk
Owns a dog	30	20
Does not own a dog	20	30

A chi-square is certainly one possible option here. But perhaps we're interested in an effect size; a correlation is a useful way to gauge an effect size, since it's a statistic that generalizes across different scales and sample sizes (this is not the case for a chi-square test value). But the phi value is very easy to get if you've already done a chi-square test of independence:

$$\phi = \sqrt{\frac{\chi^2}{n}}$$

For this data, we obtain a chi-square value of 4.00 and an n of 100, so the equation works out to:

$$\phi = \sqrt{\frac{4.00}{100}} = .20$$

Phi values can be interpreted mostly like a correlation, with values closer to +/-1 indicating a stronger relationship, so this is a fairly small relationship. With Pearson correlations, the direction matters; with phi, it doesn't really matter, because it's arbitrary which group we put in which order, as there is no amount attached to the category. So, it's smart to also provide a table of your results so your audience can understand the nature of the relationship.

But what about if we have a chi-square where we have more than two categories in one or both variables? In this case, the appropriate calculation is a **Cramer's V**:

$$V = \sqrt{\frac{\chi^2}{n \times \min(r-1, c-1)}}$$

What this formula does is in the denominator, we multiply the sample size by whichever value is larger, the number of rows minus one, or the number of columns minus one. If we take the data from Section 11.3 on parental and child smoking behaviors:

Table 12.3: Table of youth and parental smoking behaviors; expected values, rounded to the nearest whole number, are provided in parentheses.

	Never-smoker	Early Regular	Early Experimental	Late Experimental
Non-smoker	16	16	38	37
Former smoker	36	3	28	59
Current smoker	69	5	26	67

We obtained a chi-square value of 48.20, and we can see that we have fewer rows than columns, so:

$$V = \sqrt{\frac{48.20}{n \times (3-1)}} = .25$$

So, again, here we have a fairly small effect. Note also that if we have a chi-square with two groups in each variable, you basically end up with the formula for phi. Again, Cramer's V is best shared with a table so the audience can understand the nature of the relationship between the variables.

12.3.2 Point-Biserial Correlation

Another helpful type of correlation is the point-biserial correlation, where we are correlating a dichotomous variable with a continuous variable. A common use of this statistic is in test development to examine the quality of an item. So, imagine that we have an item on a test that we've scored 1 for correct and zero for incorrect. We also have each participants total score on the test. We can correlate the correctness with the total score:

Table 12.4: Participant score on item and test

ID	Score on Item	Score on Test
1	0	44

ID	Score on Item	Score on Test
2	0	37
3	0	24
4	0	37
5	0	34
6	0	51
7	0	66
8	0	42
9	0	36
10	1	52
11	1	68
12	1	79
13	1	85
14	1	60
15	1	58
16	1	90
17	1	84
18	1	44
19	1	87
20	1	92

The good news is, mathematically, the formula for the point-biserial is the same as the formula for a Pearson; if we do the calculations, we'll see a r_{pb} of .74. This is good news for this test item! What it's saying is that people who score well on my test tend to get this question correct, and those who score poorly on my test get this question wrong, which is what we'd expect. If we saw a negative correlation here, it means my students who score well on the exam get this question wrong, that likely means I have a bad item (either it is "tricky" so my strong students get confused and answer wrong, or perhaps I've made a mistake on my answer key).

You might be asking "If it's mathematically the same as a Pearson correlation, why do we need a special statistic?" The main reason is that because one variable is dichotomous, it breaks some of the assumptions of a Pearson correlation (such as the assumption we have roughly normal distributions in both variables). It's also a good reminder that we probably need to provide an interpretation for our audience. For example, if we did a point-biserial correlation on men and women and their scores on a test, the way we code the data is arbitrary (i.e., there's no reason why we'd have to code men = 1 and women = 2, we can choose whatever numbers we want for each group). So, we need to add in some extra interpretation when sharing our result to clarify for our audience which group has a higher score on our continuous variable.

You might also be asking "Wait, if we're looking at a dichotomous variable and a continuous variable, can't we just do an independent t-test?" And... yes, you can! Really, the point-biserial is just the effect size of a t-test, so they both tell you the same thing, but sometimes putting it

on a -1 to +1 scale can be helpful (like for my test development example, where test developers are very used to looking at point-biserials).

One final question you might have is that I underlined the word dichotomous a few paragraphs ago, so you may wonder “Can I use a point-biserial correlation if my variable is dichotomized?” Nope! There is another type of correlation called a biserial correlation and that is more appropriate for a dichotomized variable, because there is a distribution that underlies the dichotomous variable. I used to teach this formula to my students, but it’s not something people use much. Usually it is better to leave a continuous variable alone rather than dichotomize it; plus, biserials are fiddly and a lot of statistical programs (like SPSS) don’t even offer it as an option, so you might end up having to do the math by hand. The information is out there if you ever need it, but I’ve stopped teaching this statistic to my students because it’s so rare to use it (I have never personally needed it!)

12.3.3 Spearman’s Rho and Kendall’s Tau

The last set of alternative correlations I like to cover applies to ordinal data. Ordinal data is a little tricky to work with, and I largely avoid talking about it in introductory statistics courses, but these two values can come in useful in case you want to see if a ranked order remains the same or if it changes. As an example, let’s say that I have some students and I am ranking them on their skills in some classes:

Table 12.5: Student rankings on skills

Student	Math	Art	Logical Reasoning	Reading
Adam	1	12	2	11
Betty	2	11	1	12
Calvin	3	10	4	3
Danielle	4	9	3	4
Edgar	5	8	6	5
Fern	6	7	5	6
George	7	6	8	7
Harriet	8	5	9	8
Igor	9	4	10	9
Jackie	10	3	9	10
Kevin	11	2	12	1
Liz	12	1	11	2

Compare each set of scores to the math scores, and you’ll notice some trends:

- Art scores are perfectly inverse of the Math scores, with the worst students Math students scoring the best on Art, and vice versa.

- Logical Reasoning scores are mostly in the same order, but with lots of inversions – Betty is better than Adam in Logic, but not on Math, and so on.
- Reading scores in the middle are the same as math scores, but the top and bottom scores are out of order.

I set these up to help highlight the difference between Spearman’s Rho and Kendall’s Tau. They both are useful, but they are sensitive to different things. The formula for Spearman’s Rho is the same as a Pearson correlation. So, it is sensitive to the overall order and whether it is preserved between the two measures. Kendall’s tau is a bit different. Here, it counts the number of inversions in the data, and compares it to the number of pairs of objects:

$$\tau = 1 - \frac{2(\text{Number of inversions})}{N(N-1)/2}$$

So, when we look at the data and think about this, we can expect that the rho between Math and Logical Reasoning will be pretty good (the data order is mostly preserved) but our tau estimation will be lower, because there are lots of inversions. Meanwhile, the rho between Math and Reading will drop quite a bit because the 4 most extreme scores have swapped, but tau will look a little better because the other 8 have no inversions.

Here are the actual values we obtain if we do the math:

Table 12.6: Rho and Tau calculations for data

Variables	Rho	Tau
Math and Art	-1.00	-1.00
Math and Logical Reasoning	.96	.82
Math and Reading	-.40	-.09

Our expectations bear out here – Math and Art have a perfectly inverse relationship, so we see a perfect -1.00 correlation for both rho and tau. For Math and Logical reasoning, we do see that the flip-flops in the data drop Tau much more than they do Rho. And for Math and reading, we see that the four swapped scores lead to a much lower rho than tau.

So which one is the right one to use? It ultimately depends on what you care more about in your research. If the general order of rankings matters most, use rho; if inversions matter, use tau. When in doubt, rho is conceptually similar to a correlation, so that tends to be used more commonly.

12.4 Writing the Results of a Correlation

The good news here is that writing correlations is typically very brief and straightforward for the most part. When sharing the results of a correlation, you want to convey the following information to your audience:

- What type of correlation you are using (if you just say “correlation” your reader will assume it’s a Pearson correlation)
- The correlation value.
- Some journals also like to have degrees of freedom for Pearson correlations.
- For Pearson, point-biserial, Spearman, and Kendall correlations, you typically provide significance values. For phi and Cramer’s V, significance values are not typically given, but if you want to provide them, they will be the same value as the related chi-square significance.
- In the case of phi, Cramer’s V, and point-biserials, an interpretation of the direction of the results – again, this is because the direction of these statistics is arbitrary so you must interpret them for your audience.

The correlation between meditation and happiness was moderately strong, $r(18)=.49$, $p<.001$.

A phi correlation suggests a moderate correlation, $\phi=.20$. Specifically, people who own a dog are more likely to take a daily walk.

A point-biserial correlation indicates the test item is not functioning appropriately, $r_{pb} = .74$, $p<.001$. Specifically, students who tend to score higher on the overall test tend to do poorly on this item.

A Spearman correlation suggests that there is a strong association between ranked scores for Math and Logical Reasoning, $\rho=.96$, $p<.001$.

13 Regression

13.1 Learning Objectives

- Know what a simple regression is and what types of data it requires.
- Know the assumptions of a regression.
- Know how to interpret regression values like b-values, beta weights, and R^2 .
- Know how a multiple regression differs from a simple regression, and know how to interpret its values
- Know the different approaches to regression (entry, hierarchical, forward and backward) and their pros and cons.
- Know what multicollinearity is and how it can cause problems in regressions
- How to write up the results of a regression.

13.2 Simple Regression

I often like to say that correlations and regressions are sisters – they are very similar procedures, but they have a slightly different vibe in terms of what information they provide to you. In many cases, you are better off running a regression instead of a correlation because it gives you more information, and it tends to be more flexible than correlations in terms of the types of data it can handle. But correlations do have their place in statistics – you see them all the time, especially in tables providing a general overview of how variables in a study are related aside from the specific hypotheses a researcher is investigating.

13.2.1 Assumptions of a Regression

Regression assumptions are quite similar to what we've seen with other parametric tests:

1. The relationships between variables are linear (i.e., not curvilinear).
2. Continuous variables should be roughly normally distributed and we don't see major outliers
3. The relationship between variables indicates homoscedasticity

4. The data demonstrate independence (i.e., we don't have participants influencing one another's responses).

Regression tends to be less robust than t-tests and ANOVAs when these assumptions are violated, so if you have funky data, you may need to adjust it (addressing outliers, transforming variables) or consider an alternative approach like bootstrapping. It's also worth taking a look at your data to see if there are any data points that might be influential. For example, consider the scatterplot below:

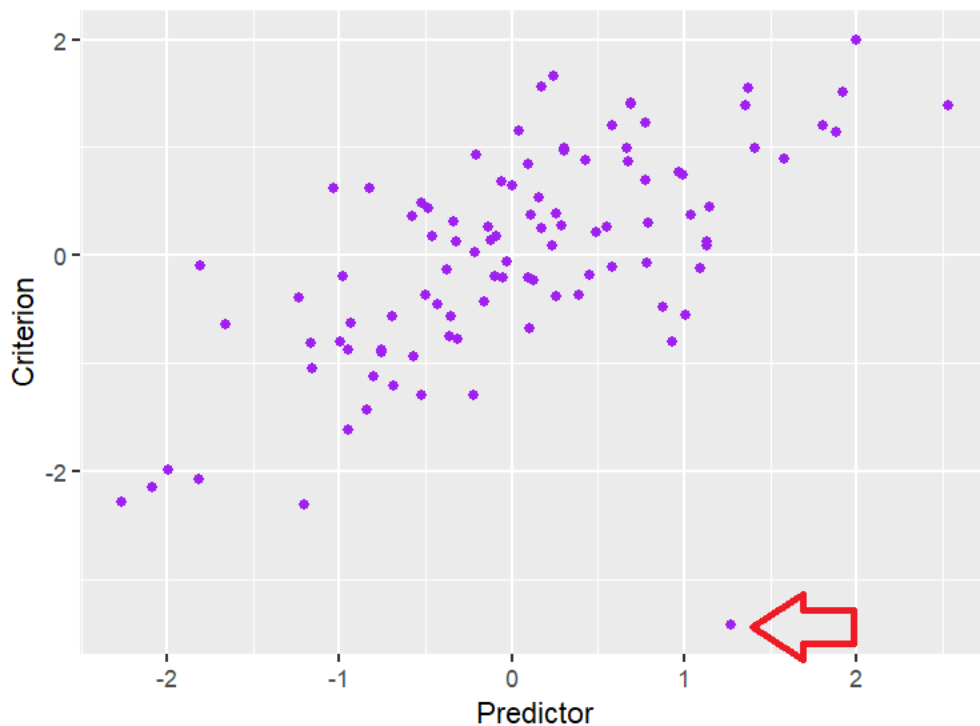


Figure 13.1: A scatterplot with an outlier

As you can see in the graph, the large arrow is pointing to a data point that is far outside of the scatterplot (and is extraordinarily low on the criterion). This data point would have an outsized effect on the results, so we may want to recheck our data to make sure there's not a mistake or that these responses aren't from a different population. There are some helpful diagnostic tools we can use to search for influential data points:

- Adjusted predicted value
- Deleted residual
- Studentized deleted residual
- Cook's distance

- Leverage
- Mahalanobis distances

So, it's worth checking to make sure your data is clean and that you are aware of any influential data points before you start your analysis.

13.2.2 A Simple Regression Between Two Continuous Variables

Let's go back to our example from Section 12.1, where we explored the relationship between meditation and happiness using Table 12.1. [Here's](#) the data set.

As you may recall, we looked over a scatterplot (Figure 12.1) to confirm that the data was linear and met the assumptions of a parametric test.

What a simple regression does, in a nutshell, is it plots a “best fit” line to the data. There is one, and only one, line that minimizes the vertical distance between each data point and a straight line. I won't cover the math of how a statistical program determines what this line is here. But you may have learned in other math classes the basic formula for a line, often written as:

$$mx + b = y$$

In most statistical notation, this formula is written as:

$$\hat{y} = b_0 + b_1(x)$$

But it's ultimately the same thing despite the change in the notation. In this formula, $\hat{y}$ indicates our predicted value for the dependent variable, and this is based on the b_0 , which is the y intercept (that is, if $x=0$, what would we predict someone's score on y would be?) and b_1 , which is the slope of the line (that is, for a one-point change in x, how much do we expect y to change?)

This ability to predict is one feature that is an advantage of regression – while with a correlation, all we can do is state the relationship between variables, with a regression, this line allows us to provide a predicted value on the dependent variable if we have someone's score on the independent variable. So, let's plot a regression line on our meditation data:

We can see right away that, while our prediction is not perfect, there is certainly a positive relationship between variables. But perhaps we want to know something more specific – if someone meditates for 60 minutes a week, what can we expect their level of happiness to be, based on these data?

When we conduct a regression, we get quite a bit of information from the output. Most statistical programs will provide some unstandardized coefficients (sometimes referred to as b-values). These values are what we can use in the formula to figure out where a point might

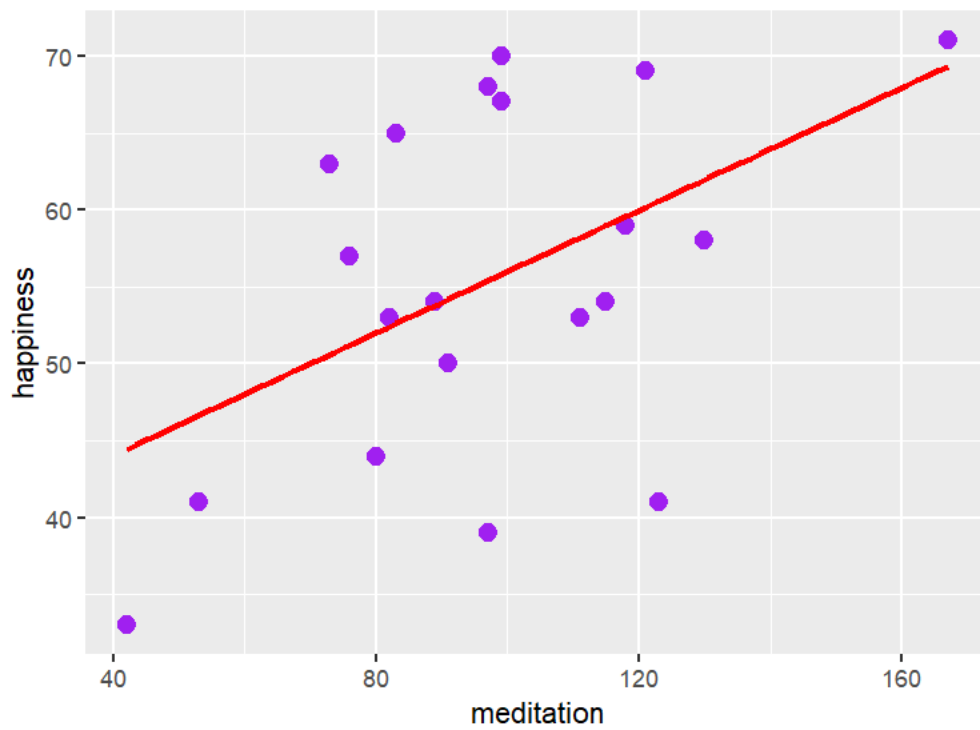


Figure 13.2: A scatterplot with a best fit line.

lay on that best fit line (often known as a “fitted value”). Take, for example, the outputs below from SPSS:

Coefficients ^a					
Model		Unstandardized Coefficients		Standardized Coefficients	
		B	Std. Error	Beta	t
1	(Constant)	36.067	8.417		4.285
	meditation	.199	.083	.491	2.392

a. Dependent Variable: happiness

Figure 13.3: Regression output from SPSS

While we don’t tend to report unstandardized coefficients, they have a lot of practical use. For example, let’s say that we do, indeed, want to make a prediction about how stressed someone is if they meditate for 80 minutes a week. We can simply use the unstandardized slope (.20) and the y-intercept (36.07) to create the formula for a line, and then we input 80 in the place of x:

$$\hat{y} = (.20)(80) + 36.07 = 52.07$$

Sure enough, if you go back to the scatterplot, you’ll see on the regression line that for a score of 80 minutes of meditation, we see the line is right at about 52. Of course, this is just a guess – you can see that one participant who spends 80 minutes meditating only gets a happiness score of about 44, but we know at this point to expect some error in any statistic.

While the unstandardized coefficients are practically useful, more often, in scientific writing researchers report the standardized values, often referred to as β values. What is happening behind the scenes is that the statistical program transforms both the predictor and the criterion into z-scores, and then calculates the regression and provides these standardized values. These are beneficial because they are on a roughly -1.00 to +1.00 scale¹, so they are more comparable. Here, we see that $\beta = .49$, which is a decently strong relationship. If you have an amazing memory, you may also remember that in Section 12.1, we found that $r = .49$ too! So for a simple regression, the β value is the same as the Pearson correlation. This will change as we move into multiple regression, but it’s a nice way to see the connection between regression and correlation.

Most programs also provide a measure of the effect size, often represented as R^2 , which is an indication of how much variance is explained by the model. For these data, we find that

¹If you have every heard of Item Response Theory (IRT), this is closely related to logistic regression – basically, we are hoping that the latent variable we are trying to measure relates to whether someone gets an item about that variable right or wrong, and that theory uses an S-shaped curve to model this relationship.

$R^2 = .24$ (which is .49 squared). So, in other words, about 24% of the differences we see in stress levels can be explained by meditation, which is quite a large amount indeed!

13.2.3 A Simple Regression Between a Dichotomous Predictor and a Continuous Criterion

Regressions are also good about handling a dichotomous predictor. Let's take, as an example, the item data from Section 12.1, which you can see in Table 12.4.

Check out [this data set](#) about how people perform on an exam item. If we wanted to, we can use regression to answer the question "If someone gets this item correct, what would their score be on the exam?" We can run a regression in the same way we would for a continuous predictor, and we also get output that provides us with some unstandardized and standardized coefficients. The interpretation is a little bit different here, but it's conceptually the same. The unstandardized values still indicate the y-intercept and the slope, but if we code our predictor using 0 and 1, we'll notice something interesting. Here's the formula we get from conducting this regression:

$$\hat{y} = 31.41(x) + 41.22$$

You'll notice if we solve for people who didn't get the question correctly, we find:

$$\hat{y} = 31.41(0) + 41.22 = 41.22$$

Aha! The solution is the y-intercept, which makes perfect sense if you think about it: The y-intercept is what y equals when x equals zero, and x equals zero when someone doesn't solve the item correctly, so 41.22 is, in fact, the average for the group that didn't answer the item correctly. When we solve for the equation for people who did get the question right, we find:

$$\hat{y} = 31.41(1) + 41.22 = 72.63$$

This also makes sense, because the slope tells us how much of a change in y occurs for a one-point change in x, and a one point change in x just means that someone got the item correct. Sure enough, if you look at the data, you'll find the exam average for people who got this item correct is also 72.63.

As you think this over, you might find yourself asking "Wait, isn't this regression basically the same as an independent t-test?" Yep! It is the same, it's just a different way of thinking about it. You might also find yourself saying "How is this different from a point-biserial correlation?" It's not different, actually! If you look at the β value, you'll find $\beta = .74$, which is the same correlation we found for this data back in Section 12.3.2! So, now you have three tools you can use to understand the difference in means between two groups.

We can also understand variance explained by looking at the R^2 . Here, $R^2 = .55$, which means about 55% of the variance in student scores is explained by whether they get this question correct.

13.2.4 Wait... can I do a regression with a polytomous variable?

You can! But it's a bit fiddly to do. Essentially, you'd need to dummy code your variable into dichotomous yes/no categories. For example, if we wanted to include religion in a regression, and people choose among 4 categories (Christian, Muslim, Buddhist, Hindu), you would need to set up 3 dichotomous variables with 0=no and 1=yes:

Table 13.1: A table demonstrating how different religions would be represented using dummy codes.

ID	Christian	Muslim	Buddhist
1	1	0	0
2	0	1	0
3	0	0	1
4	0	0	0

In this example table, we have one of each religion I listed. You'll also notice that one category (Hindu in my example) doesn't have a column. Basically, someone who is Hindu is listed as a "none of the above" category. This is necessary to do; if you included a Hindu category, you'd end up with something called "Perfect Multicollinearity" (I'll explain this in the next section) and your regression won't run. So, we can think of the Hindu group here as more of a "baseline" group.

We can also do this if a variable is not mutually exclusive. Take, for example, when we assess race in a survey – if we use a "check all that apply" instruction, someone who is multiracial might check several boxes. That's not a problem – we can make a decision if we want to use the dummy coding approach and just allow them to have a 1 in multiple categories, or if we want to create an additional "multiracial" group in our data that would be a 1 if someone checks more than one box. This depends on how that variable is typically treated, the nature of that variable, and the nature of your question.

But to analyze this, you now have more variables, so you wouldn't be able to use simple regression to examine a polytomous variable – you'd need a multiple regression. Let's talk about that next.

13.3 Multiple Regression

Simple regression shares a lot of DNA with correlations. However, remember that in Section 12.1, we were only able to do correlations between two variables, so as a statistical analysis tool, it is limited in that way. But multiple regressions are not – we can put in several variables of different types into our model to see how they all work together to predict some outcome. But

in order to make the jump from simple to multiple regression, we need to discuss something called “shared variance.”

13.3.1 Shared Variance and Multicollinearity

In most social sciences, we recognize that nearly all variables will have some relationship with one another. For example, imagine I’m trying to predict happiness using minutes of meditation and stress levels. But almost certainly, my predictors (meditation and stress) share a relationship. So, when we try to use both variables at the same time, their correlation with each other means that they probably also both explain some aspects of variance in meditation for people.

In fact, in this data these predictors share a decent correlation with one another ($r=-.41$, $p=.08$); this is visually represented in Figure 13.4 by the mauve-colored overlap between the stress and meditation circles. This means that when we attempt to predict happiness, these two variables will compete with one another in our model, because both of them explain some aspects of happiness (represented by the grayish purple triangle where all circles overlap). If we were to run separate simple regressions with just one predictor, we’d find significant results. But when we put them in a model together, they may look like they are not as useful as predictors because they share this variance.

When predictors have a high correlation with each other (and by high, I mean $r=+/- .80$ or stronger) we have a situation known as **multicollinearity**. This causes problems with regression because they make our estimates unstable (the β values bounce around as you change the model, or we start to find β values that are larger than $+/-1.00$). It also makes it hard to know which variable is important.

You can imagine that at the extreme end where two predictors share a perfect $+1.00$ correlation, there’s no point in putting them both into our regression, because they tell us the same thing. At the less extreme end, predictors with a $.80$ or stronger correlation will overlap a lot, and you won’t get a lot of **incremental validity** out of using both predictors (incremental validity is represented by the greenish overlap between meditation and happiness, and by the purpleish overlap between stress and happiness in Figure 13.4; it is the unique variance explained by each predictor).

So, before you build a model, it’s a good idea to check for high intercorrelations among predictors. Some statistical programs will provide a variance inflation factor (VIF); a VIF of greater than 10 indicates multicollinearity, so you might want to reconsider what predictors you are using in your model.

In our example, our predictors share a moderate correlation, so we don’t have multicollinearity, but as we will see, they do compete over variance in an interesting way.

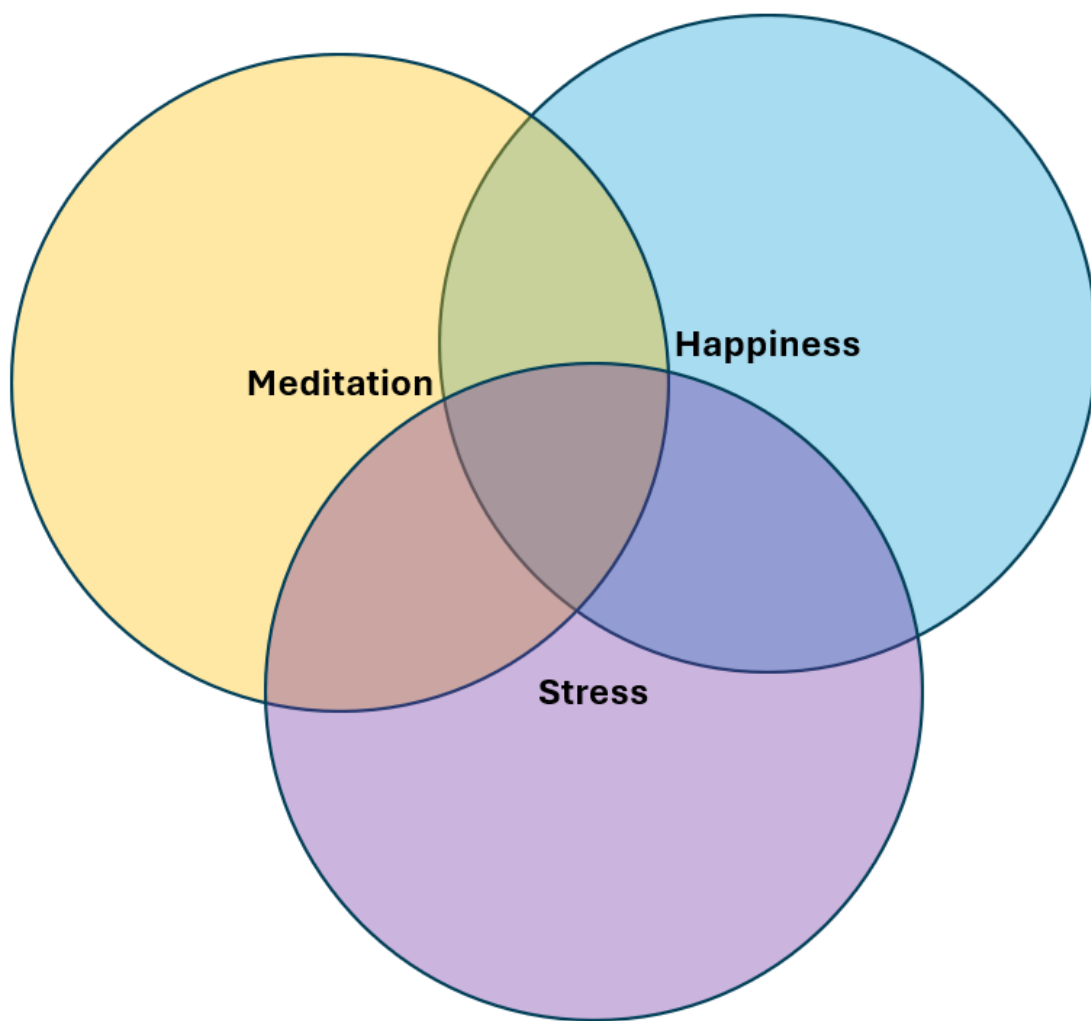


Figure 13.4: An illustration of how our predictors and criterion might overlap

13.3.2 An Entry Multiple Regression with Two Predictors

As we move from simple to multiple regression, we're still just charting a best fit line, but now the line is taking place in three or more dimensions, so it gets a little tricky to illustrate.

There is still one, and only one, best fit line that tries to minimize the distances between each point and the prediction line. The equation for our line gets a bit more complex, as we now need to add an additional variable:

$$\hat{y} = b_0 + b_1(x_1) + b_2(x_2)$$

When we conduct our regression, we get these results:

Coefficients ^a						
		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
Model		B	Std. Error	Beta		
1	(Constant)	57.908	17.142		3.378	.004
	meditation	.147	.088	.363	1.664	.114
	stress	-.343	.237	-.316	-1.449	.165

a. Dependent Variable: happiness

Figure 13.5: SPSS Output of multiple regression

We still have a y intercept, but now we have a slope assigned to each predictor in our model. Again, the output provides us with b-values, which allow us to make a prediction on someone's score on the outcome given their scores on the predictors. It also gives us β values which help us understand the relative "importance" of each variable in the model.

In this particular model, we see that whereas meditation time predicted happiness in our simple regression, it no longer is a significant predictor once we add stress into the model. Because our two predictors are (negatively) correlated, it seem that they are fighting over variance in the happiness variable, which leads them to drop out of significance (we also are working with a very small data set for this example, so as you've learned, we don't have a lot of power here, which means it's easy for large effects to be non-significant). Despite the lack of significance, we can still use the equation to guess what someone's happiness level would be if they spend 40 minutes meditating and rate their stress as a 57:

$$\hat{y} = 57.91 + .15(40) + -.34(57) = 44.53$$

And given the β values, we can see that they are relatively similar in how important they are in predicting outcomes, but in opposite directions (meditation time is positively related to happiness, while stress is negatively related to happiness).

The regression we’ve just done here is often known as an **entry regression**; we start with a specific model we want to test, and just test it all at once. However, this is not your only option!

13.3.3 Hierarchical Regression

Another option we have with regression is we can add more predictors to our model in several steps. This is a great option if there’s a variable we know probably relates to our outcome, and we want to know whether another variable predicts above and beyond that first variable. This gets right at the question of incremental validity – if we have one variable in our model and then add two more, this tells us whether the addition of those variables is at all helpful for predicting outcomes. We can compare models to see how our equation and the beta weights change as we add predictors. We can also look at a change in R^2 (sometimes notated as ΔR^2) to see how much more variance we can explain with an additional variable.

To illustrate how this works, it’s helpful to have a larger data set, so you can check out the data set [meditation example 2](#) to understand what’s going on in our analyses. In this example, imagine we still want to study happiness, but now we’ve collected some data on gender, household income, meditation, stress, and motivation. The first five lines of the data are provided below:

Table 13.2: Table of the first five lines of the dataset Meditation 2

id	meditation	happiness	stress	liking	gender	hhincome	motivation
1	107	54	43	41	0	57	6.61
2	13	42	69	48	1	120	2.15
3	93	58	43	51	0	111	6.37
4	90	48	51	41	0	78	8.71
5	76	37	61	62	0	70	7.73

In our research, perhaps we recognize there is some relationship between demographics like gender and income on happiness, and we’re interested in whether meditation will help above and beyond those factors. So, we can put gender and income into the first step of our model (Model 1), then add meditation in the second step (Model 2). We can also generate information about how much of a change we see in variance we can explain once we add meditation in the second step. Output might look something similar to this:

Table 13.3: Regression coefficients for Model 1

Variable	B	β	p
Intercept	49.86	–	<.001

Variable	B	β	p
gender	3.03	.12	.06
hhincome	.03	.07	.28

Table 13.4: Regression coefficients for Model 2

Variable	B	β	p
Intercept	29.98	–	<.001
gender	-.37	-.01	.79
hhincome	.02	.03	.54
meditation	.25	.59	<.001

Table 13.5: R2 changes between Model 1 and Model 2

Model	R^2
Model 1	.02
Model 2	.34
ΔR^2	<.001

These results tell us a few things. First, we can see that in our first model, gender and household income don't provide much prediction on happiness. When we add meditation in, we can see that our coefficients in the model change a little, but the outcome is the same, and we see that meditation is significant. Indeed, if we compare the R^2 between models, we see it jumps up quite a bit (and an ANOVA conducted on these models indicates a significant change, $p < .001$). So, this tells us that gender and income are not helpful predictors, but that our model improves once we add meditation.

13.3.4 Exploratory Regressions: Forwards and Backwards Regression

There is one additional way we can use regression, but I want to emphasize here that this is not a great approach in terms of scientific rigor. Ideally, when we conduct research, we have a hypothesis in mind, and we collect data to see if there's support for that hypothesis. But sometimes we just have data that's been collected for other purposes, and we might be curious about what different factors might predict an outcome. In this case, we can take advantage of exploratory regressions that examine what relationships exist, and use that to test several models to come up with the best model to explain the outcome. The problem with this approach is that it takes advantage of chance – if we had 20 variables in our initial model, we may accidentally identify one of them as important just because of the 5% chance of a Type

I error. So, in order to feel confident in our findings, we really should replicate our study with a sample drawn specifically to retest our model. Hopefully we'd find the same outcome, but there certainly is a chance we won't!

There are several approaches we can take to these exploratory regressions, but two are popular enough in the literature to merit an example. Sometimes, we start with an empty model and allow our statistical program to choose variables that perform the best and add them to the model until adding variables no longer improves our prediction. This approach is known as a forward regression. We can also do the opposite – we start with all possible variables in the model, and the statistical program removes them one by one until removing one produces a significant change in the variance it can predict. This approach is known as backward regression.

Sometimes, you will get slightly different findings if you do a forwards versus backwards regression. I generally recommend people use a backward regression if they want to explore data, because it's better at discovering something called a suppressor variable. A suppressor variable is when it makes the relationship between two variables stronger when it's added to the model. Take for example in our Meditation 2 data, imagine I am trying to predict happiness, and I don't have a good theory for what might be happening, so I conduct a backward regression with all other variables as a predictor. My initial model will include all variables, and each subsequent model removes the least useful predictor. My final model looks like this:

Table 13.6: Regression coefficients for a backward regression

Variable	B	β	p
Intercept	74.46		
meditation	.15	.36	<.001
stress	-.53	-.43	<.001
liking	-.21	-.17	<.001
gender	3.77	.15	.003

Here, we see that household income and motivation don't end up predicting anything significantly, so they drop out of the model. However, this certainly might just be due to chance, so whenever you use an exploratory method like this, it's best practice to do a follow up study using these variables and an entry regression to confirm that your findings are dependable.

13.4 Writing the Results of a Regression

When writing the results of a regression, you typically want to indicate the following:

- Provide information about the regression – what variables are included, and explain whether you’ve used an entry, hierarchical, or exploratory approach. If you don’t specify anything, most people will assume an entry regression.
- Although b values are helpful in most applied situations, we don’t tend to report them in scientific reports.
- β values are more helpful in a scientific context, so we typically report those, with their related significance value.
- The R^2 value is helpful to understand how much variance is explained by your regression.
- If you are doing a hierarchical regression, you must provide all the β weights and their significance at each step. It also is helpful to either provide an R^2 value or a ΔR^2 value so that your audience understands how adding predictors changes how each variable contributes to the prediction, and whether the addition of new predictors helps us better predict the outcome. Often, tables make it much easier to convey this information briefly and clearly.

A multiple regression indicates that meditation does not significantly predict happiness ($\beta=??$, $p=.??$) and ??? ($\beta=??$, $p=.??$) .

I conducted a hierarchical regression with xxx in the first step, and xxx in the second step. Results are presented in Table 1. The final model suggests that the addition of ??? :::::

14 Logistic Regression

14.1 Learning Objectives

- Know what a binary logistic regression is, and what kinds of research questions it can address.
- Know the ways that a logistic regression differs from a regular linear regression.
- Know the assumptions of a logistic regression, and how violations might affect your results.
- Know how to interpret the results of a logistic regression.

Section 13.1 has hopefully demonstrated that regressions can be very powerful tools for analyzing multiple predictors on a continuous outcome. However, you can certainly imagine many contexts in which our outcome is binary – dead/alive, successful/unsuccessful, pass/fail, right/wrong. However, a simple graph shows that when we try to map a continuous predictor to a binary outcome, a regression line is not a great fit!

This is where a procedure known as logistic regression comes into play. Logistic regression fits an S-shaped curve (sometimes known as an ogive curve or a sigmoidal curve) to the data. As you can see, this is a better approximation of what's going on in the data:

At the leftmost side of this S-curve, the line is very close to zero, but it never quite reaches zero. In the middle of the curve, we see a fairly steep curvy line, and that leads to the rightmost side where our line creeps closer and closer to 1, but never quite reaches it. In plain language, when we're discussing human behavior, we can't ever really say there is a perfect zero chance someone will do something, nor can we say there's 100% chance they will do something. Probabilities can approach these values, but never quite reach them. That probability is what we are trying to model with a logistic regression.

Logistic regression can be a bit mathematically intimidating, but it is a very helpful procedure that shows up in a lot of contexts. One good example we discuss is the issue of whether someone gets an item correct or incorrect on an exam¹. In this chapter, we'll talk a lot about

¹If you have every heard of Item Response Theory (IRT), this is closely related to logistic regression – basically, we are hoping that the latent variable we are trying to measure relates to whether someone gets an item about that variable right or wrong, and that theory uses an S-shaped curve to model this relationship.

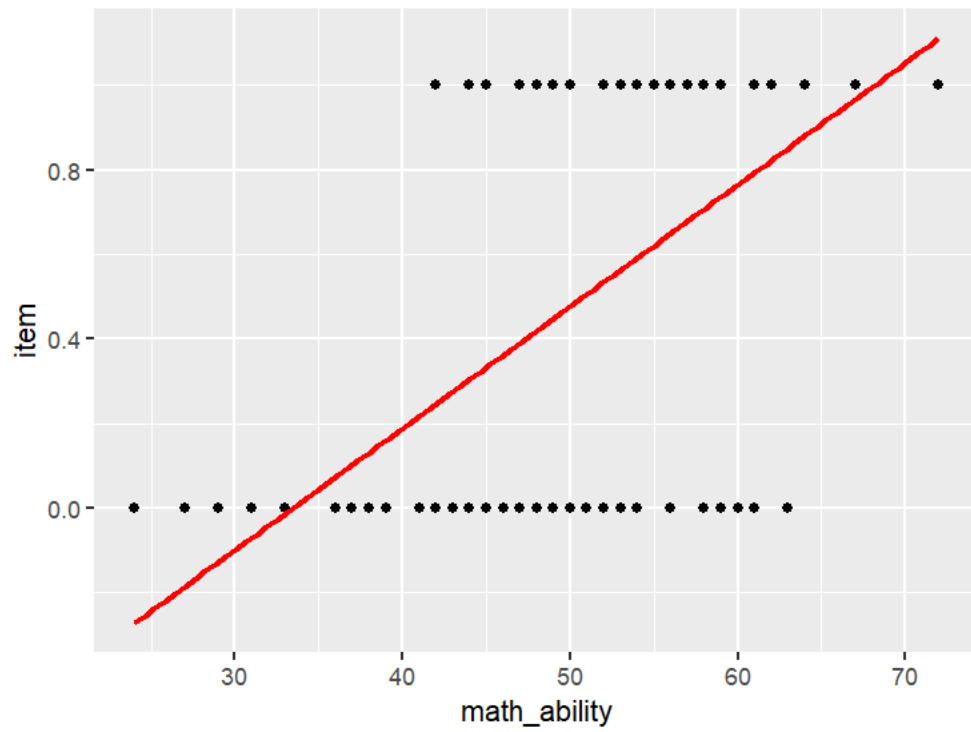


Figure 14.1: A straight regression line mapped onto a binary outcome

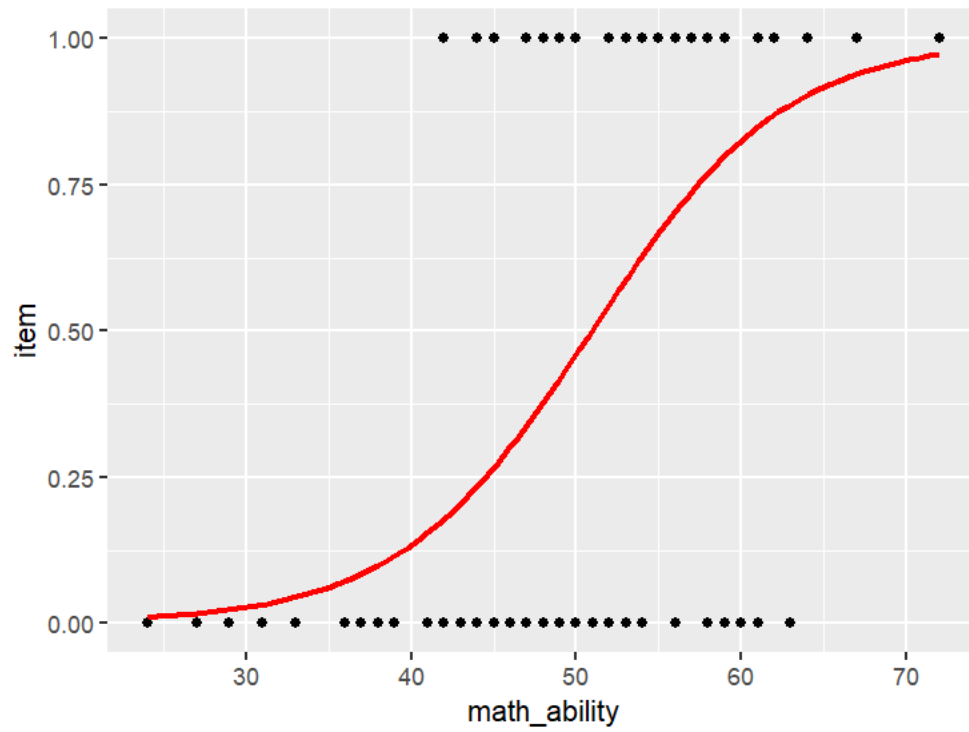


Figure 14.2: The same distribution fit with an s-curve

whether an “event” happens. Usually in data sets, we use a 0 and 1 coding to establish our binary outcome.

Take, for example, the data set [test item](#), where we have information about students from two different classes, their math ability on a standardized test, whether they are an international student, and whether they got a particular item correct on an exam. The item is coded as 1 if they get the question right, so we can think of the “event” we’re interested in as “answers the question correctly”:

Table 14.1: The first few rows of the test item data set

id	class	math_ability	international	item
1	1	52	1	1
2	2	59	1	1
3	1	57	1	1
4	1	47	0	0
...	...	...	...	...

Imagine that this test item is a word problem intended to assess math ability. So, maybe our first question is whether a student’s math ability assessed with another test predicts whether a student will get this item right or wrong. A logistic regression is a good way to answer this question, but first, we need to touch on the topic of probability, odds, and logit.

14.1.1 Probability, Odds and Logit

Hopefully at this point, you are getting a little more comfortable with the concept of probability. Probability is all about the likelihood of something happening based on the total of events. If we grab a table way back from [Section 11.3](#) where we looked at the association of violent media and aggression:

Table 14.2: Contingency table of exposure to violent media and aggression

	Aggression	No Aggression	TOTAL
Exposed to Violent Media	90	10	100
Not Exposed to Violent Media	60	40	100
TOTAL	150	50	200

As you learned in [Section 4.1](#) , we can calculate the probability of an event by looking at the event compared to the total. So, if we want to know the probability of aggression among children exposed to violent media, we calculate:

$$p(Aggression|Exposed) = \frac{90}{100} = .90$$

However, odds are different than probabilities. Odds are all about the comparison between something happening versus it not happening (rather than the total). This is calculated using probability:

$$odds = \frac{p}{1-p}$$

Note how the denominator is different here – Instead of using the total as our denominator, we’re using the probability of the event *not* happening. So, for our example, the odds that a participant will be aggressive given that they are exposed to violent media is:

$$\frac{90}{10} = 9.0$$

As you can see, whereas probabilities are squeezed into a limited range (they can only be between .00 and 1.00) odds can range from 0 to infinity. When an event is quite common in a group, the odds can be quite large.

It will also be helpful to talk about risk ratios and odds ratios. These are essentially effect sizes for probabilities and odds. To calculate the risk ratio (also known as relative risk) of a child being aggressive when exposed to violent media, I take a ratio of the two probabilities (that is, the probability of aggression given a child is exposed to violent media, and the probability of aggression if a child is not exposed to violent media:

$$p(Aggression|Exposed) = \frac{90}{100} = .90$$

$$p(Aggression|Not Exposed) = \frac{60}{100} = .60$$

$$RR = \frac{.90}{.60} = 1.50$$

Practically speaking, I can state “Children with exposure to violent media are 1.50 times more likely to show aggression than those who are not exposed to violent media.”

The odds ratio works the same way, except we use odds instead of probabilities for the comparison:

$$odds_{exposed} = \frac{.90}{.10} = 9.0$$

$$odds_{not\ exposed} = \frac{.60}{.40} = 1.5$$

$$OR = \frac{9.0}{1.5} = 6.0$$

Here, I can state “The **odds** of aggression among children who are exposed to violent media are 6 times the **odds** of children not exposed to violent media.” Note that we have to say “odds” here – **not** likelihood or probability! This also shows that when a certain outcome is common, odds and odds ratios are larger than probabilities and risk ratios, so it can sometimes mislead people into thinking an effect is larger than it actually is.

OK, I need to add one more wrinkle to this discussion before we dive into logistic regression, and that is something known as a logit. This sometimes is a little difficult for students to absorb, because we don’t deal with logarithms often so it’s easy to forget how they work.

You know how we have certain math procedures that are opposites? Like addition and subtraction, or like multiplication and division – these procedures “undo” each other. Logarithms are basically this except for exponents. So, if:

$$5^3 = 125$$

then

$$\log_5(125) = 3$$

The log returns the exponent we’d need to take the base number (5) and get it to equal the power (125). One specific type of log is the natural log (ln), where the base number is e (roughly equal to 2.718... don’t worry too much about what e is, it has to do with how numbers compound over time). The logit, is the natural logarithm of the odds, which we calculate like this:

$$\text{Logit}(Y) = \ln(\text{odds})$$

So for our example, if we need to get the logit of our odds, we would calculate:

$$\text{logit}(9) = \ln(9.0) \approx 2.20$$

I admit that this can be a bit of a mindbender, so why are we doing this to ourselves?? The answer is twofold:

1. Using the logit essentially takes our curvy S-curve and maps it onto a regular line, which makes it a little easier for us to interpret. More on this in a little bit!
2. Using the logit also helps move our probabilities and odds into a more symmetrical scale, which allows us to understand direction and magnitude more clearly. The table below translates different probabilities into odds and odds ratios. You’ll see that the logit rescales our values into something that has zero as its midpoint, and gives positive and negative values meaning.

Table 14.3: Probability, Odds, and Log Odds

Probability	Odds	Logit	Interpretation
.01	.01	-4.59	Event is unlikely to occur
.50	1	0	Event is equally likely to occur versus not occur
.99	99	+4.59	Event is very likely to occur

I personally find looking at the probability of .50 helpful here – if we flip a coin, we have a 50% chance of getting tails. We have odds of 1 (sometimes known as even odds or 1:1). And we have a logit of zero. All of these values mean the same thing – the likelihood of heads and tails are equal. We are just re-translating that information to a different scale. It’s a little complicated, but very useful for logistic regression.

14.1.2 Conducting a simple Logistic Regression

Back to the research question at hand: “Does a student’s math score predict whether they will get this item correct?” Conducting a logistic regression allows us to fit an S-curve to the data and provides us with some estimates of how likely it is a student will answer the item correctly, given their score on the math exam. There are several pieces of information that can be very helpful in exploring this question. First, I find it very helpful to look at some contingency tables, which help us understand how our model can improve our accuracy.

First, it’s very helpful to start with a baseline measure of how accurate we can be without using a predictor. When we had a continuous outcome, we would just use the average as our most basic model. However, with a binary outcome, this won’t work all that well – instead, what we do is we take the most common outcome, and assume this occurs for everyone. If we do that, how accurate are we?

Here are the results we should get from our data:

Table 14.4: Predicted and Observed Outcomes; bold values indicate where we have guessed the outcome correctly.

Observed Values	Predicted Values	
	Incorrect answer	Correct answer
Incorrect answer	54	0
Correct answer	46	0

In the case of the test item data, we can see that out of 100 participants, 54 participants answered the question incorrectly. So, if we went ahead and assumed that nobody answered the question correctly, we would only be right 54 out of 100 times, so we’d have 54% accuracy. Not great! But this is good news for us – this basic model is pretty lousy, so we have lots of room for improvement. If we had 90% of people answering the item incorrectly, just guessing that everyone got it wrong would mean we’d be right 90% of the time, and it would be difficult to improve on that accuracy.

Next, we can use our statistical program to use our predictor to predict whether people get the answer right or wrong. Remember in Section 13.1 when we could use B values in a regression to provide a fitted value (i.e. where the point would be on our best fit line)? Our statistical

program is doing basically the same thing here, except it's calculating a probability and using that to guess whether they got the answer right – if the probability is over 50%, it guesses that they got the answer right, and if it's below 50%, it guesses that they got it wrong. We can then use those fitted values to see if our accuracy improves.

Here are the results we should get from our data:

Table 14.5: Predicted and Observed Outcomes; bold values indicate where we have guessed an outcome correctly.

Observed Values	Predicted Values	
	Incorrect answer	Correct answer
Incorrect answer	40	14
Correct answer	13	33

Using information from students' math scores, we can see that our model accurately predicts incorrect answers in 73 cases, and accurately predicts correct answers in 33 cases. Our model isn't perfect – it made an inaccurate prediction in 27 cases, but this means we have 73% accuracy, which is much better than our initial accuracy of 54%. So, in terms of practical implications, it seems like our model is useful!

We also have some values known as pseudo-R squares that function mostly like a typical R^2 – one of the better statistics for this is known as the Nagelkerke R square (or some programs call it the Cragg and Uhler R square... it's the same thing). In our example, we find Nagelkerke R square is .39, so about 39% of the differences in whether students get the question right is explained by their math score.

If I'm trying to explain logistic regression to laypeople, this is usually where I stop, because from here on out, things get pretty technical. Unfortunately for you, I consider you to be a technical audience, so we're going to dig into the results a little more. In most programs, you'll get a table that might look something like this:

Variables in the Equation							
		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	math_ability	.171	.037	20.935	1	<.001	1.187
	Constant	-8.711	1.898	21.063	1	<.001	.000

a. Variable(s) entered on step 1: math_ability.

Figure 14.3: SPSS output from a logistic regression

One useful value we get from most programs is something known as a Wald value². This is a type of chi-square value, and this is the value we typically use to determine whether the predictor's influence is significant. In research, Wald values are often reported, along with their respective p-value.

Just like with a regular regression, we also get an unstandardized coefficient (i.e. a B-value). But remember, our model isn't linear, it's an S-curve! So, we can't treat these values the same way as we do in a normal regression. Back when we were doing a linear regression, we were able to predict someone's score by using this equation:

$$\hat{y} = b_0 + b_1(x)$$

But, there's an important difference in logistic regression! We can still input a y-intercept and a slope, but the outcome is no longer someone's score. Instead, it is the logit of an event happening:

$$\text{logit} = b_0 + b_1(x)$$

Let's walk through an example to understand how this works. Imagine that we have a student who has earned a score of 81 on the math test. We can input that score into this equation to figure out the logit of them getting that item correct:

$$\text{logit} = b_0 + b_1(x) = -8.71 + .17(81) = 5.06$$

So, if someone has a score of 81 on the math test, the logit of them getting this item correct is 5.06. If we have someone with a point higher on the test, an 82, we know their logit increases by .17 (remember, one reason we use logit is to put our data on a straight line, so the logit increases in a linear way).

I don't know about you, but I would prefer not to work in logits because they are confusing and hard to explain. Instead, I'd prefer to talk about odds or probability. The good news is that it's pretty easy to calculate the odds – we just have to “exponentiate” this value:

$$e^{(b_0+b_1(x))}$$

Going back to our example, if we want to know the odds of someone getting this question right if they have a math score of 81, we take e to the 5.06 power:

$$e^{5.06} \approx 157.59$$

This means this person's odds of getting the item right are over 157 to one – in other words, it seems reasonable to predict they will get the item correct. If you prefer probability instead of odds, that's easy enough to convert:

$$\text{probability} = \frac{\text{odds}}{1+\text{odds}}$$

²SPSS provides a Wald test, but other programs, like R, provide a z-test instead. If you prefer to provide a Wald test value, the good news is you just square the z value to get the Wald test. The p-values are identical between the two tests, so you can report them as is.

$$probability = \frac{157.59}{158.59} \approx .994$$

The probability that this person will get that item correct are very high. Again, this value will never quite reach 100%, but it can get very close. If I were going to make a bet, I'd feel very comfortable betting that this person would get this test item correct.

Odds and probability are really helpful for explaining how we can predict an outcome given a value on the predictor. However, one downside here is that once we move away from logits, we are back to dealing with a curvilinear relationship. So we can no longer say that a one point change on our predictor leads to a certain change in the odds or the probability of an outcome. If you look at the red line on Figure 14.2, the line is steep in the middle, but flattens out at the ends. This means that for a one-point change in math ability, the probability changes a lot when a student's math ability is between 40 and 60. When it is below 30 or above 70, the probability changes very little.

The main point here is that just like with a linear regression, those B values are very helpful for estimating what someone's outcome is likely to be, but we don't tend to report them much in reports or articles. We generally stick to the Wald value, or the odds ratio, which I'll cover next.

Most programs will provide you with an odds ratio (OR), sometimes labeled as $\text{Exp}(B)$. This value is just the exponentiated slope in our equation ($e^{.17} \approx 1.19$). This tells us for a one point change in our predictor variable, how many times higher the odds are the event will happen. So this means that if a student is one point higher on their math test, the odds they will get this question correct go up by 1.19. And remember that because we're talking about odds, the "neutral point" for the odds ratio is a 1. Here, we see this value is a bit above 1, so this means that as math scores go up, the odds of getting this question right also increase. Odds ratios can be a little confusing when they are new to you, but researchers who deal with a lot of binary outcomes are used to seeing them and understanding them. For example, topics around healthcare and health behavior have a lot of binary outcomes, so those types of journals often have lists of odds ratios in their tables.

14.1.3 Multiple Logistic Regression

Just like with a standard linear regression, the beauty of logistic regression is that it's quite flexible, so we can use it with multiple variables, and with multiple types of variables (e.g., continuous, dichotomous). We can also do things like hierarchical or exploratory regression. I'll leave aside the hierarchical and exploratory pieces, but I do think walking through a multiple logistic regression can help solidify what you are learning in this chapter.

Let's return to the test item data set. Let's redo our logistic regression, but let's add the two additional predictors in – which class a participant is in, and whether or not they are international students. When I run a logistic regression, here is the output I see:

Table 14.6: Predicted and Observed Outcomes; bold values indicate where we have guessed an outcome correctly.

Observed Values	Predicted Values	
	Incorrect answer	Correct answer
Incorrect answer	47	7
Correct answer	7	39

Table 14.7: Logistic Regression Coefficients

	B	SE	Wald	df	Sig.	OR
class	.71	.74	.92	1	.34	2.04
math_ability	.27	.07	15.20	1	<.001	1.31
international	-4.14	.88	22.27	1	<.001	.016
constant	-12.89	4.34	15.42	1	<.001	.000
R^2	.73					

Let's start with the easier findings. First, we can see from the significance values that class doesn't appear to predict whether someone will answer this item correctly, but math ability and international status do. We can also see from our contingency table that we have continued to improve our predictions – with no predictors, we know that we'd only get it right about 54% of the time. With all three predictors, that value jumps up to 86%, so we also are going to be more accurate in making guesses about our participants outcomes if we have these predictors in our model. So, this is all good news for us so far!

The odds ratios also tell us that as these values go up, the odds of getting the answer correct also go up. We already know from before that as someone's math ability is higher, they are more likely to get the item correct. We also see that people in the class 2 are more likely to get the item correct (though it is too small of a difference to be statistically significant). Finally, and perhaps more worrisome here, is that our international students are unlikely to answer this item correctly (the B value is negative, and our odds ratio is less than 1). So, maybe we need to revisit the item to ensure it doesn't have a cultural bias of some sort that's preventing our international students from doing well.

Finally, if I wanted to predict whether someone was apt to get a question correct if they were in class 1, they were an international student, and they had a math ability of 60, I could do so:

$$\text{logit} = b_0 + b_1(x_1) + b_2(x_2) + b_3(x_3) = -12.88 + (1 \times .71) + (.27 \times 60) + (1 \times -4.14) = -0.11$$

$$\text{odds} \approx e^{-.11} \approx .90$$

$$\text{probability} \approx \frac{.90}{1+.90} \approx .47$$

In simple terms, it's slightly more likely this student will get the question wrong than they will get it right.

14.2 Writing the Results of a Logistic Regression

There's a little less consistency in how people share the results of a logistic regression.

- For a non-scientific audience, I recommend providing information about the contingency table, the pseudo-R square, and information about the odds or odds ratio.
- For a scientific audience, I recommend providing the Wald value, its significance, odds ratios, the pseudo-R square and information about the contingency table.
- Although b values are helpful in most applied situations, we don't typically share them in scientific reports.

A logistic regression indicates that the model improves prediction from 56% accuracy to 86% accuracy. Ultimately, we see that participants with high math scores and domestic students are more likely to answer the item correctly. About 73% of the differences in students answers can be explained by the variables in our model.

A logistic regression indicates that international status and math ability significantly predict how participants answer the question ($p < .001$). Class, however, is not predictive ($p = .34$). The model explains approximately 73% of the variance in responses. Refer to Table x for details about Wald values and odds ratios.

15 Appendix 1: Tables

15.1 Z-score Table

A z-score table shows the percentage of values to the left of a given z-score on a standard normal distribution. For negative z-scores, find the proportion for the absolute value of the z-score, and subtract that proportion from 1. As an example, for a z-score of -1.11, we find that proportion of +1.11, (which is .8665) and we subtract that value from 1 (which is .1335).

Table 15.1: Zscore table

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5081	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
.09	.8159	.8186	.8212	.8238	.8238	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

15.2 T-Table

This table provides critical t values for a t-test. To find the critical t-value for a t-test, locate the degrees of freedom on the left, and choose the column that identifies how many tails your test has, and what level you've set α at (.05 or .01).

Table 15.2: Student T Distribution

Degrees of Freedom	.01 (one-tail test)	.05 (1-tail test)	.01 (2-tailed test)	.05 (2-tailed test)
1	31.821	6.314	63.657	12.706
2	6.965	2.920	9.925	4.303
3	4.547	2.353	5.841	3.182
4	3.747	2.132	4.604	2.776
5	3.365	2.05	4.032	2.571
6	3.143	1.943	3.707	2.447
7	2.998	1.895	3.500	2.365
8	2.896	1.860	3.355	2.306
9	2.821	1.833	3.250	2.262
10	2.764	1.812	3.169	2.228
11	2.718	1.796	3.106	2.201
12	2.681	1.782	3.054	2.179
13	2.650	1.771	3.012	2.160
14	2.625	1.761	2.977	2.145
15	2.062	1.753	2.947	2.132
16	2.584	1.746	2.921	2.120
17	2.567	1.740	2.898	2.110

Degrees of Freedom	.01 (one-tail test)	.05 (1-tail test)	.01 (2-tailed test)	.05 (2-tailed test)
18	2.552	1.734	2.878	2.101
19	2.540	1.729	2.861	2.093
20	2.528	1.725	2.845	2.086
21	2.518	1.721	2.831	2.080
22	2.508	1.717	2.819	2.074
23	2.500	1.714	2.807	2.069
24	2.492	1.711	2.797	2.064
25	2.485	1.708	2.878	2.060
26	2.479	1.706	2.779	2.056
27	2.473	1.703	2.771	2.052
28	2.467	1.701	2.763	2.048
29	2.462	1.699	2.756	2.045
Large	2.327	1.645	2.575	1.960

15.3 F-Table

This table provides critical values for an F-test. Each column provides critical values for the numerator degrees of freedom (i.e., your within degrees of freedom, which should be the number of groups minus 1). The rows provides critical values for the denominator degrees of freedom. So, if you are testing $F(2,20)$, the critical value is 3.49.

Table 15.3: F Distribution Table

		Numerator df			
Denominator df		1	2	3	4
5		6.61	5.79	5.41	5.19
10		4.96	4.10	3.71	3.48
15		4.54	3.68	3.29	3.06
20		4.35	3.49	3.10	2.87
25		4.24	3.39	2.99	2.76
30		4.17	3.32	2.92	2.69
40		4.08	3.23	2.84	2.61
60		4.00	3.15	2.76	2.52
Infinity		3.84	3.00	2.60	2.37

15.4 Chi-square-Table

This table provides critical values for a chi-square test. Find your degrees of freedom in the leftmost column, and the associated p value indicates the critical value.

Table 15.4: Chi-Square Distribution Table

df	P
1	3.84
2	5.99
3	7.82
4	9.49
5	11.07
6	12.59
7	14.07
8	15.51
9	16.92
10	18.31

16 Summary

In summary, this book has no content whatsoever.

1 + 1

[1] 2

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